D00258.00017

Structural Airport Pavement NDT and Evaluation

BAKFAA Version 3.4 User's Manual

Prepared for:

Prepared by:

Applied Research Associates, Inc.
Mid-Atlantic Division
2628 Fire Road

Suite 300

Egg Harbor Township, NJ 08234

March 3, 2023

Contract No. 692M15-21-D-00004

Order No. 692M15-22-F-00129
Task Order 17
NDT for Structural Evaluation of Airport Pavements
Deliverable: 5.2.1.1 Update Upgraded BAKFAA User Manual, Help Files, and Presentation

Table of Contents

Chapter 1. Introduction to BAKFAA 1

Chapter 2. Installation of BAKFAA 2

2.1 Installation Requirements 2

2.1.1 Operating System Requirements 2

2.1.2 Hardware Requirements 2

2.2 Installation Process 2

2.3 How to Uninstall the BAKFAA Application 4

2.4 Running the Program 4

Chapter 3. BAKFAA Interface 6

3.1 BAKFAA Modular Interface 6

3.1.1 Interface Definitions 7

3.1.2 Configuring the Modular Interface 10

3.2 BAKFAA Version and FWD File Location 13

3.3 Pavement Layer Structure 14

3.3.1 Seed Modulus 18

3.3.2 Backcalculated Modulus 19

3.3.3 Poisson's Ratio 19

3.3.4 Interface Parameter 20

3.3.5 Pavement Layer Thicknesses 21

3.3.6 Moduli Changeable 21

3.4 Saving and Loading Pavement Section 21

3.5 Units 22

3.6 FWD Test Information 27

3.6.1 Load FWD Files 30

3.6.2 View FWD File 33

3.6.3 Convert FWD File to PDDX Format 34

3.6.4 FWD File Parser 35

3.7 Selection of FWD Drops for Backcalculation 39

3.8 Backcalculation Analysis 40

3.8.1 Iteration Tolerance 41
3.8.2 Iteration Number 42
3.8.3 Root Mean Square Error 42
3.8.4 Backcalculated Modulus 42
3.8.5 Calculated Deflections 42
3.8.6 .CSV Output File 43
3.8.7 Show Output 43
3.8.8 Abort an In-Progress Backcalculation 44
3.9 Batch Graph 45
3.10 Load Transfer Efficiency (LTE) Calculation 46
Chapter 4. Select Load and Run LEAF 49
4.1 Load and Gear Configuration 50
4.2 LEAF Output 53
Chapter 5. Sample BAKFAA Analysis 55
5.1 Flexible Pavement Example 55
5.2 Rigid Pavement Example 59
5.3 Composite Pavement Example 64
5.4 LTE Calculation Example 69
5.5 Error Messages 70
5.5.1 User Input Errors 70
5.5.2 FWD File Error 71
Chapter 6. Advice and Reminders 72
Chapter 7. References 73

List of Figures

Figure 2-1. BAKFAA installation welcome dialog box 3
Figure 2-2. BAKFAA installation dialog 3
Figure 2-3. BAKFAA in the Windows Start button program menu 5
Figure 2-4. BAKFAA Windows desktop shortcut icon 5
Figure 3-1. Default BAKFAA interface 6
Figure 3-2. Interface showing window types and floating tab 8
Figure 3-3. Interface showing the header, toolbar, and modules 9
Figure 3-4. Interface showing document and tab types, and grouping 9
Figure 3-5. Adjusting interface module size 10
Figure 3-6. Positioning a module in the GUI (a) adding to group of tabs (b) placing as adjacent tab (c) placing as tab on edge of window (d) collapsing a tab (e) floating window/tab 13
Figure 3-7. Changing title order in module group 13
Figure 3-8. a) Default interface displaying BAKFAA version and b) location/name of the file in use after loading the file 14
Figure 3-9. Defining pavement structure in BAKFAA 15
Figure 3-10. Selecting pavement type in BAKFAA 15
Figure 3-11. Adding or deleting a pavement layer in BAKFAA 16
Figure 3-12. Layer Property menu used to add a pavement layer 17
Figure 3-13. Pavement structure submodule 18
Figure 3-14. Saving and loading pavement sections 21
Figure 3-15. Example of pavement section .XML file containing FWD test and pavement structure data 22
Figure 3-16. Selection of Units 23
Figure 3-17. Default FWD test setup shown in English units 24
Figure 3-18. Default FWD test setup shown in Metric units 25
Figure 3-19. FWD file sub-window shown in English units 26
Figure 3-20. FWD file sub-window shown in Metric units 27
Figure 3-21. Default FWD test setup 28
Figure 3-22. Exclusion of sensor warning 29

Figure 3-23. Calculated deflections and updated deflection graph after backcalculation analysis 30
Figure 3-24. (a) Loading a FWD file and (b) Selecting a drop 32
Figure 3-25. Viewing details of an FWD station 33
Figure 3-26. FWD file sub-window 34
Figure 3-27. Example of delimited data file and the extraction of software edition 36
Figure 3-28. Extraction of software edition from Dynatest R32 file with fixed-width format 36
Figure 3-29. Example a two-level split extraction 37
Figure 3-30. Example of functionality of "Translate" section of the XML file 38
Figure 3-31. Example of functionality of "AssignedData" section of the XML file 38
Figure 3-32. FWD drop selection in the "FWD Drops" module and the "Select Drops", "Select All", and "Clear All" buttons 40
Figure 3-33. FWD drop selection using the "Select Drops" button and the "Drop No." menu 40
Figure 3-34. Start batch backcalculation analysis 41
Figure 3-35. "Show Output" sub-window displaying backcalculation results of a batch analysis 44
Figure 3-36. "Batch Graph" sub-window displaying measured and calculated deflections 46
Figure 3-37. "Batch Graph" sub-window displaying backcalculated modulus values 46
Figure 3-38. LTE cases 47
Figure 3-39. LTE calculation in BAKFAA 48
Figure 4-1. Pavement structure and evaluation depth settings for LEAF analysis 49
Figure 4-2. Load Library for LEAF analysis 51
Figure 4-3. Summary of items in the LEAFAircraft.Ext file for a B-747-400 example 52
Figure 4-4. Schematic of B-747-400 Landing Gear Parameters in the LEAFAircraft.Ext file 53
Figure 5-1. Backcalculation results for flexible pavement example (first attempt) 57
Figure 5-2. Backcalculation results for flexible pavement example (second attempt) 58
Figure 5-3. Backcalculation results for flexible pavement example (third attempt) 58
Figure 5-4. Backcalculation results for rigid pavement example (first attempt) 60
Figure 5-5. Backcalculation results for rigid pavement example (second attempt) 61

Figure 5-6. Backcalculated layer modulus values for all test points for drop no. 3 in rigid pavement example 62
Figure 5-7. Deflection basin at DMI of 6,600 ft. showing unusual deflections in the rigid pavement example 63
Figure 5-8. FWD data for DMI of 6,600 ft. removed using the “View FWD File” sub-window in the rigid pavement example 63
Figure 5-9. Backcalculation analysis using default seed modulus values in the composite pavement example 66
Figure 5-10. Backcalculated layer modulus values for drop no. 2 at all test points using default seed modulus in the composite pavement example 66
Figure 5-11. Backcalculation analysis using temperature-adjusted P-401 modulus in the composite pavement example 67
Figure 5-12. Backcalculated layer modulus values for drop no. 2 at all test points using temperature-adjusted P-401 seed modulus in the composite pavement example 67
Figure 5-13. Backcalculation analysis using fixed P-401 modulus value in the composite pavement example 68
Figure 5-14. Backcalculated layer modulus values for drop no. 2 at all test points using fixed P-401 modulus value in the composite pavement example 68
Figure 5-15. LTE calculation example 70

List of Tables

Table 2-1. CPU and RAM requirements for running BAKFAA 3.4 2
Table 2-2. List of installed application files 4
Table 3-1. Typical modulus values and ranges for paving material 19
Table 3-2. Default seed modulus and Poisson's ratio values for pavement layers in BAKFAA 20
Table 3-3. English and Metric units used for illustration of different entities in BAKFAA 23
Table 3-4. File types supported by BAKFAA 31
Table 3-5. Available specifications for KUAB and CarlBro file types 39
Table 4-1. Critical locations in analysis of pavement structure responses 50
Table 5-1. Composition of pavement layers for the flexible pavement example 55
Table 5-2. Flexible pavement example backcalculated layer modulus values and statistics..... 59
Table 5-3. Composition of pavement layers for the rigid pavement example..... 59
Table 5-4. Rigid pavement example layer modulus values and statistics..... 64
Table 5-5. Composition of pavement layers for the composite pavement example..... 64
Table 5-6. Composite pavement example backcalculated layer modulus values and statistics for three approaches 69

List of Acronyms

BAKFAA	The FAA software to perform backcalculation of pavement material properties using Heavy Weight Deflectometer/Falling Weight Deflectometer data
CPU	Central Processing Unit
CSV	Comma Separated Values
GUI	Graphical User Interface
LEAF	FAA Layered Elastic Analysis program
LTE	Load Transfer Efficiency
FWD	Falling Weight Deflectometer
HMA	Hot Mix Asphalt
HWD	Heavy Weight Deflectometer
PCC	Portland Cement Concrete
RAM	Random Access Memory
RMSE	Root Mean Square Error
XML	Extensible Markup Language

CHAPTER 1. INTRODUCTION TO BAKFAA

BAKFAA is a computer program developed by the Federal Aviation Administration (FAA) for backcalculation of pavement layer moduli using heavy weight deflectometer/falling weight deflectometer (HWD/FWD) data. BAKFAA uses the FAA Layered Elastic Analysis Program (LEAF) as the forward method to simulate the FWD load and to calculate pavement deflections. The backcalculation process is based on an iterative mathematical process to find a combination of layer moduli such that the root mean square (RMS) error of the measured and LEAF computed deflection basins is minimized. BAKFAA employs a downhill multi-dimensional simplex as the optimization method. The minimization procedure is a direct implementation of sets of routines in Section 10.4 of "Numerical Recipes in Fortran" by Press et al, 1992. The routines were translated from Fortran into Visual Basic to be applied in the BAKFAA program. BAKFAA Version 3.4 (BAKFAA 3.4) was developed in Microsoft Visual Studio 2019 and compiled under .NET Framework 4.5.

CHAPTER 2. INSTALLATION OF BAKFAA

2.1 Installation Requirements

BAKFAA is developed in Microsoft Visual Studio 2019 using the Visual Basic .NET programming language (VB>NET) and compiled under .NET Framework 4.5. BAKFAA installation must meet .NET Framework 4.5 basic installation requirements.

2.1.1 Operating System Requirements

- Windows XP Home or Windows XP Professional, both with Service Pack 2 or later.
- Windows Server 2003 family with Service Pack 1 or later.
- Windows Vista.
- Windows Server 2008.
- Windows 7.
- Windows 8.
- Windows 10.

2.1.2 Hardware Requirements

Table 2-1 lists the Central Processing Unit (CPU) and Random Access Memory (RAM) requirements for running an application under the .NET Framework 4.5.

Table 2-1. CPU and RAM requirements for running BAKFAA 3.4

	CPU Required	RAM Required
Minimum	Pentium 400 MHz	96 MB
Recommended	Pentium 1 GHz or higher	256 MB or more

2.2 Installation Process

Use the following link to download and save the BAKFAA application setup package from the FAA website:

https://www.airporttech.tc.faa.gov/Products/Airport-Pavement-Software-Programs

Administrative rights are required to install BAKFAA. Once installed, the user does not require administrative rights to run BAKFAA.

Double click on BAKFAA Installer.msi to launch BAKFAA installation. The installation process consists of a series of dialog boxes similar to the one shown in Figure 2-1. Follow the prompts to install the software. A system dialog box asking if the user wants to allow the program to make changes to the computer may appear. Click “Yes” to continue the installation process. The default installation directory is **C:\Program Files\FAA\BAKFAA**. The installer will create this directory if it does not already exist. The user can change the default directory by providing a different address. The user may be required to enter an administrator password at this point (Figure 2-2). Click the “Finish” button and exit the setup wizard.



Figure 2-1. BAKFAA installation welcome dialog box

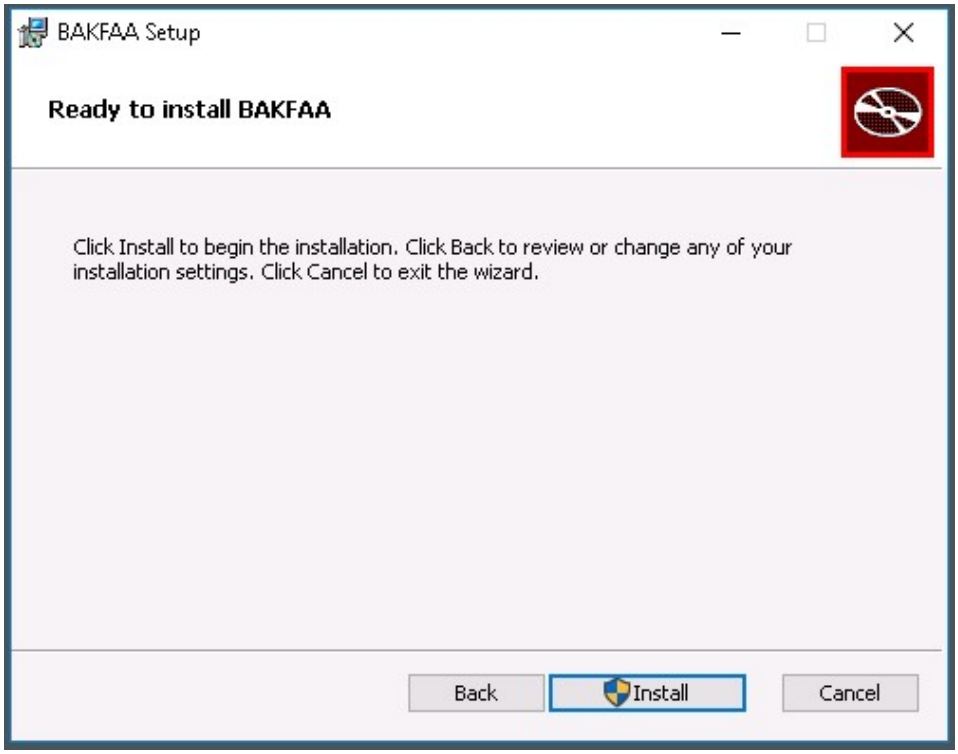


Figure 2-2. BAKFAA installation dialog

Upon installation completion, the application files listed in Table 2-2 are created in the BAKFAA installation folder.

Table 2-2. List of installed application files

No.	File Name	Description
1	BAKFAA.pdb	Program Debug Database containing BAKFAA debug info
2	BAKFAA.xml	XML document containing identification & summary of functionality of subroutines and classes added to BAKFAA in Phase II
3	LEAFClassLib.dll	Class Library containing LEAF
4	LEAFClassLib.pdb	Program Debug Database containing LEAF debug info
5	LEAFAircraft.Ext	ASCII Text file containing aircraft gear configurations
6	System.Windows.Controls.DataVisualization.Toolkit.dll	Reference for user interface
7	System.Windows.Controls.Input.Toolkit.dll	Reference for user interface
8	System.Windows.Controls.Layout.Toolkit.dll	Reference for user interface
9	System.Windows.Interactivity.dll	Reference for user interface
10	WPFToolkit.dll	Reference for user interface
11	XMLOutput_template.xml	Template for saving application data

2.3 How to Uninstall the BAKFAA Application

To uninstall the BAKFAA application, click the BAKFAA Installer.msi file and select “Remove” when the BAKFAA setup dialog box appears. Confirm the selection by clicking “Next” and the removal process will begin.

2.4 Running the Program

BAKFAA can be run from the start menu or desktop. By default, the installer creates a start menu item called BAKFAA. To run the program from the start menu, click the Windows Start button, locate BAKFAA in the Windows program menu as shown in Figure 2-3, and click the BAKFAA menu item to start the program.

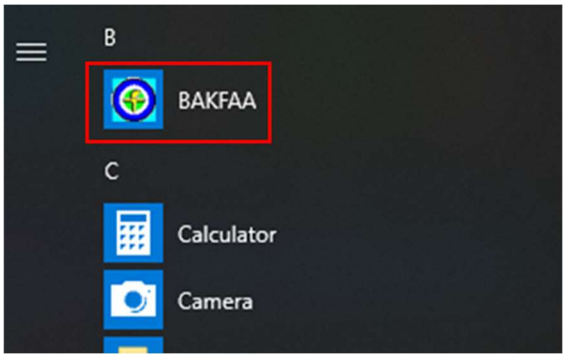


Figure 2-3. BAKFAA in the Windows Start button program menu

The installer also creates a shortcut icon on the desktop called “BAKFAA” by default, as shown in Figure 2-4. To run from the desktop, double click the BAKFAA icon to start the program.



Figure 2-4. BAKFAA Windows desktop shortcut icon

CHAPTER 3. BAKFAA INTERFACE

The BAKFAA interface (Figure 3-1) consists of modules that display pavement section, FWD data file, and backcalculation analysis information in a user-configurable and easily understood format. Upon opening the software, a sample pavement structure is loaded with default layers and material properties. The user can edit this structure and save it for later use or load a previously saved structure from a file. The user can load an FWD data file, review its information, edit the data, and save the updated file. When performing an analysis, the user can select an individual FWD drop and perform a single backcalculation analysis or select multiple drops and perform batch analysis. The Load and Run LEAF option allows the user to perform layered elastic analysis.

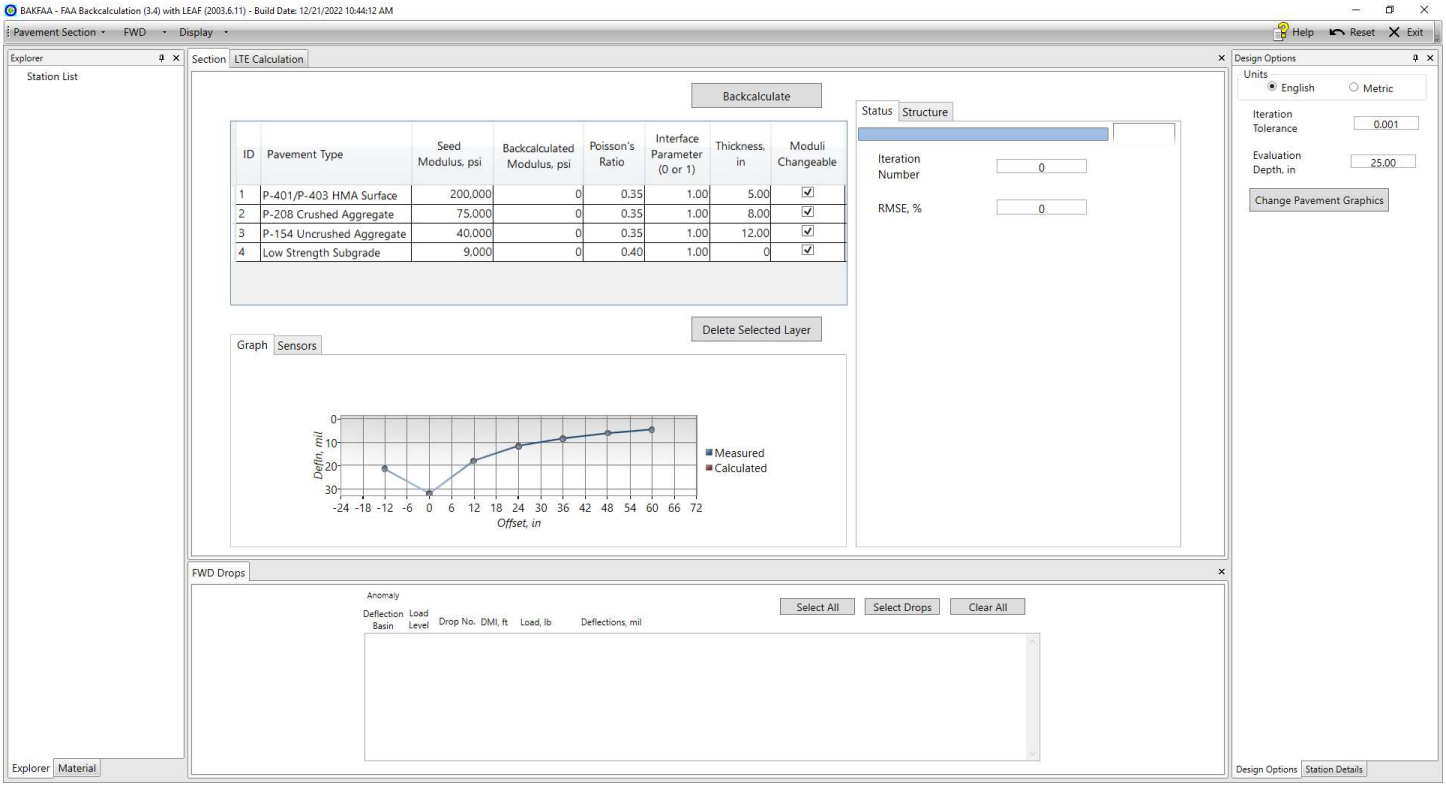


Figure 3-1. Default BAKFAA interface

3.1 BAKFAA Modular Interface

BAKFAA's graphical user interface (GUI) received an overhaul for version 3.4. Instead of a single main window, program functions are divided into 'modules' that each serve a specific purpose. The configuration of the modules is user adjustable. Each module can be resized, moved, and even removed to suit the user's preferences using an intuitive control set. The GUI is unified by a top toolbar which includes main program controls. The "Reset" button in the top right corner resets the GUI to the default layout (including position and size) while all input and output information is retained. The modular interface only changes how BAKFAA looks; the program's functions remain the same.

The following modules are included in BAKFAA:

- Section – shows pavement structure, deflection information, and backcalculation controls
- FWD Drops – displays the FWD file data (if a file is loaded) and drop selection controls
- Select Load – displays the LEAF load library for various loading configurations
- Design Options – includes design and display controls
- LTE Calculation – performs Load Transfer Efficiency (LTE) calculations
- Explorer – lists FWD stations and drops
- Material – presents available pavement materials
- Station Details – shows information for all drops at the selected location

3.1.1 Interface Definitions

The user has a great deal of freedom to alter the GUI, allowing for easier and faster workflow according to individual preference. To describe the available options, the relevant terms for the BAKFAA user interface are defined in this section.

- *Window* – Contains the entire BAKFAA program (see Figure 3-2).
    - *Main Window* – Contains the header and toolbar, at minimum, and contains all modules by default.
    - *Floating Window* – Auxiliary window containing at least one module.
    - *Sub-Window* – A single window used in certain display and output functions, ex. the "Batch Graph" sub-window. This cannot be controlled like a module.
- *Header* – Information shown at the top of the main window. This cannot be moved.
- *Toolbar* – Contains options and drop-down menus in main window. This cannot be moved.
- *Module* – A single interface unit that corresponds to a certain function. This can be adjusted by the user (see Figure 3-3).
    - *Title* – The name of the module. The location of the title depends on the module position type.
    - *Submodule* – Tabbed interface item within a module that cannot be adjusted.
- *Module Position Type* – How a module is positioned, which controls how it is displayed. All module position types are defined below. Position has no effect on module function.
    - *Document* – Embedded in the main window. Its title is positioned at the top of the module in a tabbed format (see Figure 3-4).
    - *Tab* – Shows title at the top, as well as in a tabbed format at the bottom when grouped (see Figure 3-4). Can be configured in the different ways described below.
        - *Docked Tab* – Embedded in the main window.
        - *Collapsible Tab* – Embedded on the edge of the main window. The tab pops out when its title is clicked, and auto-hides when unselected.
        - *Floating Tab* – A single tab or multiple tabs in a floating window (see Figure 3-2).
- *Module Group* – A set of modules that are configured to use the same area in a window. A module that does not share space with any other module is called ungrouped (see Figure 3-4).

- *Menu* – Separate or drop-down box that appears for further user input when using some functions. For example, the FWD "Select Drops" button opens the drop selection menu.

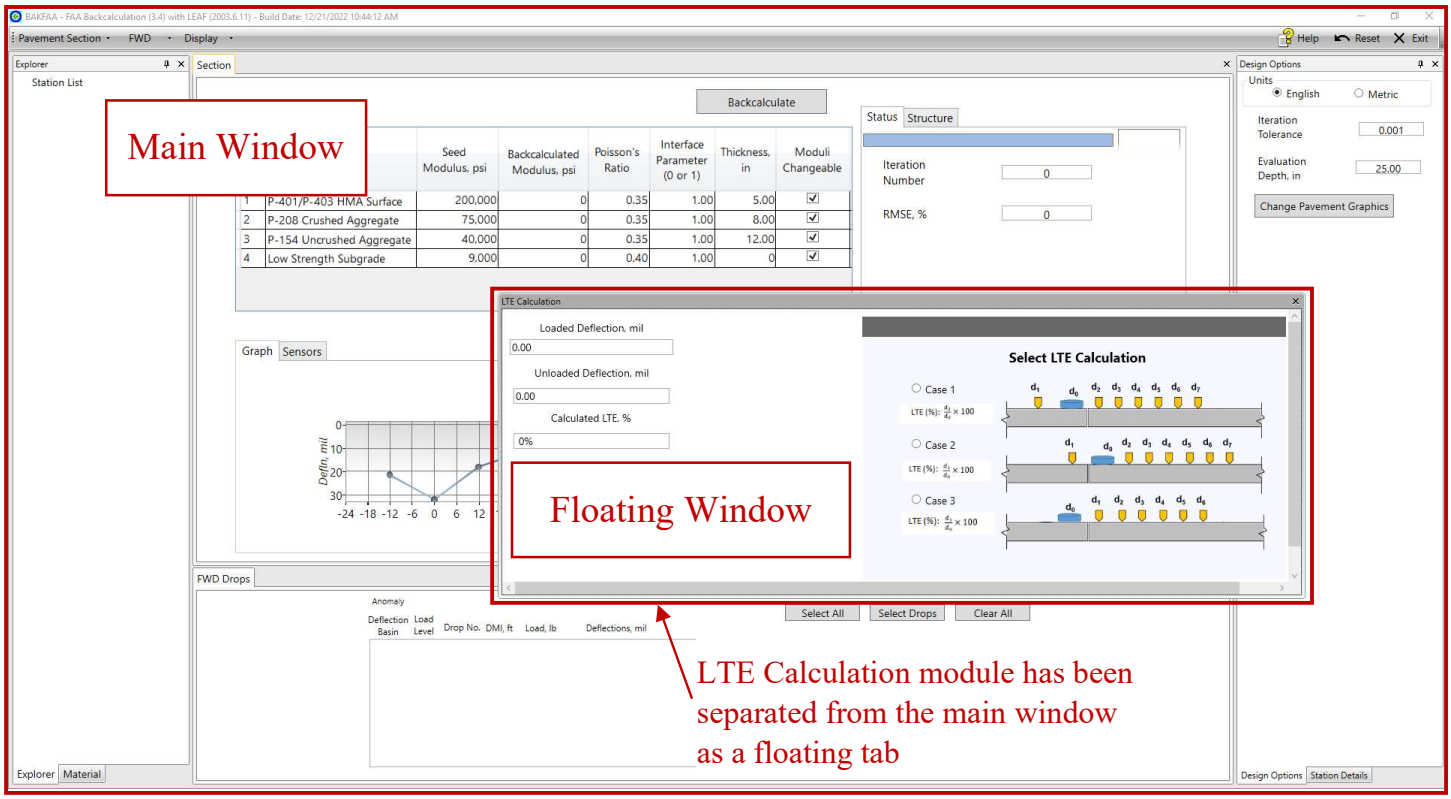


Figure 3-2. Interface showing window types and floating tab

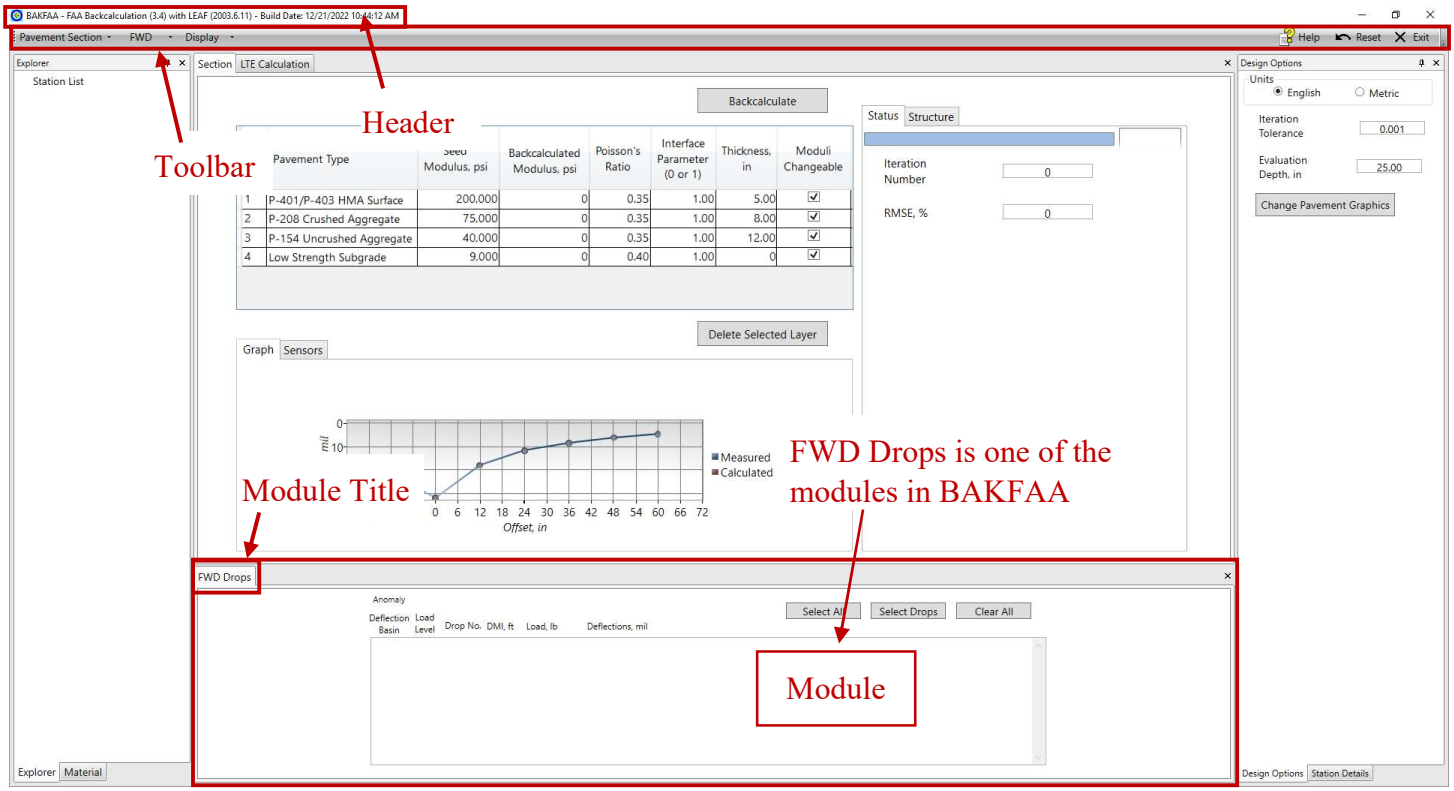


Figure 3-3. Interface showing the header, toolbar, and modules

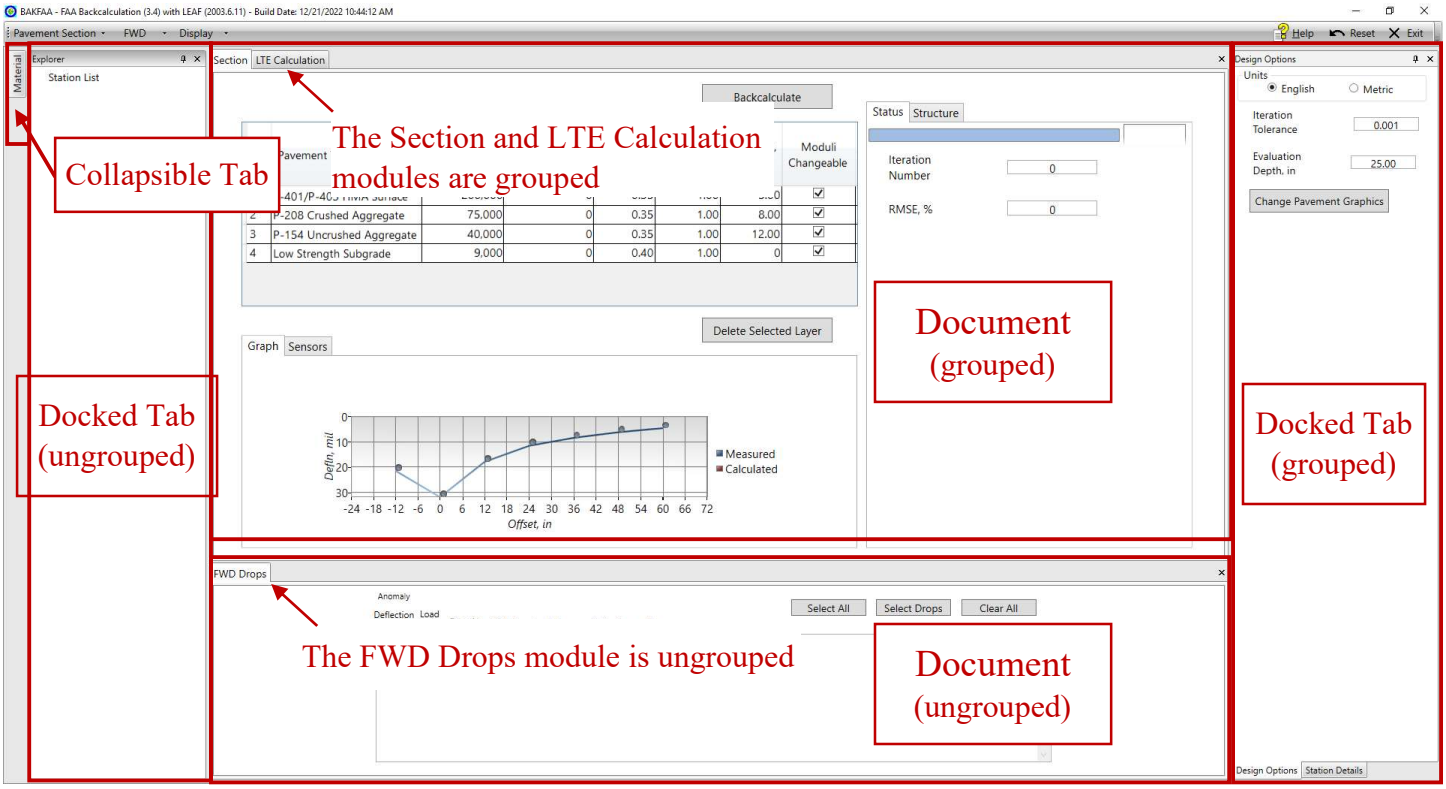


Figure 3-4. Interface showing document and tab types, and grouping

3.1.2 Configuring the Modular Interface

To resize a module, click and drag the module border. Other modules will change size to fit as necessary. An example is shown in Figure 3-5 increasing the size of the left-side tab, which causes the centrally located modules to shrink.

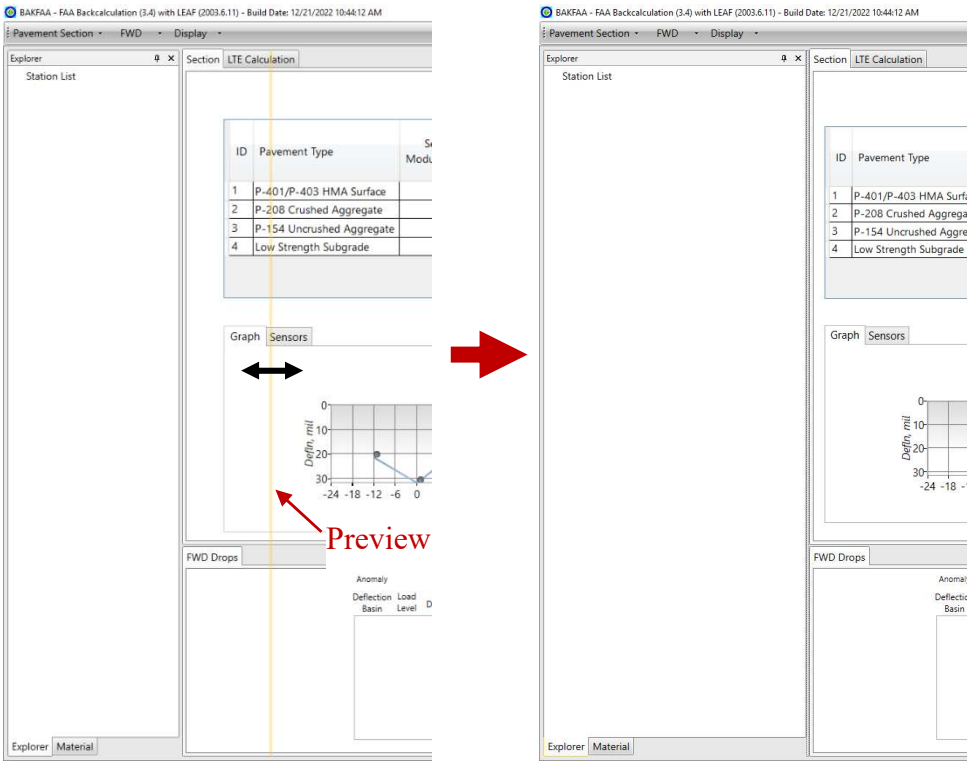
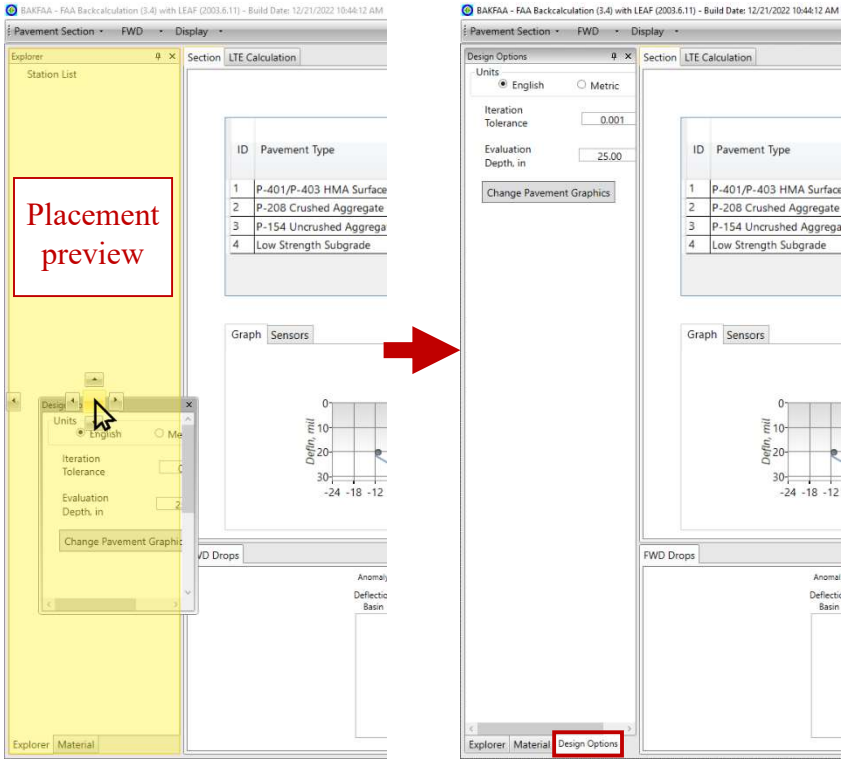


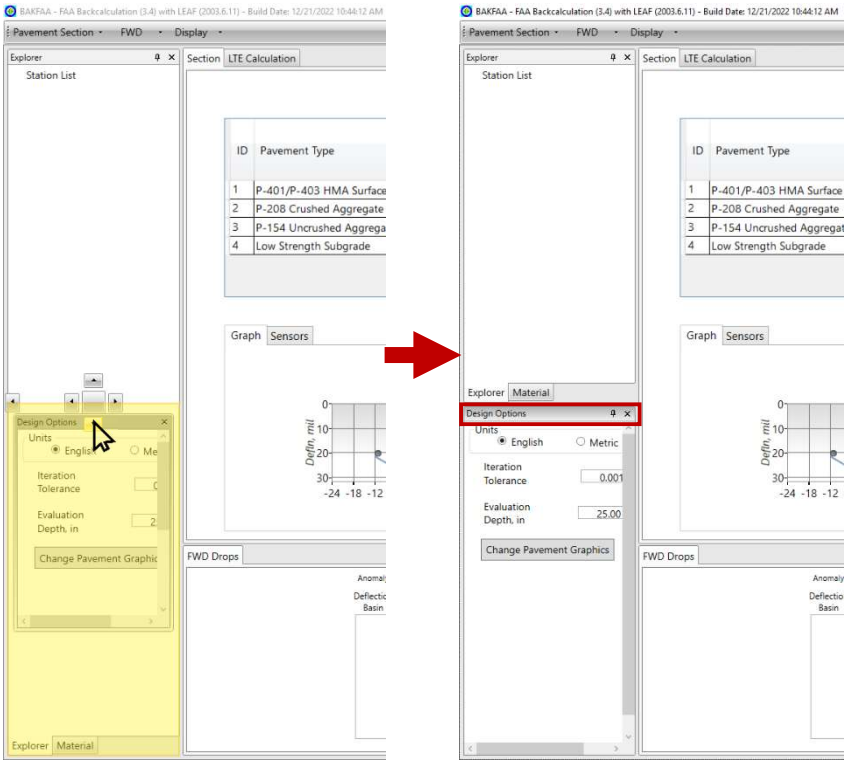
Figure 3-5. Adjusting interface module size

Modules have several possible position types as defined in Section 3.1.1: an embedded document, a docked tab, a collapsible tab, or a floating tab. All types are adjusted with the same controls. To change the position type of a module, click and drag the module title and move it to the desired location. Hovering over a location will show a preview of the module placement in yellow. Alternatively, the position type of a tab can be change using the drop-down menu in the top-right corner of that tab. In addition, the user can toggle between a docked tab and a collapsed tab using the "pin" button. Modules can be closed by clicking the "X" in the upper right corner, which will remove them from view. To restore closed modules to default view, click the "Reset" button.

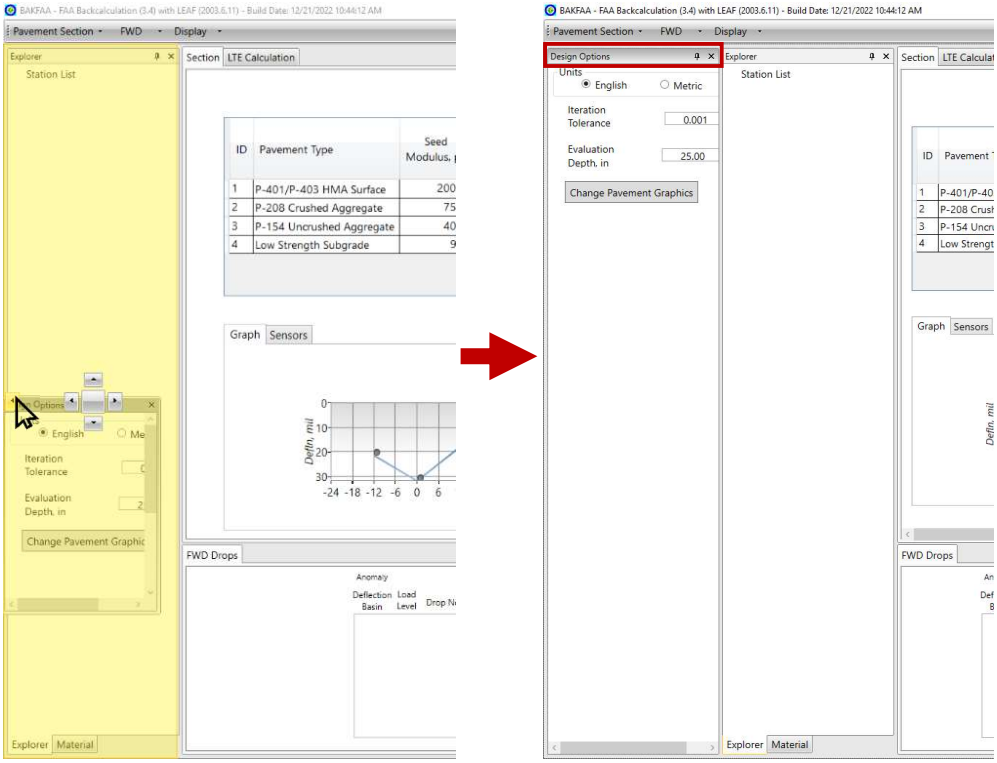
When moving a module, there are location markers in each module's area as shown in Figure 3-6 which correspond to the position types. Dropping a module on the center square adds it to the module group. The arrows around the center square place the module as an adjacent module. Adjacent modules are placed as the same type as the parent module; a module placed from a document will be a document, and the same is true for tabs. Arrows are also available around the edges of the window which will place the module as a tab on that edge. Examples are shown in Figure 3-6 (a) through (d) moving the "Design Options" module from the right side of the GUI (default position) to various locations on the left side. A module can be configured as a floating tab by releasing it anywhere other than on a location marker, as is shown in Figure 3-6 (e).



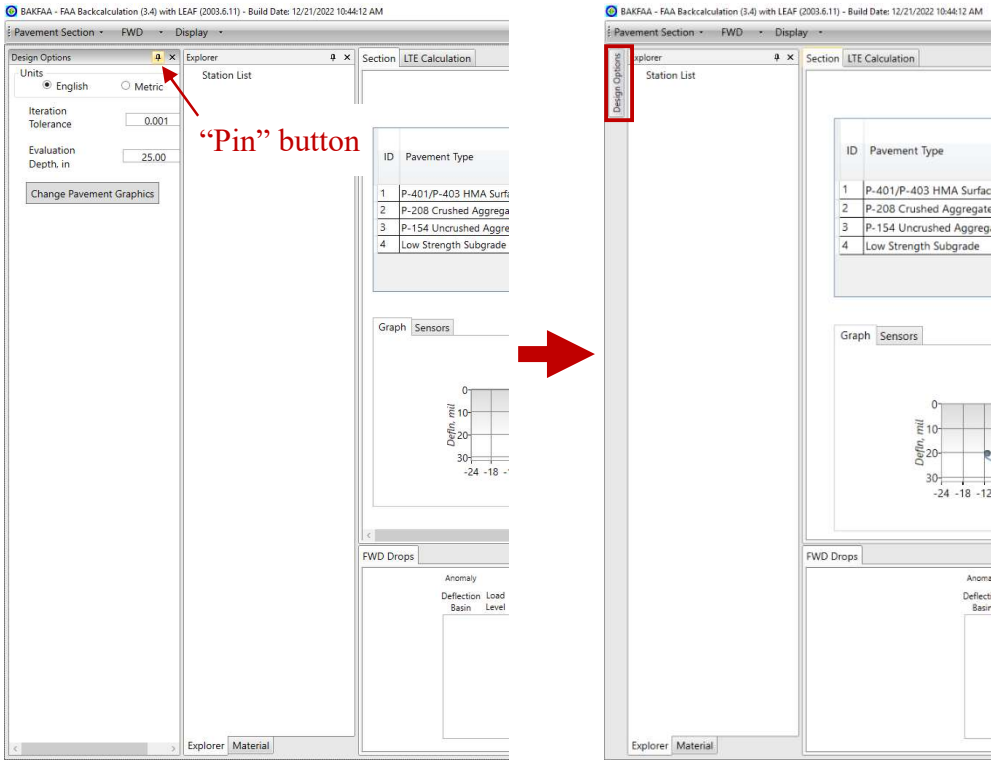
(a) Adding to group of tabs (center square)



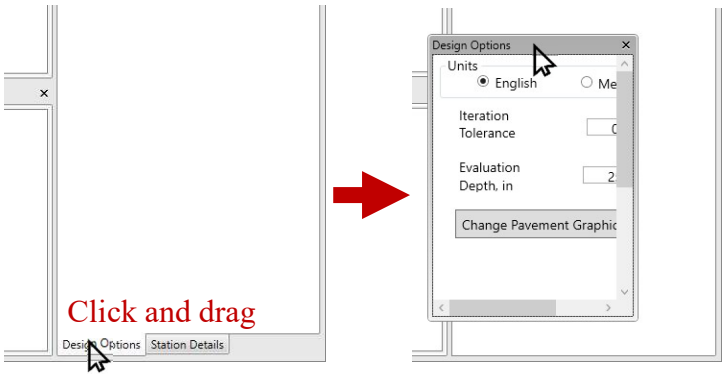
(b) Placing as adjacent tab (surrounding arrows)



(c) Placing as tab on edge of window (edge arrows)



(d) Collapsing a tab (using the "pin" button)



(e) Floating window/tab

Figure 3-6. Positioning a module in the GUI (a) adding to group of tabs (b) placing as adjacent tab (c) placing as tab on edge of window (d) collapsing a tab (e) floating window/tab

The order of modules in a group can be changed by clicking and dragging the titles to the desired order as seen in Figure 3-7.

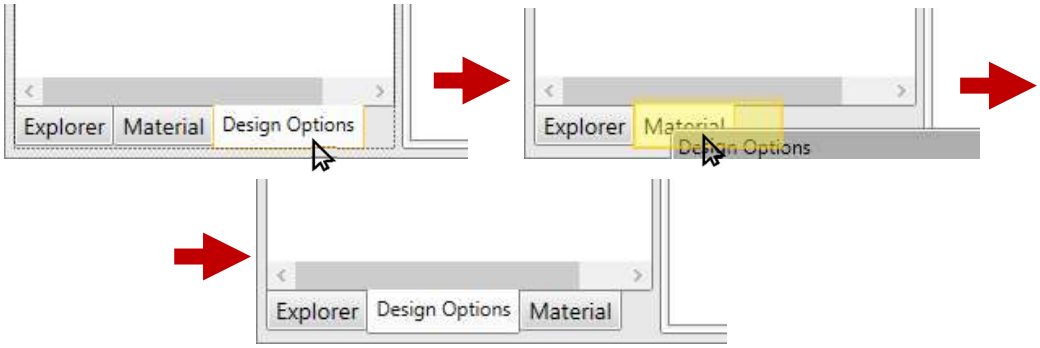
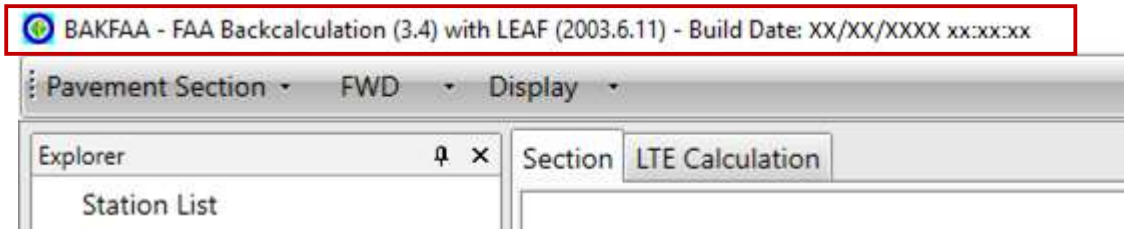


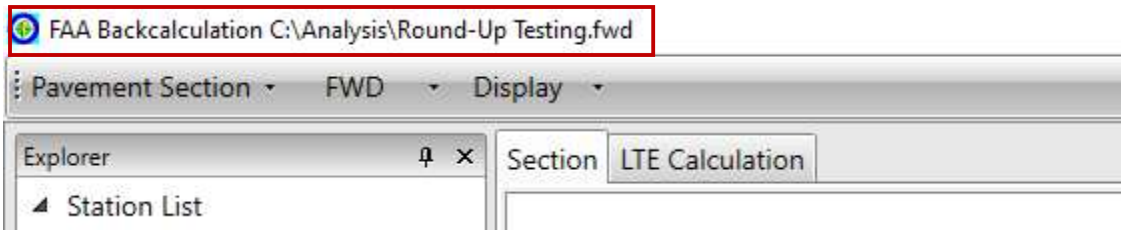
Figure 3-7. Changing title order in module group

3.2 BAKFAA Version and FWD File Location

The program header at the top of the window displays the version and build date of BAKFAA that is currently in use, highlighted with a red box in Figure 3-8. After loading an FWD file or .XML file, it displays the location and name of the file. The address is provided for user's information only. The user cannot change the appearance.



(a) Version name (default)



b) After loading file

Figure 3-8. a) Default interface displaying BAKFAA version and b) location/name of the file in use after loading the file

3.3 Pavement Layer Structure

The user must define the pavement layer structural information for use in backcalculation and LEAF analysis. The pavement structure is a set of layers with individual thicknesses and material properties that make up the pavement section. Defining the pavement structure requires selecting a number of layers and setting the pavement layer type, seed modulus, Poisson's ratio, thickness, and the interface parameter for each in the area highlighted with a red box in Figure 3-9. BAKFAA includes a library of pre-defined standard layer materials with default material properties for customizing pavement structures, but also allows the user to define a layer directly with the 'User Defined' type. Most of the pre-defined material types correspond to FAA specification items found in AC 150/5370-10H, "Standard Specifications for Construction of Airports". The library also includes four subgrade types ranging from ultra-low to high strength material. It should be noted that BAKFAA does not distinguish between flexible and rigid pavements. The user must account for this when defining the pavement structure and its layer attributes.

The user can change the pavement layer type from a drop-down menu under the "Pavement Type" column in the "Section" module, as shown in Figure 3-10. The user can add a layer from the drop-down menu under the "Pavement Type" column (see Figure 3-11) which will open the "Layer Property" menu shown in Figure 3-12. The material and position in the structure relative to the selected layer must be chosen. Alternatively, the user can drag and drop material names from the "Material" module to add layers to the structure. The user can delete a layer using the "Delete Selected Layer" button as illustrated in Figure 3-11. The minimum number of layers is three and the maximum is six. Most layer types can be placed at any position in the pavement structure, with the exception of a subgrade layer which can only be set as the bottom of the structure. The user must assign pavement structure properties for each layer. BAKFAA will provide a warning "Non-numerical values are not acceptable" if any of the property boxes are left blank. Regardless of the layer type assigned to the last layer in the pavement structure, BAKFAA treats the last layer as a semi-infinite layer and assigns 0 for the thickness.

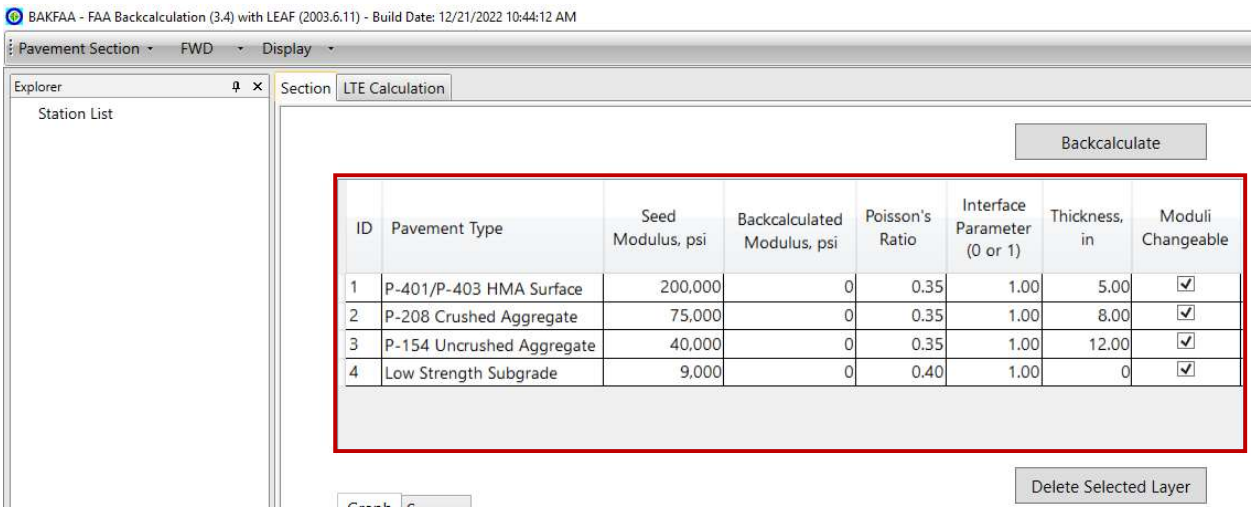


Figure 3-9. Defining pavement structure in BAKFAA

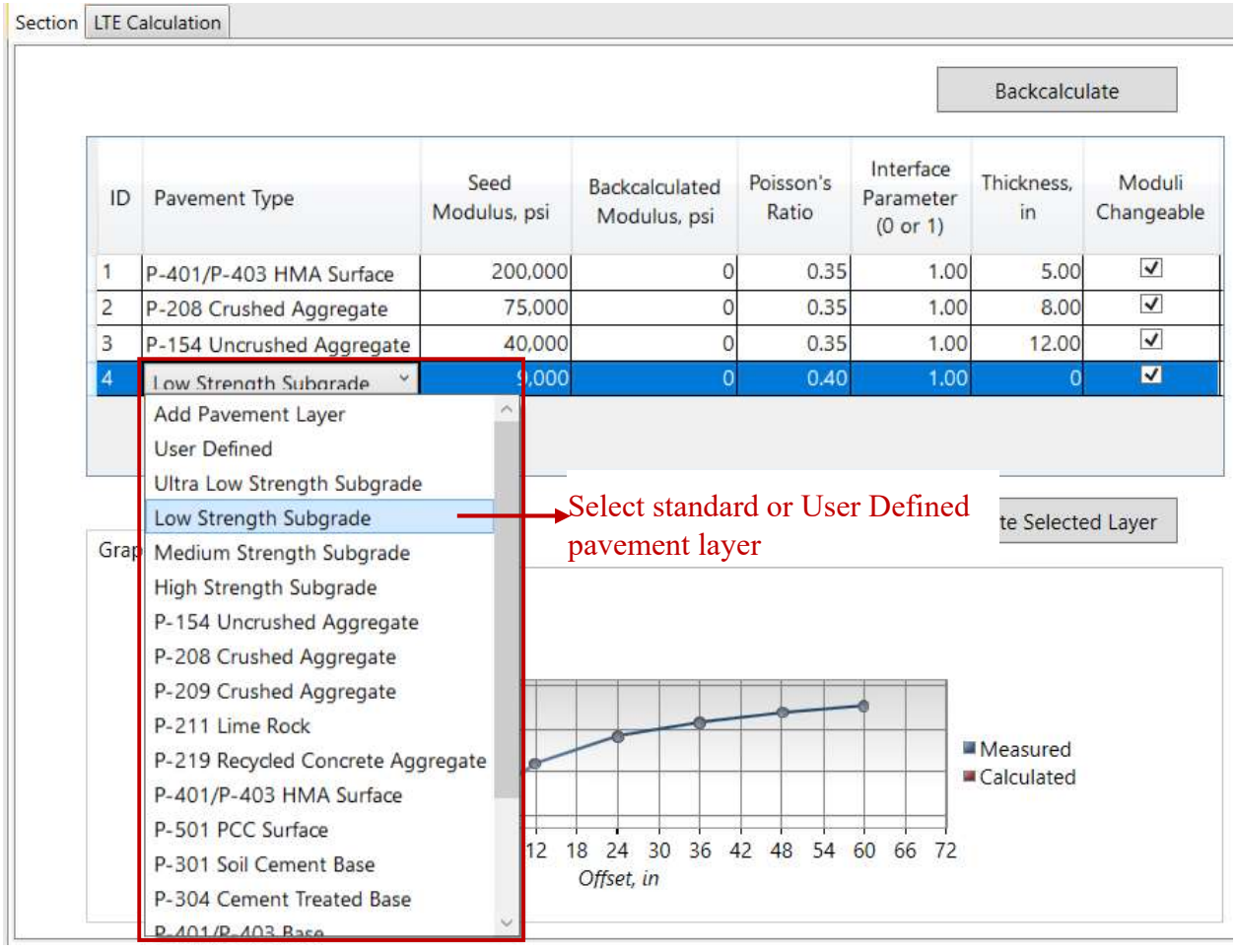


Figure 3-10. Selecting pavement type in BAKFAA

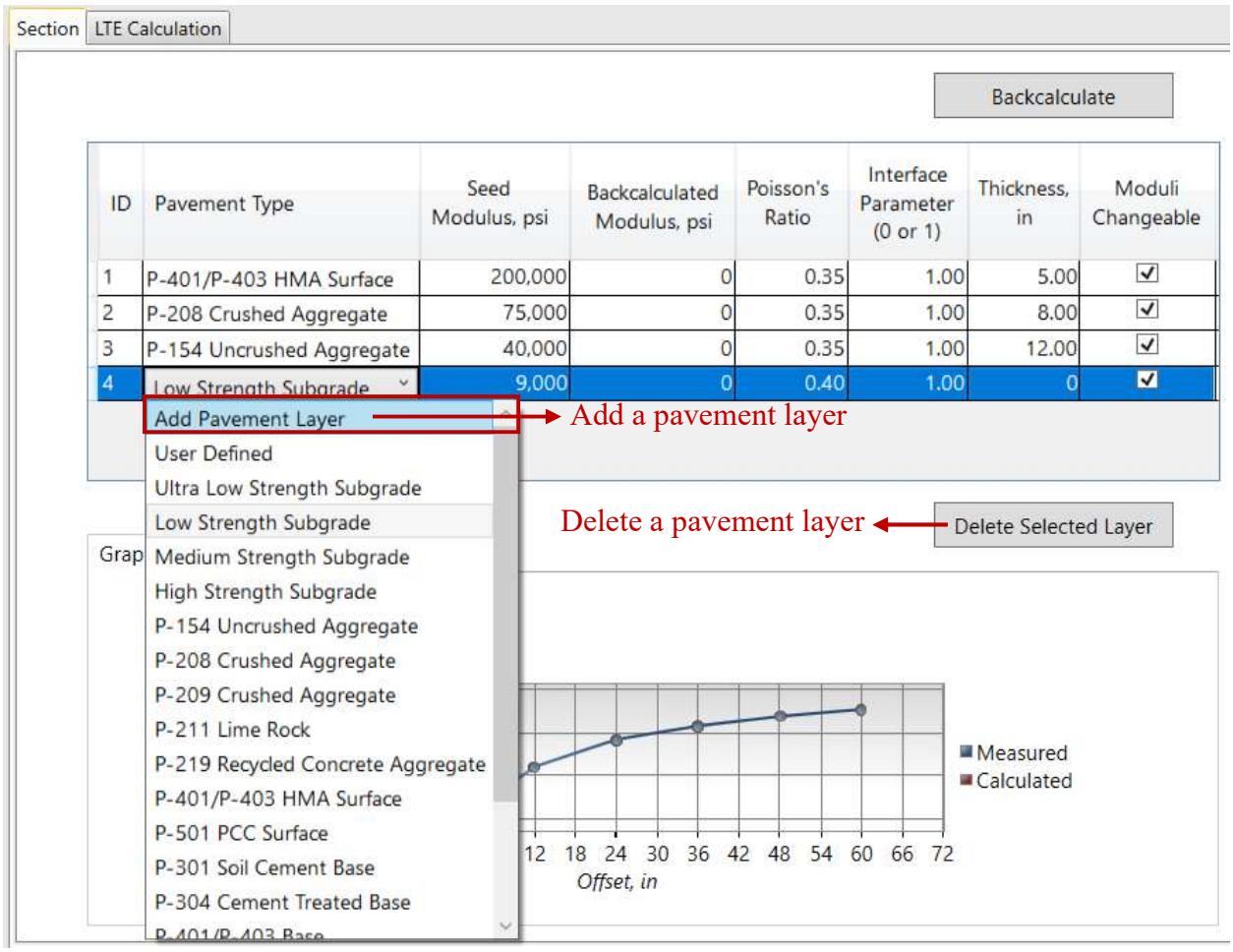


Figure 3-11. Adding or deleting a pavement layer in BAKFAA

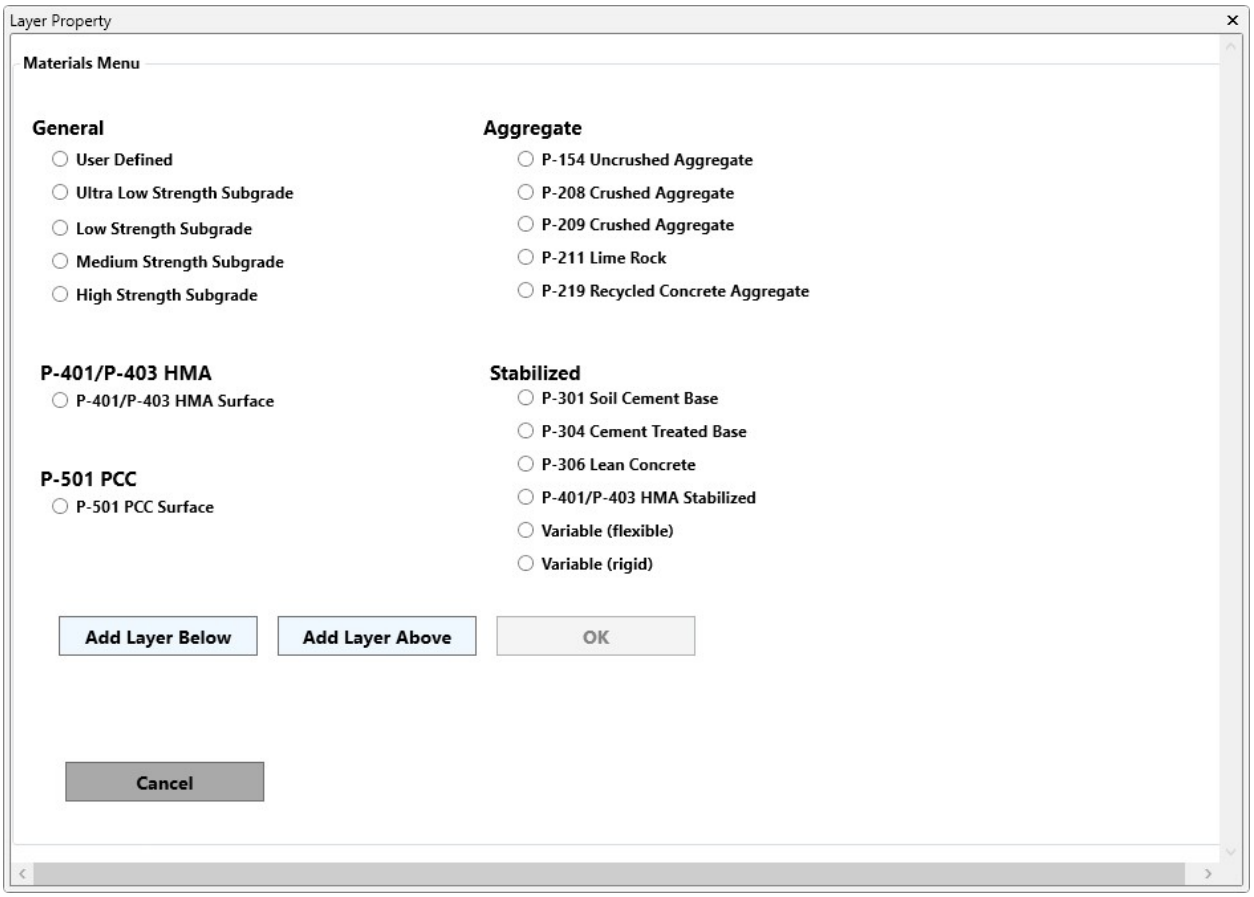


Figure 3-12. Layer Property menu used to add a pavement layer

To supplement the pavement layer table, the structure is also depicted in a graphical format as a pavement cross section in the "Structure" submodule as seen in Figure 3-13. It presents the material type as well as the thickness and seed modulus for each layer. Graphics are applied according to the general type of material. The user can toggle between hatched (default) graphics and photorealistic graphics with the "Change Pavement Graphics" button in the "Design Options" module. Layer thicknesses are reflected proportionally in the figure, with the infinite bottom layer displayed as 12 inches (305 mm) thick. Clicking on a pavement layer opens the "Layer Property" menu (see Figure 3-12) which can be used to change the material or add a pavement layer.

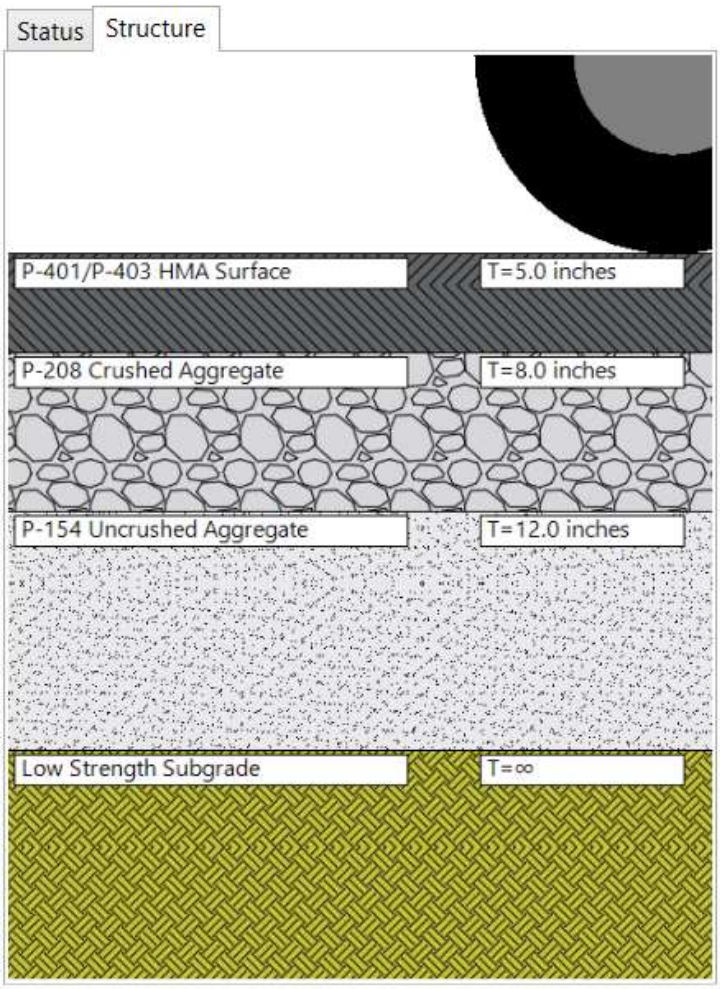


Figure 3-13. Pavement structure submodule

3.3.1 Seed Modulus

BAKFAA requires a seed modulus value for each pavement layer. Seed moduli are initial modulus values used in the backcalculation iterative procedure. The pre-defined materials have default seed values, but these can be edited by the user to any value greater than zero. BAKFAA will provide a warning if the user inputs zero or a negative value. If the user edits the seed modulus value, BAKFAA will automatically reclassify the material as 'User Defined'. Seed modulus values are generally assigned based on the best information available from the material properties, such as historic data, material aging, level of damage to each layer, temperature of asphalt concrete, and moisture or freezing conditions of the unbound material. Other laboratory or nondestructive testing data can also be used to estimate seed modulus values.

Table 3-1 provides typical modulus values and ranges for paving materials based on FAA Advisory Circular 150/5370-11B, "Use of Nondestructive Testing in the Evaluation of Airport Pavements". Table 3-2 presents default seed modulus values for pavement layers in BAKFAA.

Table 3-1. Typical modulus values and ranges for paving material

Material	Item Number	Low Value		Typical Value		High Value	
		psi	MPa	psi	MPa	psi	MPa
Asphalt Mix Pavement	P-401/P-403	70,000	500	500,000	3,500	2,000,000	14,000
Cement Concrete Pavement	P-501	1,000,000	7,000	5,000,000	35,000	9,000,000	60,000
Lean Concrete Base Course	P-306	1,000,000	7,000	2,000,000	14,000	3,000,000	20,000
Asphalt Treated Base		100,000	700	500,000	3,500	1,500,000	10,000
Cement Treated Base	P-304	200,000	1,400	750,000	5,000	2,000,000	14,000
Aggregate Base Course	P-208/P-209	10,000	70	30,000	200	50,000	350
Granular Subbase Course	P-154	5,000	30	15,000	100	30,000	200
Stabilized Soil		10,000	70	50,000	350	200,000	1,400
Cohesive Soil		3,000	20	7,000	50	25,000	170

3.3.2 Backcalculated Modulus

The "Backcalculated Modulus" column displays the backcalculated modulus value for each pavement layer following the completion of backcalculation analysis. If the user selects a single FWD drop and runs backcalculation, the "Backcalculated Modulus" column will include the result of the backcalculation for that single selected drop. If the user selects multiple FWD drops and runs backcalculation, the geometric mean (written as 'Geo. Mean') of the backcalculated layer modulus values for the selected drops will be displayed.

3.3.3 Poisson's Ratio

BAKFAA requires a Poisson's ratio for each pavement layer. While the pre-defined standard materials have default values, they and those of user defined materials can be assigned to any numerical value between 0 and 0.5 by clicking on the cell and inputting a new value. The default Poisson's ratio values for pre-defined materials available in BAKFAA are presented in Table 3-2.

Table 3-2. Default seed modulus and Poisson's ratio values for pavement layers in BAKFAA

No.	Material	Item Number	Default Seed Modulus		Default Poisson's Ratio
			psi	MPa	
1	Ultra-Low Strength Subgrade		5,000	35	0.40
2	Low Strength Subgrade		9,000	60	0.40
3	Medium Strength Subgrade		15,000	100	0.40
4	High Strength Subgrade		20,000	140	0.40
5	Granular Subbase Course	P-154	40,000	275	0.35
6	Aggregate Base Course	P-208	75,000	515	0.35
7	Crushed Aggregate Base Course	P-209	75,000	515	0.35
8	Lime Rock Base Course	P-211	60,000	410	0.35
9	Recycled Concrete Aggregate Base Course	P-219	75,000	515	0.35
10	Soil Cement Base	P-301	250,000	1,720	0.20
11	Cement Treated Aggregate Base Course	P-304	500,000	3,445	0.20
12	Asphalt Mix Pavement Surface Course	P-401	200,000	1,375	0.35
13	Asphalt Mix Pavement Base Course	P-403	400,000	2,755	0.35
14	Cement Concrete Pavement	P-501	4,000,000	27,560	0.15

3.3.4 Interface Parameter

The Interface Parameter represents the bonding condition between two pavement layers and has a value of 0 or 1, with 0 indicating no bond and 1 indicating fully bonded layers. The default value for the interface parameter is 1 in BAKFAA since complete bonding between pavement layers is commonly the case in pavement structures. However, the user should assign 0 to the interface parameter where no bonding exists between adjacent pavement layers, such as with a concrete slab on top of an asphalt layer.

3.3.5 Pavement Layer Thicknesses

BAKFAA requires a thickness for each pavement layer. BAKFAA treats the last layer as a semi-infinite layer, so its thickness must be zero. Zero is not an accepted thickness for other layers.

3.3.6 Moduli Changeable

The "Moduli Changeable" checkbox allows the user to determine whether a layer's modulus will be calculated in the backcalculation analysis or treated as a fixed value based on the assigned seed modulus. The checkbox is toggled on for all layers by default. If the "Moduli Changeable" checkbox is checked, the layer modulus will be calculated during the backcalculation. If it is not, modulus value will be fixed to the assigned seed modulus value and will not change during the iteration process. The backcalculated layer modulus will be the same as the seed modulus.

3.4 Saving and Loading Pavement Section

The user can save a defined pavement structure for later use by clicking "Pavement Section" on the top toolbar and selecting "Save Pavement Section" as highlighted in Figure 3-14. If an FWD file is not loaded, the pavement structure will be saved in an .XML file format. If an FWD file is loaded, BAKFAA will provide a popup box and ask if the user wants to save the FWD deflection data along with the pavement structure data. Click the "Yes" button to save both FWD and pavement structure data. Click the "No" button to only save the pavement structure data. The user will be asked to provide a file path to save the .XML file.

The user can load any saved pavement section by clicking the "Pavement Section" drop-down from the toolbar and selecting "Load Pavement Section". If the saved XML file contains only the pavement structure data, only the pavement structure data will be loaded and the user will be required to load an FWD file separately. If the saved .XML file contains both the pavement structure and FWD data, both data types will be loaded to the BAKFAA interface.

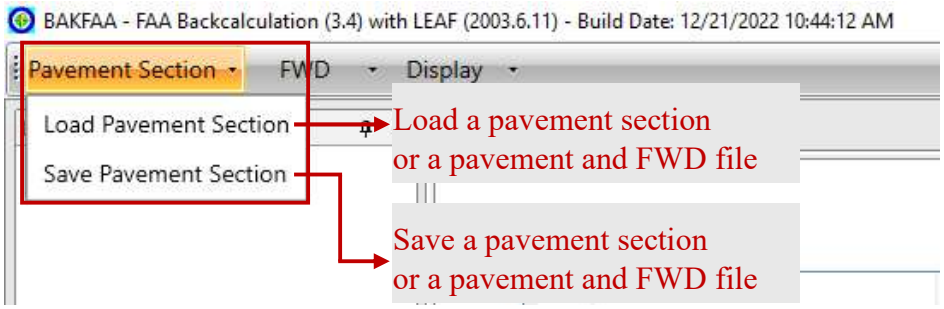


Figure 3-14. Saving and loading pavement sections

A pavement structure file consists of .XML tags. There are two sections in the file: "BasinList" and "Structures". The "BasinList" section contains the tags associated with the FWD tests. This includes FWD drops, sensor offsets, and recorded deflections for each FWD drop. The "Structures" section contains tags associated with each pavement layer, including layer thickness, seed modulus, Poisson's ratio, interface parameter, and moduli changeable values. Figure 3-15 presents an example of a "BasinList" section and a "Structures" section of the .XML file.

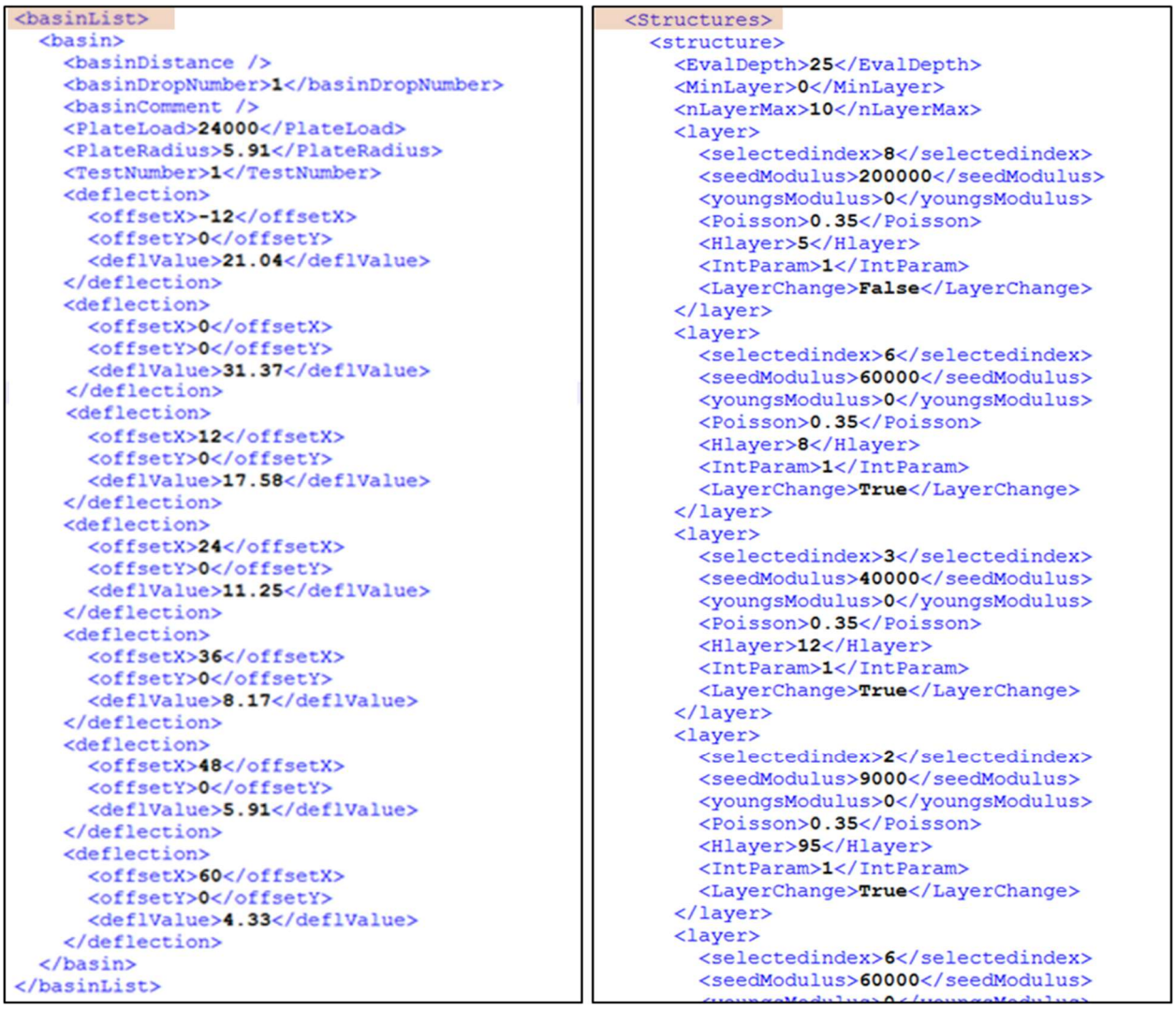


Figure 3-15. Example of pavement section .XML file containing FWD test and pavement structure data

3.5 Units

Throughout the BAKFAA program, the values of items can be displayed in either English or metric units by selecting the appropriate radio button under the "Design Options" module as highlighted in Figure 3-16. The default unit setting is English (US Customary). Changing the unit will automatically update the units of all FWD and pavement information as shown in Figure 3-17 through Figure 3-20. The output files will also conform to the user's unit selection.

Table 3-3 lists the English and metric units used for different items in the BAKFAA GUI and output files.

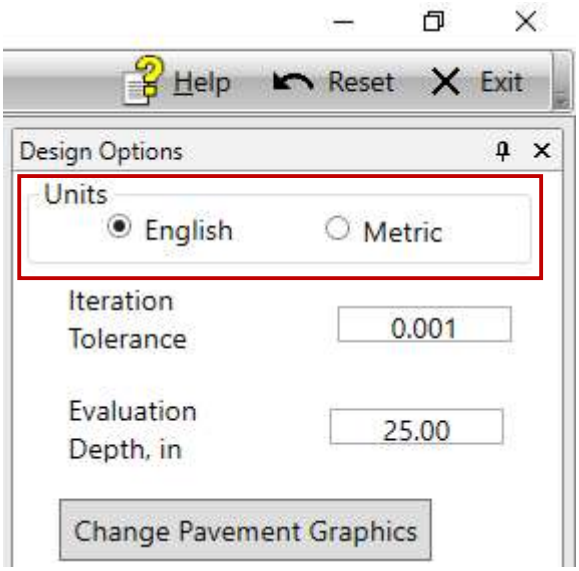
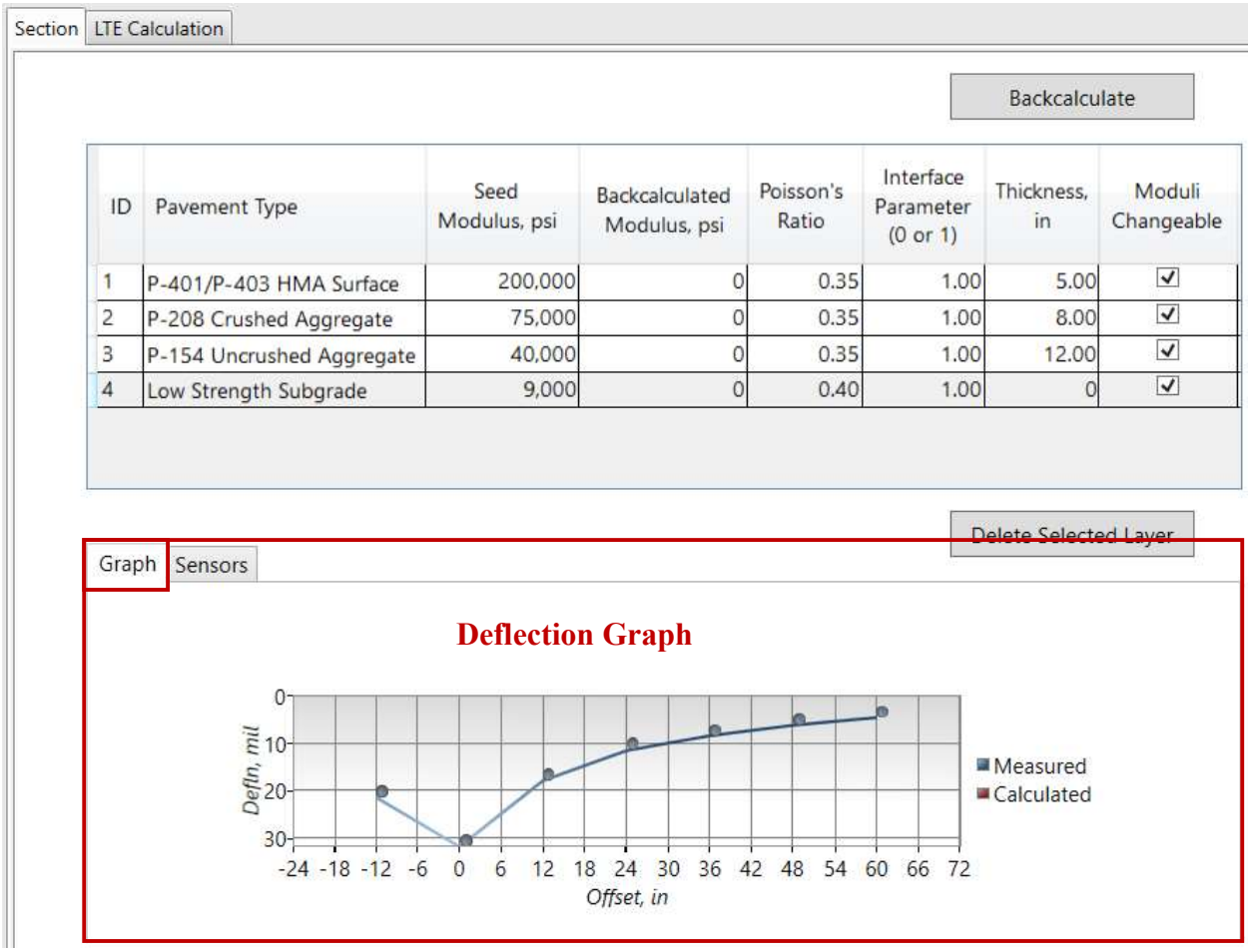


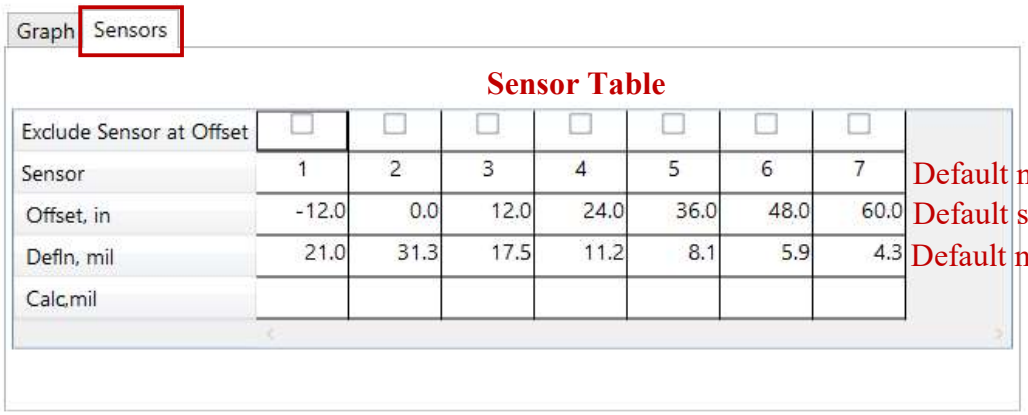
Figure 3-16. Selection of Units

Table 3-3. English and Metric units used for illustration of different entities in BAKFAA

Entity	**Unit**	
	English	**Metric**
Pavement Layer Thickness	Inch (in.)	Millimeter (mm)
Distance	Feet (ft.)	Meter (m)
FWD Sensor Offset	Inch (in.)	Millimeter (mm)
FWD Sensor Deflection	Mil	Micron (mc)
FWD Plate Radius	Inch (in.)	Millimeter (mm)
FWD Plate Contact Area	Square inch (in^2)	Square millimeter (mm^2)
FWD Plate Contact Pressure	Pound per square inch (psi)	Megapascal (MPa)
Load	Pound (lbs.)	Kilonewton (kN)
Modulus	Pound per square inch (psi)	Megapascal (MPa)

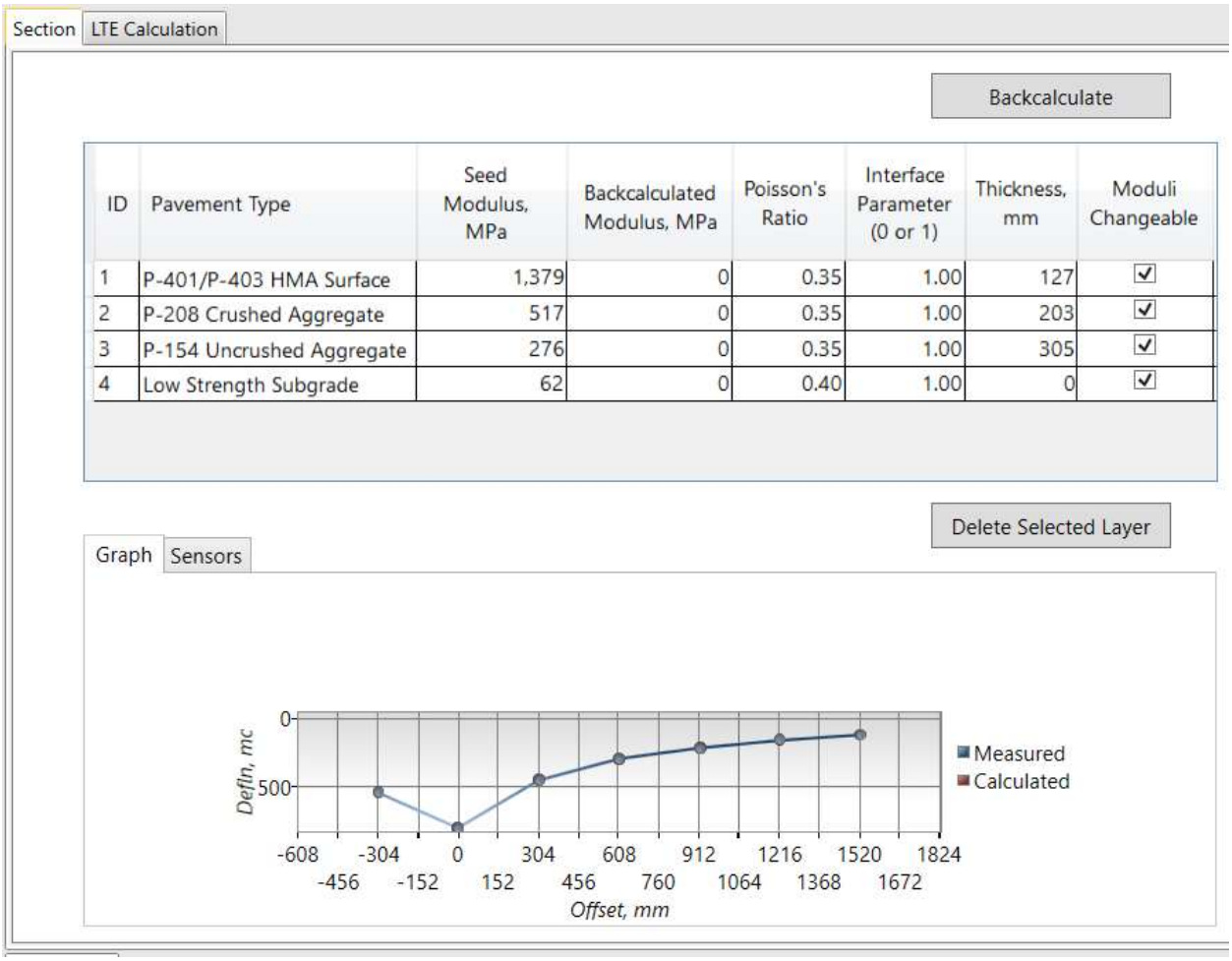


(a) deflection graph

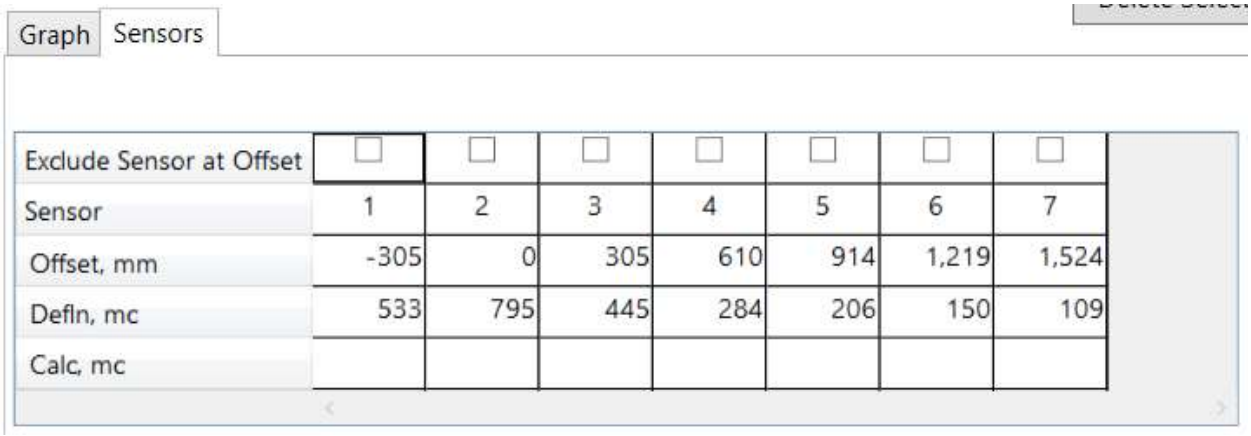


(b) sensor table

Figure 3-17. Default FWD test setup shown in English units



(a) deflection graph



(b) sensor table

Figure 3-18. Default FWD test setup shown in Metric units

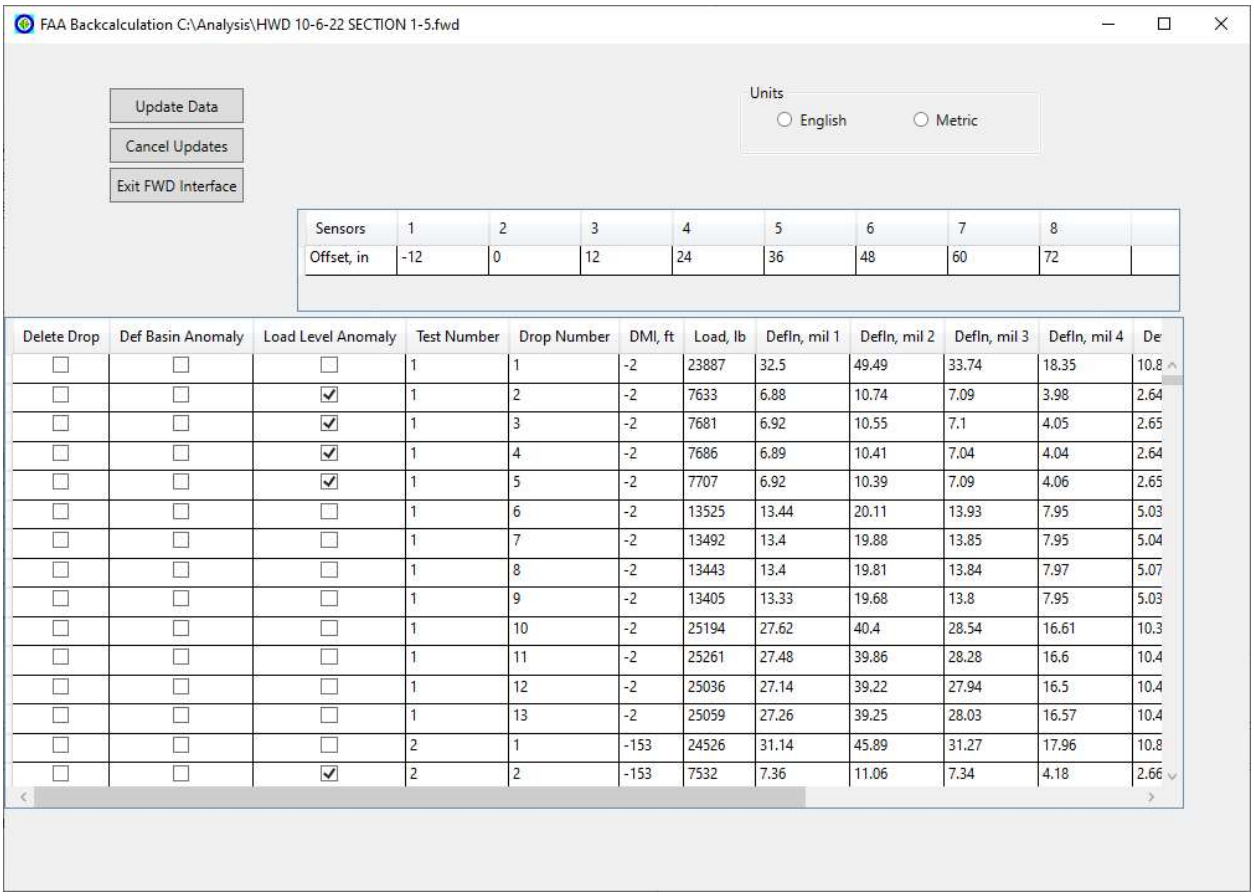


Figure 3-19. FWD file sub-window shown in English units

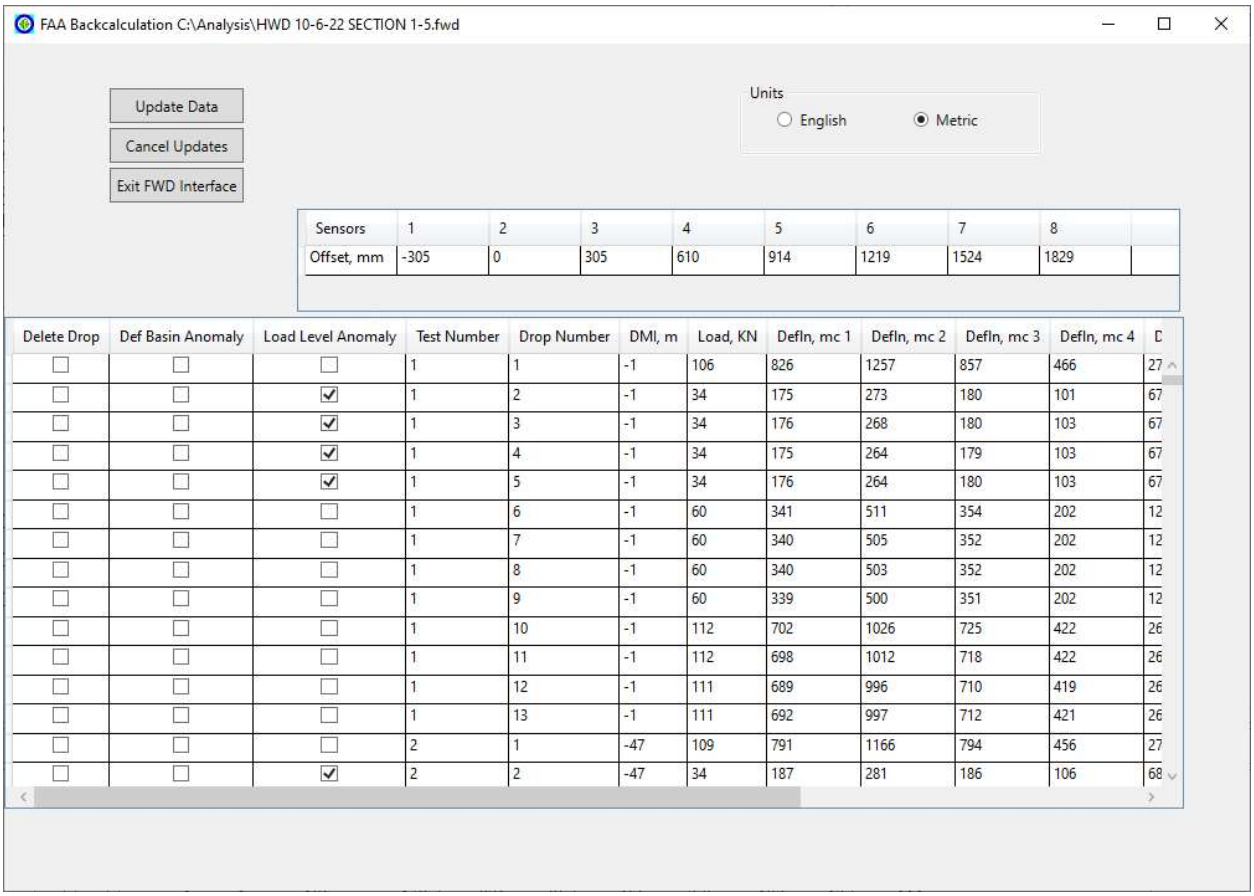


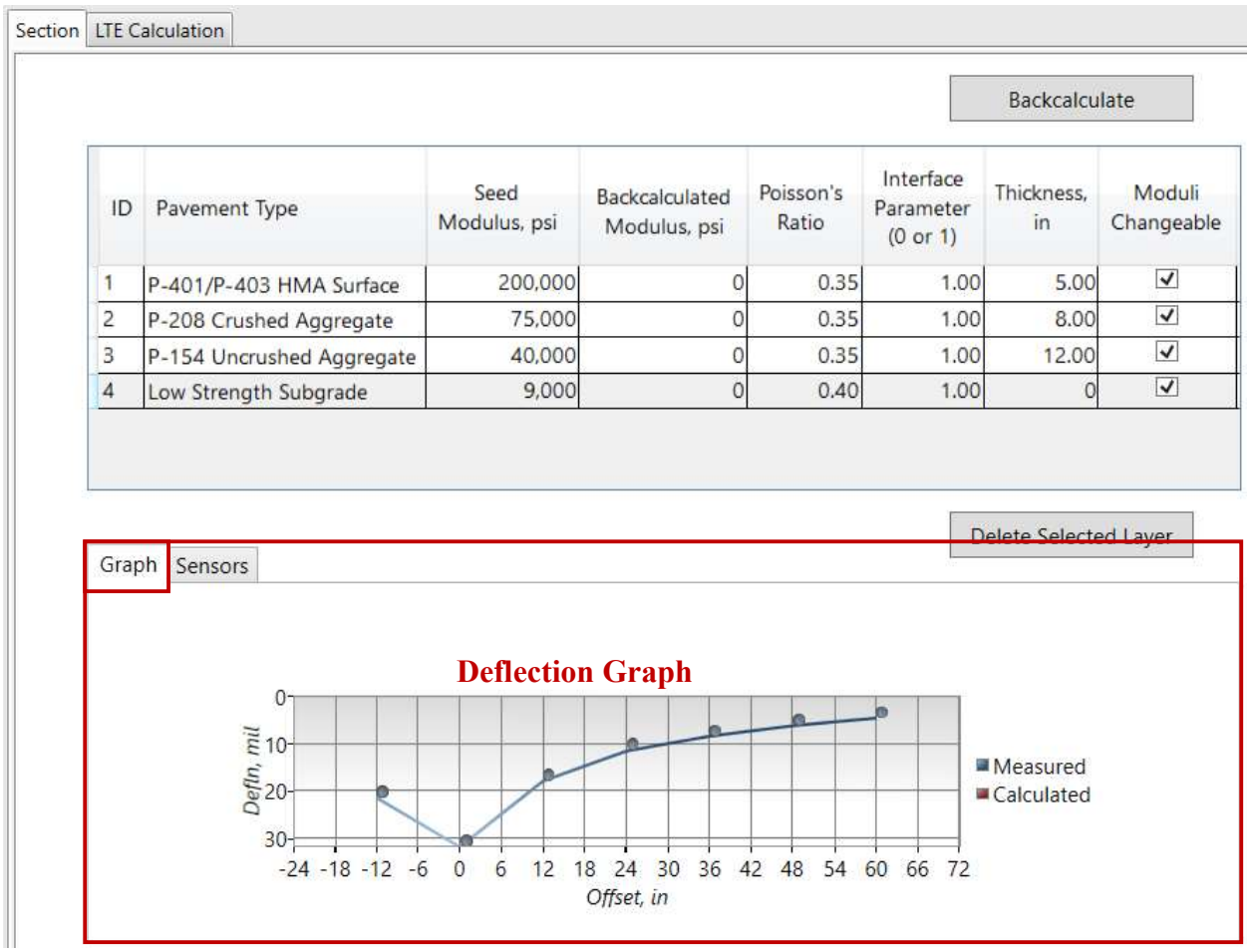
Figure 3-20. FWD file sub-window shown in Metric units

3.6 FWD Test Information

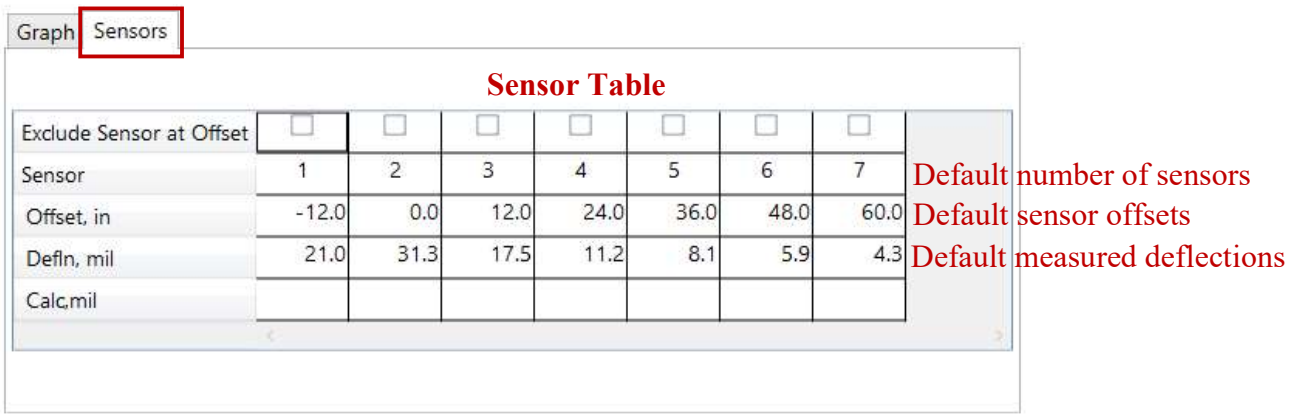
BAKFAA's "Section" module provides a deflection graph and a sensor table that display FWD information, as highlighted in Figure 3-21 (a) and (b) respectively. The sensor table includes FWD sensor number, sensor offset, and measured deflection at each sensor (from the FWD file data). The graph and table also include calculated deflections at each sensor from the BAKFAA analysis, which will update after the backcalculation is completed. The values in the deflection table are not editable.

The "Exclude Sensor at Offset" row in the sensor table allows the user to exclude any of the sensors from the backcalculation analysis by clicking the checkmark box. BAKFAA will provide a popup message warning indicating that to regain the excluded sensor, the user must reload the original FWD or pavement structure .XML file (see Figure 3-22).

When the BAKFAA software is opened, a default FWD test setup with placeholder values is displayed. After the user loads a file, the values will update based on the information in the file. The deflection graph displays a plot of both measured and calculated deflections. Figure 3-23 (a) and (b) show the calculated deflections and updated deflection graph after completion of the backcalculation. This example uses the default deflection basin and pavement structure settings and excludes the sensor at the -12-inch offset.



(a) Deflection graph



(b) Sensor table

Figure 3-21. Default FWD test setup

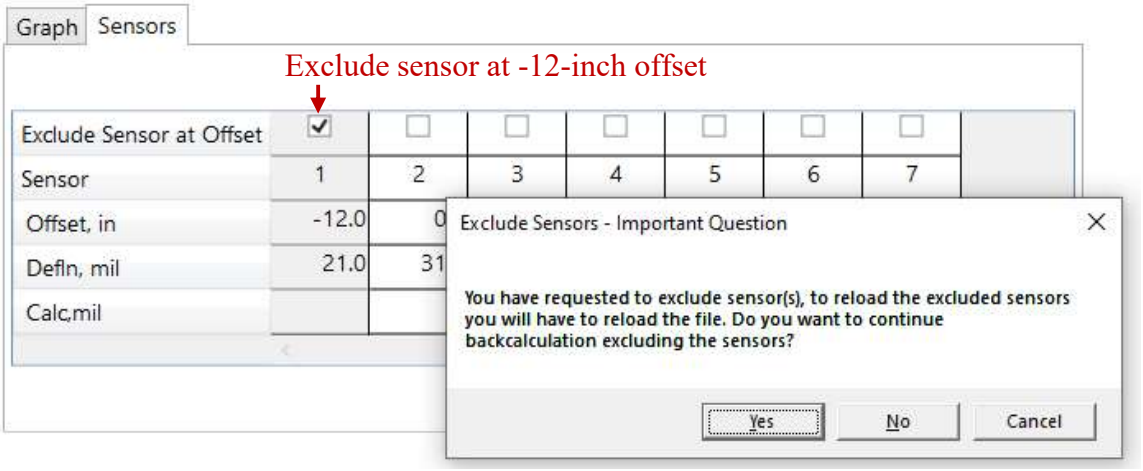
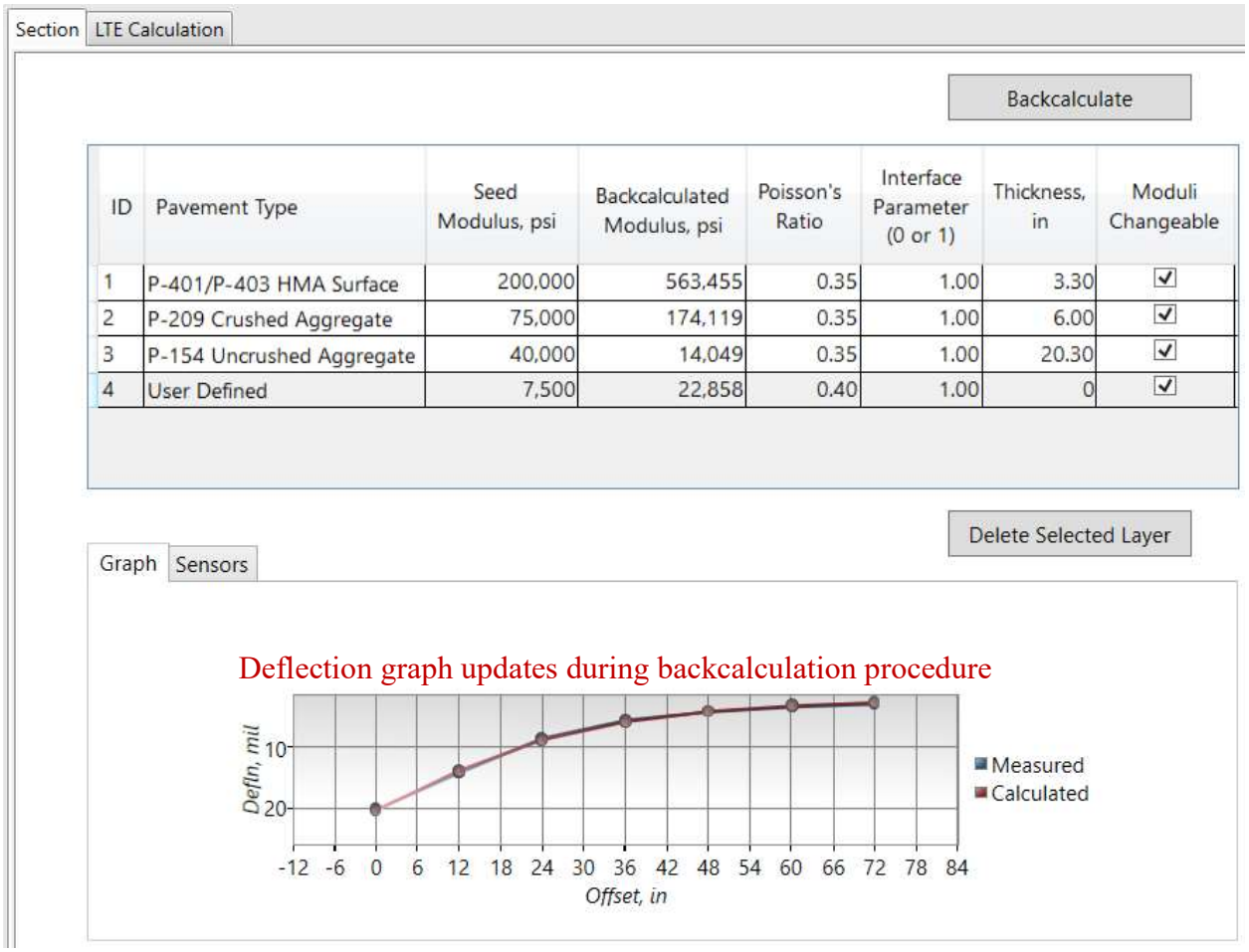
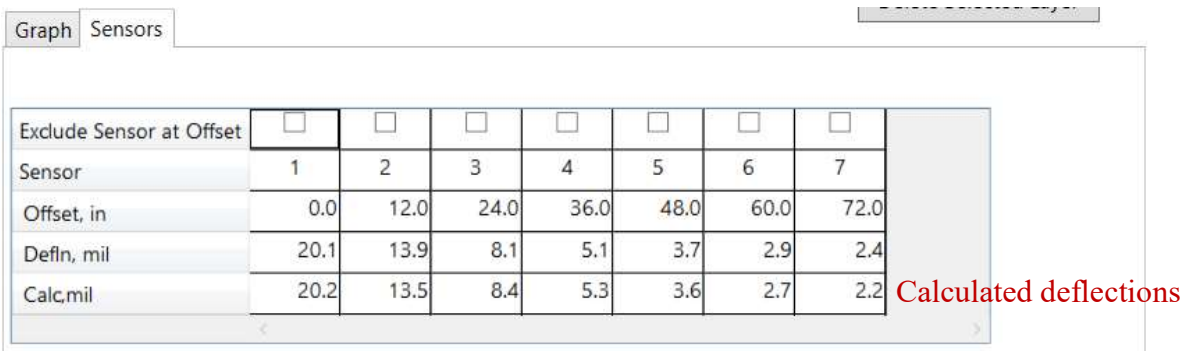


Figure 3-22. Exclusion of sensor warning



(a) Deflection graph



(b) Sensor table

Figure 3-23. Calculated deflections and updated deflection graph after backcalculation analysis

3.6.1 Load FWD Files

Several manufacturers produce FWD devices and each device uses a unique FWD file format. BAKFAA can process proprietary file formats used by some of the leading FWD manufacturers, including Dynatest, KUAB, Foundation Mechanics (JILS division), Pavetest, and CarlBro. Table 3-4 lists all the file types supported by the current version of BAKFAA.

Table 3-4. File types supported by BAKFAA

Equipment Manufacturer	Associated Data Format	File Extensions
Dynatest	R80, R32	.fwd, .hwd
	F20, F25	.fwd, .F20, .F25
	PDDX, XML	.ddx, .xml
KUAB	KUAB	.fwd, .hwd
JILS	M3, M5	.fwd, .hwd, .dat
PaveTest	PaveTest	.csv
CarlBro	CarlBro	.fwd, .txt
AASHTO	Pavement Deflection Data Exchange	.ddx

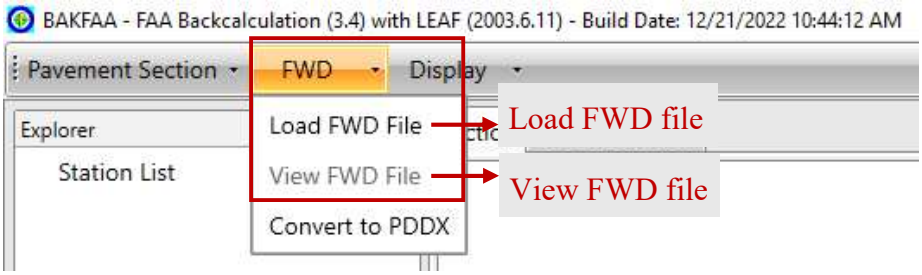
BAKFAA allows the user to import an FWD file directly into the interface by clicking the "FWD" drop-down in the top toolbar then "Load FWD File", and then browsing to select an FWD file (see Figure 3-24 (a)).

The FWD file information for each FWD drop including measured deflections, FWD plate load, distance measuring instrument (DMI) location information, drop number, and anomaly checks will be displayed in the "FWD Drops" module, as highlighted by a red box in Figure 3-24.

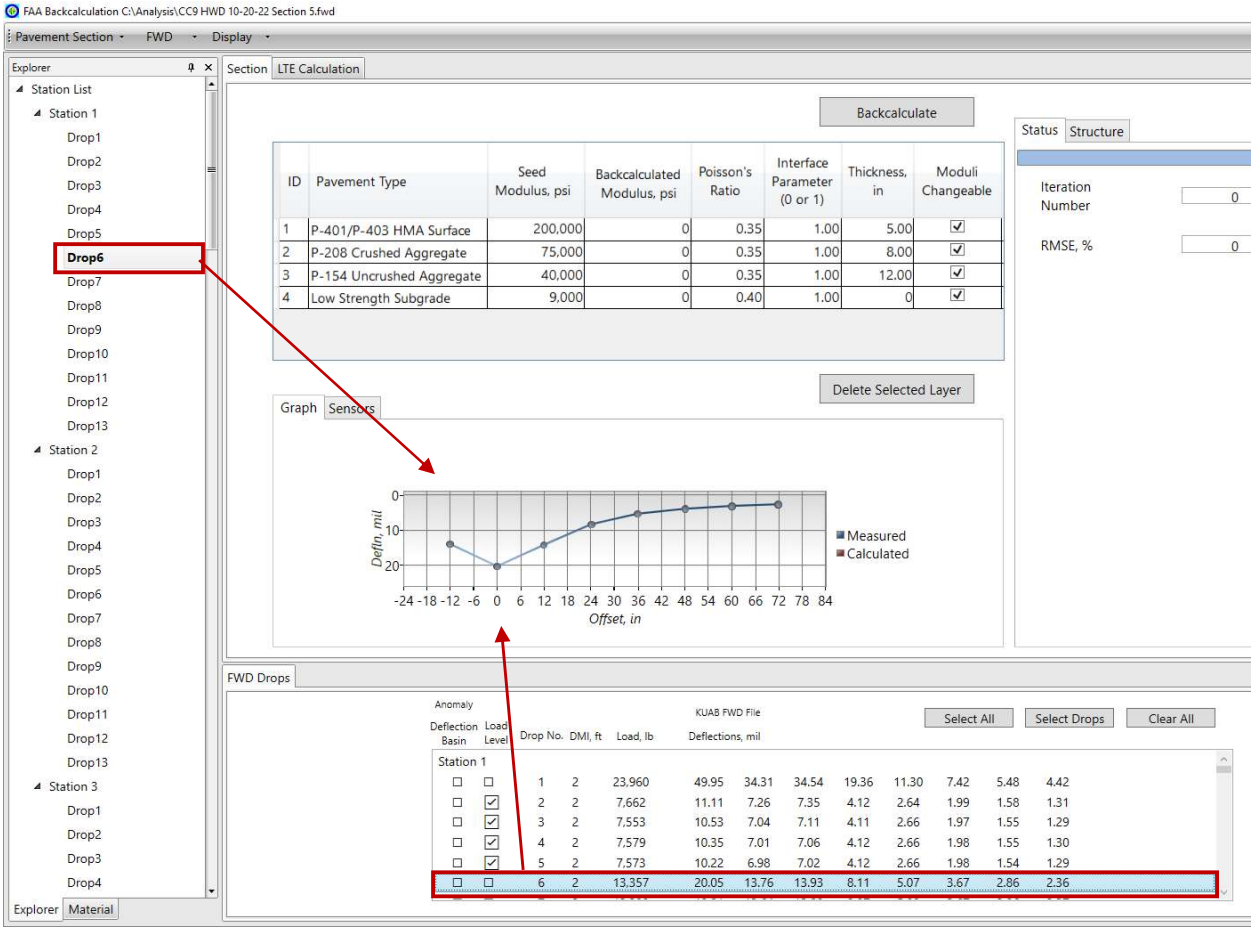
- Anomaly check boxes allow quality control of FWD data. FWD drops are flagged if anomalies are identified in load levels or deflection basins.
    - Deflection basins are flagged if the recorded deflections are 1) showing a non-decreasing sequence, 2) larger than 100 mils (2540 mc), or 3) recorded zero. Deflections larger than 100 mils are flagged because they are beyond the linear range of most deflection sensors.
    - Load levels are flagged if the recorded FWD plate load is smaller than 9,900 lbs. or larger than 60,024 lbs. (44 kN – 267 kN).
- "Drop No." is the number of FWD drops in a single FWD test. BAKFAA has no limitations on the maximum number of drops in a single FWD test. Test locations can have a different amount of drop numbers.
- "DMI" identifies the location (also known as 'station') of each FWD test point.
- "Load" displays the applied FWD load for each drop.
- "Deflections" are measured deflections recorded by the FWD sensors for each drop.
- The operator comments for test locations are identified by BAKFAA and replaced with a 'Station' number, starting at 1 and increasing sequentially.

The “Station Details” module displays the FWD drop information including measured deflections, FWD plate load, DMI location information, drop number, and anomaly checks for all drops in the station of the selected drop (see Figure 3-25). The values in “Station Details” are not editable.

The deflection graph and sensor table update when the user selects an FWD drop (see Figure 3-24 (b)).



(a) Loading a FWD file



(b) Selecting a drop

Figure 3-24. (a) Loading a FWD file and (b) Selecting a drop

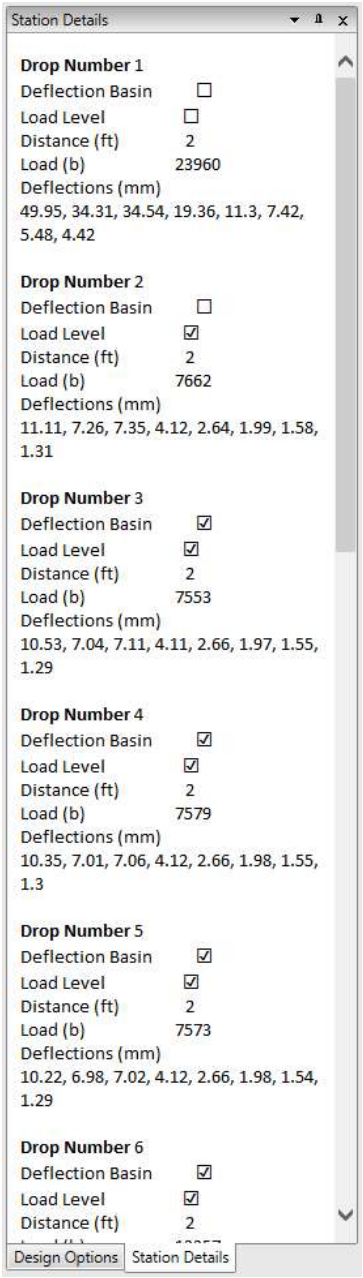


Figure 3-25. Viewing details of an FWD station

3.6.2 View FWD File

BAKFAA allows the user to view and edit a loaded FWD file by clicking the "FWD" drop-down in the top toolbar then "View FWD File" (see Figure 3-24). Figure 3-26 shows the FWD file sub-window for a sample FWD file. This sub-window is interactive, allowing the user to switch units or edit the FWD information. Click the "Update Data" button to save the changes and click the "Exit FWD Interface" button to return to the main window. Any changes made to the FWD file will be reflected in the "FWD Drops" module. This will not alter the original FWD file, however

the user can save the modified FWD file as a new .XML file by clicking on the “Pavement Section” drop-down in the top ribbon then “Save Pavement Section”.

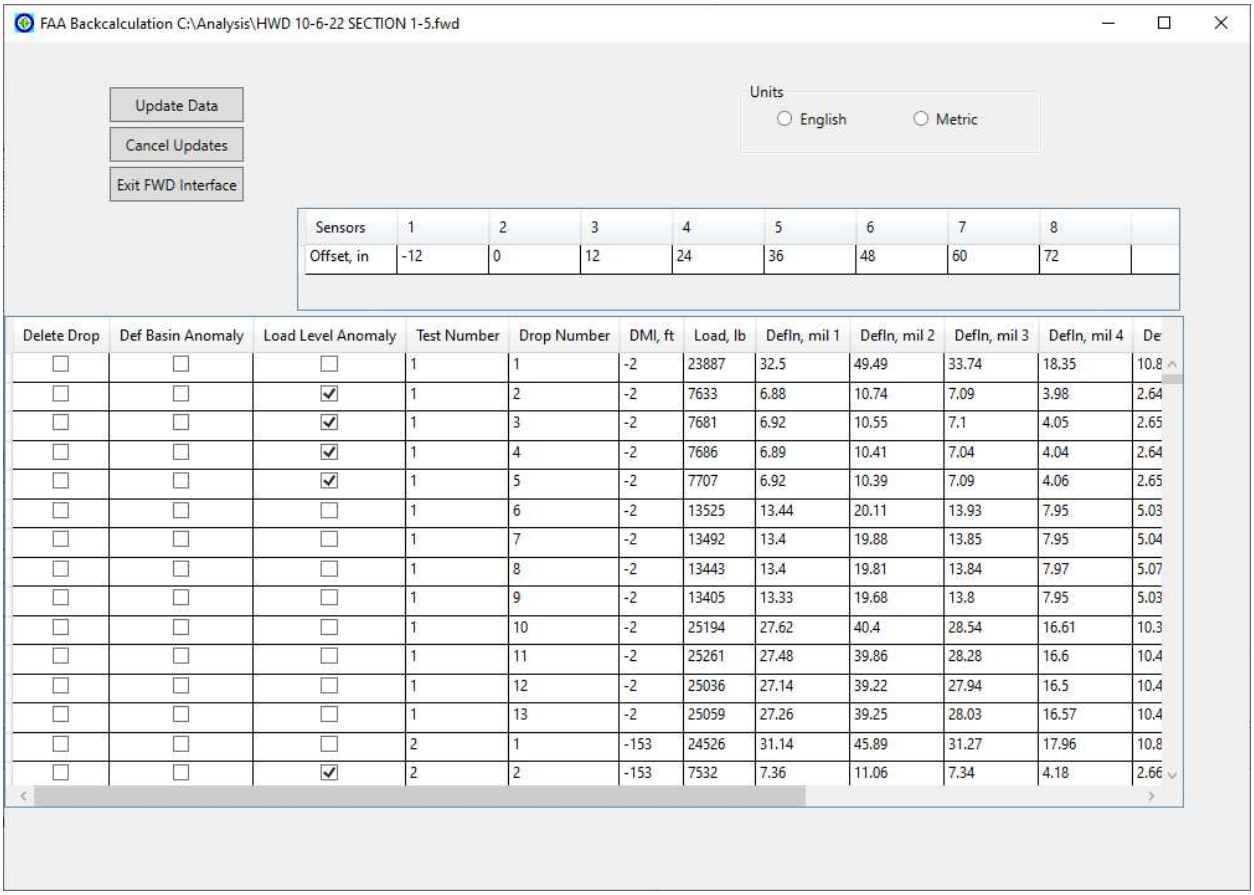


Figure 3-26. FWD file sub-window

3.6.3 Convert FWD File to PDDX Format

The user can convert any FWD file format supported by BAKFAA to the Pavement Deflection Data Exchange (PDDX) format by clicking on the “FWD” drop-down in the top toolbar then “Convert to PDDX” and browse to provide a save path for the file. The PDDX files can be loaded to the program by clicking on the “FWD” drop-down then “Load FWD File”.

Pavement deflection data files are typically structured using American Standard Code for Information Interchange (ASCII) columnar or tagged formats. Processing the deflection data in ASCII format requires a device-specific procedure for each manufacturer’s FWD device. The PDDX format is intended to address the shortcomings of dealing with multiple file formats and to facilitate easy access and data exchange capabilities for all FWD data formats. The PDDX format includes data storage in ASCII format, similar to the deflection data format used by FWD manufacturers. The ASCII file type allows the user to easily edit and organize data files using a standard text editor. However, the PDDX file format specification has a modular structure in which data elements of similar characteristics are grouped in sections. The first line of each section contains a data description demarcated with brackets. Each section contains several variables with

their associated values. Below is an example of the general format of this modular approach (AASHTO 1998):

```
[Section 1]
Entry1 = Value (s)
Entry2 = Value (s)
Entry2 = Value (s)

[Section 2]
Entry4 = Value (s)
Entry5 = Value (s)
Entry6 = Value (s)
```

The PDDX format allows for more flexibility in terms of the amount of data that can be stored and can accommodate future additional data elements that may be collected by FWD devices (AASHTO 1998). Further information about PDDX file formats can be found in the technical guide for PDDX format (version 1.0) published by the American Association of State Highway and Transportation Officials (AASHTO).

3.6.4 FWD File Parser

BAKFAA's file processing class that reads FWD files uses the file format specifications from FWD manufacturers and generates a set of properties to retrieve data required for functions in the backcalculation and LEAF analysis. The manufacturers' FWD file formats are not hard coded into the file processing class, instead the file processing class reads the data format specifications from the data source and processes the file based on the specifications. The advantage of this design is that when a new data format is introduced by a manufacturer, only a file specification needs to be created for the new layout. In this case, BAKFAA will read the specification and parse the file without requiring any reprograming in the source code.

The process uses a file specification in .XML format to determine how to read the content of data files. The file specification must provide the key information required for BAKFAA. Spec files should specify "Header" data and "Test" data. Header data is specific to the file. Test data will contain information from each test location and each FWD drop. The parser works by comparing each line in a data file against patterns specified in the .XML file. There are two main modes of operation, identified by the <SplitType> tag: delimited (D) and fixed width (F).

In the delimited mode, the software searches each line for the value of the <Line> tag. If it finds a match, it breaks the line into fields by splitting it based on the contents of the <Delim> tag. It then stores the value in the field number identified by the <Pos> tag and stores it in the variable identified by the <Name> tag. Field numbers are zero-based, that is, the first field in a line is field number 0. An example of a delimited data element (FWD file) and the required code for extracting specific data from the file is provided in Figure 3-27. This example illustrates the extraction of the FWD software edition "programVersion" from a Dynatest F25 file. This information is found in the second field (position 1) on a line that starts with "5001" and is delimited by commas (SplitType "D"). In this example, "programVersion" has the value of "25.60" in the first line.

Dynatest F25 Spec File Contents

```
1  5001,25.60,1,40, 3, 1,"FwdWin          "
2  5002,"25USN   ","8082-025","9000-074"
3  5003,"Administ","Bedford ","TWA1ROC1","F25"
4  5010,1,0,1,0,1,1,0,5,1,1,1,1,0,0,1,0,0,0,0,0,1,"MDB"
5  5011,0,1,2014,10,08,15,23,0,"Non",000
```

Extraction of software edition from Dynatest F25 file

```
<FWDData>
  <Line>5001</Line>
  <Name>programVersion</Name>
  <SplitType>D</SplitType>
  <Delim>Comma</Delim>
  <Pos>1</Pos>
  <DataType>Header-40</DataType>
</FWDData>
```

Figure 3-27. Example of delimited data file and the extraction of software edition

In the fixed-width mode, when the software finds a match to <Line> a tag, it extracts a value <Delim> characters long starting at line position <Pos>. The value is stored in the variable identified by the <Name> tag. Character positions are also zero-based. The example provided in Figure 3-28 finds the same information, "programVersion" in a Dynatest R32 file, which is a fixed-width format file.

```
<FWDData>
  <Line>R32</Line>
  <Name>programVersion</Name>
  <SplitType>F</SplitType>
  <Delim>2</Delim>
  <Pos>30</Pos>
  <DataType>Header-36</DataType>
</FWDData>
```

Figure 3-28. Extraction of software edition from Dynatest R32 file with fixed-width format

The parser has the ability to perform a two-level split. It extracts a string as described above, and then splits the extracted value according to the <SplitType2>, <Delim2>, and <Pos2> tags before storing in the variable identified by the <Name> tag. It is possible to have different types of splits (delimited or fixed-width) for the first and second levels. An example of a two-level split is provided in Figure 3-29. In KUAB files, extracting the day, month, and year of the file requires a two-level split. The line containing the date starts with "IDate Created". The information data is delimited from the label by a ":" (colon). The date "4/26/2013" is after the colon, meaning it is in position 1. The label "IDate Created" is position 0 because numbering is zero based. In the 2nd level, The parser splits the date "4/26/2013" into day, month, and year, and store the values in fileDay, fileMonth, fileYear, respectively. The date is delimited by "/" (forward slash), as such

<SplitType2> is D (delimited). The day is in position 0, the month is in position 1, and the year is in position 2.

KUAB File Contents

```
IKUAB FWD FILE    : STH169_NB_RWP.fwd
HOperator         : DDR/PR
HRoad             : STH 169 NB RWP
HWeather          : sunny, 45 F

IDate Created     : 4/26/2013
IVersion          : 2.4.15
ILoad Mode        : 2          (7+7 plates, 6 buffers)
IPlate Radius     : 5.91       (in)
IExtra Field Set  : KUAB
IDrop Sequence    : 2234
INo of drops      : 1111
IRecord Drop?     : NYYY
IDrop Height      :     1     2     3     4     5     6
```

Extraction of the day, month and year, and storing the values in fileDay, fileMonth, and fileYear

```
<FWDData>
  <Line>IDate Created</Line>
  <Name>fileDay</Name>
  <SplitType>D</SplitType>
  <Delim>:</Delim>
  <Pos>1</Pos>
  <SplitType2>D</SplitType2>
  <Delim2>/</Delim2>
  <Pos2>1</Pos2>
  <DataType>Header-0</DataType>
</FWDData>
<FWDData>
  <Line>IDate Created</Line>
  <Name>fileMonth</Name>
  <SplitType>D</SplitType>
  <Delim>:</Delim>
  <Pos>1</Pos>
  <SplitType2>D</SplitType2>
  <Delim2>/</Delim2>
  <Pos2>0</Pos2>
  <DataType>Header-0</DataType>
</FWDData>
<FWDData>
  <Line>IDate Created</Line>
  <Name>fileYear</Name>
  <SplitType>D</SplitType>
  <Delim>:</Delim>
  <Pos>1</Pos>
  <SplitType2>D</SplitType2>
  <Delim2>/</Delim2>
  <Pos2>2</Pos2>
  <DataType>Header-0</DataType>
</FWDData>
```

Figure 3-29. Example a two-level split extraction

There are two sections at the top of the .XML files: “Translate” and “AssignedData.” Translate is used to convert proprietary values and flags to the standard values used by BAKFAA. For instance, CarlBro uses the value “Metric” or “English” to indicate the units for the test location station. The translate tag converts these values to “m” or “ft”, the values used internally by BAKFAA, as shown in Figure 3-30. Translate has three XML tags: Name, InValue, and OutValue. When the value of unitsDistance in the file is “English”, BAKFAA changes the value to “ft” before storing it internally.

```
<CarlBro_Spec>
 <Translate>
     <Name>unitsDistance</Name>
     <InValue>Metric</InValue>
     <OutValue>m</OutValue>
</Translate>
<Translate>
     <Name>unitsDistance</Name>
     <InValue>English</InValue>
     <OutValue>ft</OutValue>
</Translate>
```

Figure 3-30. Example of functionality of "Translate" section of the XML file

The "AssignedData" section of the spec file is used to provide information to BAKFAA that is not available in the data file. For example, CarlBro files always store values in Metric units and therefore do not store the unit type in the data file itself. This information must be provided in the "AssignedData" section using the <Name> and <Value> tags as provided in Figure 3-31.

```
<AssignedData>
     <Name>unitsDefl</Name>
     <Value>micron</Value>
</AssignedData>
<AssignedData>
     <Name>unitsLoad</Name>
     <Value>kn</Value>
</AssignedData>
```

Figure 3-31. Example of functionality of "AssignedData" section of the XML file

For KUAB and CarlBro, the Spec files are specific to the number of sensors on the device. For these types of files, BAKFAA first counts the number of sensors and selects the Spec file for the number of sensors. Table 3-5 lists the specifications available for the KUAB and CarlBro files.

Table 3-5. Available specifications for KUAB and CarlBro file types

Manufacturer	Available Specs
KUAB	KUAB_Spec_7_Sensors.xml
	KUAB_Spec_8_Sensors.xml
	KUAB_Spec_9_Sensors.xml
CarlBro	CarlBro_Spec_4_Sensors.xml
	CarlBro_Spec_9_Sensors.xml
	CarlBro_Spec_17_Sensors.xml

Users can change the number of sensors in these files by copying an existing file and adding or removing tags that identify sensor information, such as:

- sensorDeflY
- sensorDeflX
- dropDeflections

Required values are:

- sensorDeflY (1 for each sensor)
- sensorDeflX (1 for each sensor)
- dropDeflections (1 for each sensor)
- unitsPlateRadius
- unitsLoad
- unitsDefl
- unitsOffset
- unitsDistance
- equipmentPlateRadius
- locationDistance
- dropLoad

3.7 Selection of FWD Drops for Backcalculation

There are three ways to select FWD drops for backcalculation analysis:

1. Selecting single or multiple FWD drops. The user can select a single FWD drop by clicking on any drop in the “FWD Drops” or “Explorer” modules. The FWD drop will be selected and the user can run backcalculation for the single selected drop. In the “FWD Drops” module, the user can select multiple consecutive drops by holding ‘Shift’ then clicking on the first and last drops or can select multiple distinct drops by holding ‘Ctrl’ then clicking on the desired drops (see Figure 3-32).

2. Selecting all drops by clicking the “Select All” button in the “FWD Drops” module shown in Figure 3-32.
3. Selecting specific drop number(s) by clicking the “Select Drops” button in the “FWD Drops” module. Select the desired drop numbers (drop No.1, No.2, etc.) from the “Drop No.” menu, as shown in Figure 3-33. For example, by selecting drop number 2 from the menu, the second FWD drop for each test location will be selected.

Clicking the “Clear All” button in the “FWD Drops” module (Figure 3-32) will clear any and all FWD drop selections.

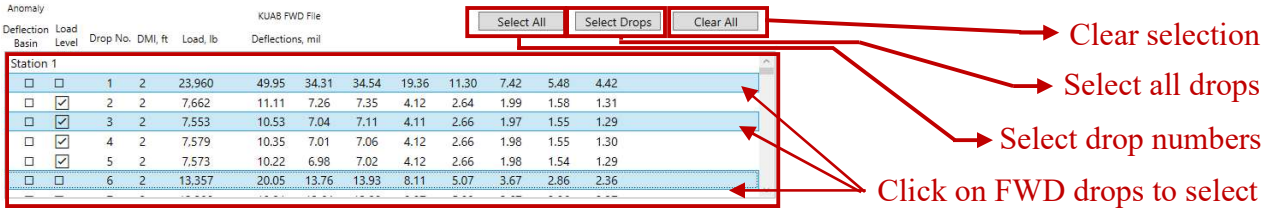


Figure 3-32. FWD drop selection in the “FWD Drops” module and the “Select Drops”, “Select All”, and “Clear All” buttons

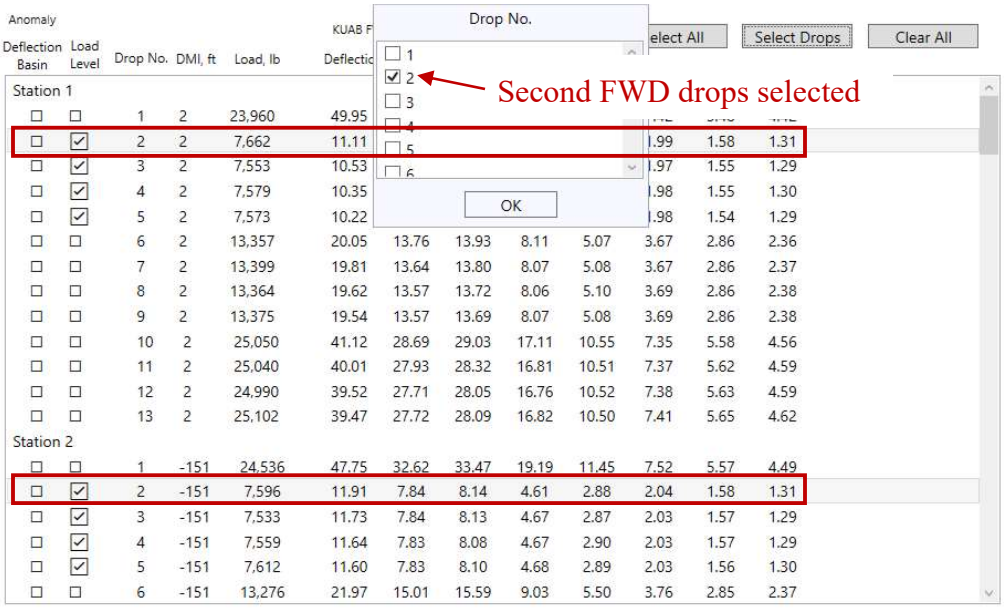


Figure 3-33. FWD drop selection using the “Select Drops” button and the “Drop No.” menu

3.8 Backcalculation Analysis

Before performing a backcalculation analysis, the user must define the pavement layer structure, load an FWD file, select the desired FWD drops, and finally assign an iteration tolerance as shown in Figure 3-34.

To begin the backcalculation, click the “Backcalculate” button in the “Section” module (Figure 3-34). Backcalculation can be performed in a single analysis mode if one FWD drop is selected or in a batch analysis mode if more than one drop is selected. In the single mode, the backcalculation results for the selected drop will appear in the main interface. In batch mode, the backcalculation

results will be saved in a .csv output file. When selecting multiple drops, a dialog box will appear asking the user to confirm that a batch analysis is desired (Figure 3-34). The user will then be asked to supply a file name and folder address where the .csv file will be saved. Selecting "Cancel" in the pop-up dialog box will abort the backcalculation process. The analysis outputs are the backcalculated modulus values, the root mean square error (RMSE), iteration number, and the calculated deflections.

The backcalculation "Status" submodule (Figure 3-34) updates the user on the progress of backcalculation actions. The progress bar in blue estimates time remaining to complete the single or batch backcalculation. The text box to the right of the bar shows the drop number.

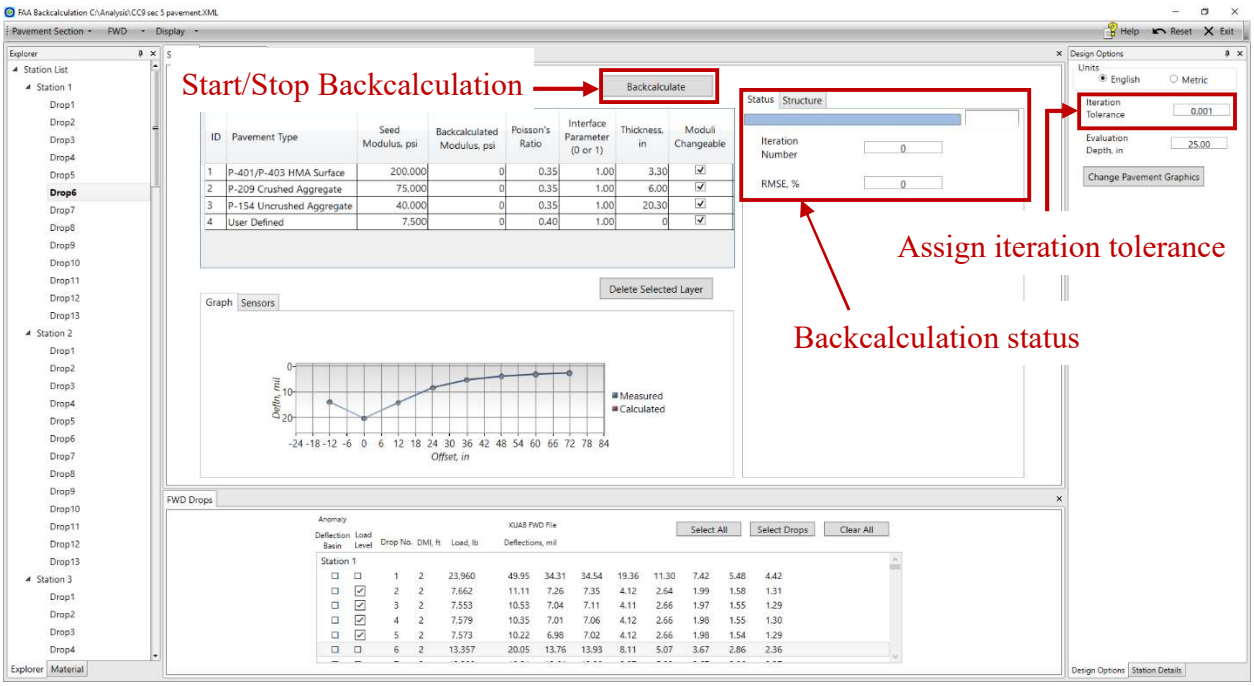


Figure 3-34. Start batch backcalculation analysis

3.8.1 Iteration Tolerance

Iteration tolerance is a user assigned dimensionless parameter that controls the stoppage criteria of the iterative process used in backcalculation analysis. At the end of each iteration, BAKFAA calculates an error term internally as defined in Equation 3-1.

$$Err = \sum_{i=1}^{N} (MD_i - CD_i)^2 \qquad (3-1)$$

where MD_i and CD_i are the measured and calculated deflections at sensor i, respectively, and N is the number of FWD sensors. The iteration tolerance *rtol* at the end of each iteration is defined in Equation 3-2:

$$rtol = 2 \times \frac{(Err)_j - (Err)_{j+1}}{(Err)_j + (Err)_{j+1}} \qquad (3-2)$$

where $(Err)_j$ and $(Err)_{j+1}$ are the errors at two consecutive iterations j and $j+1$, respectively. BAKFAA stops the backcalculation when the rtol is less than the user defined iteration tolerance.

The default value for iteration tolerance is 0.001, however the user can define another value based on the needs of the analysis being performed. The appropriate value for iteration tolerance is typically between 0.001 and 0.1. The user may choose to iterate values through this range to achieve desired accuracy, however higher iteration tolerance values may result in higher RMS error.

3.8.2 Iteration Number

Iteration number is an analysis output value that indicates the number of iterations completed in a backcalculation analysis. The user cannot change the iteration number. When multiple drops are analyzed, the iteration number for each drop is recorded and saved in the .csv output file. The interface displays the iteration number of the last drop selected.

3.8.3 Root Mean Square Error

The root mean square error (RMSE) between the measured and calculated deflections is an analysis output value and is calculated as a percentage per ASTM D5858-96 (2020) "Standard Guide for Calculating In Situ Equivalent Elastic Moduli of Pavement Materials Using Layered Elastic Theory" as defined in Equation 3-3.

$$RMSE = 100 \times \left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{d_i^c - d_i^m}{d_i^m}\right)^2\right]^{1/2} \qquad (3-3)$$

where:

d_i^c : Calculated deflection for ith sensor
d_i^m : Measured deflection for ith sensor
N: Number of sensors

In a single analysis, the RMSE is displayed in the "Section" module under the "Status" tab. In a batch analysis, the RMSE values for each selected drop are recorded in the .csv output file. The interface displays the RMSE of the last drop selected.

3.8.4 Backcalculated Modulus

In a single drop analysis, the backcalculated modulus values for each layer will be displayed in the "Backcalculated Modulus" column of the "Section" module. In a batch analysis, the "Geo. Mean Backcalculated Modulus" column displays the geometric mean of the backcalculated layer modulus values for the selected drops (see Figure 3-35) and the values for each selected drop are recorded in the .csv output file.

3.8.5 Calculated Deflections

BAKFAA reports the calculated deflections at the FWD sensor locations for all drops selected for the backcalculation analysis. In a single drop analysis, the calculated deflections are displayed in

the sensor table of the "Section" module. In a batch analysis, the calculated deflections are saved in the .csv file and the sensor table displays the calculated deflections for the last drop selected.

3.8.6 .CSV Output File

In a batch analysis, the results of the backcalculation will be saved in a .csv output file in a user specified directory. The .csv output file in BAKFAA contains the following information for each FWD drop selected:

- Pavement structure (pavement type, seed modulus, Poisson's ratio, interface parameter, thickness, and moduli changeable selection for each layer)
- FWD information (DMI, drop no., load, and operator comments or remarks)
- Iteration tolerance
- Backcalculated modulus
- RMSE
- Iteration number
- Measured deflections for each selected FWD drop
- Calculated deflections for each selected FWD drop
- LTE values (if selected by the user)

3.8.7 Show Output

Located under the "Display" drop-down in the top toolbar, the "Show Output" option opens a sub-window that displays a summary of the backcalculation results as shown in Figure 3-35. In a single drop analysis, the sub-window displays the backcalculation results for the specified drop, including the backcalculated modulus values, RMSE, iteration number, and calculated deflections. "Show Output" also displays FWD information and calculated Impulse Stiffness Modulus (ISM) as defined in Equation 3-4:

$$ISM = \frac{L}{D_0} \qquad (3-4)$$

where L is the applied FWD load and D_0 is the deflection under FWD load plate.

In a batch analysis, the FWD information and analysis results (calculated deflections, RMSE, iteration number, and ISM) are only displayed for the last test point selected.

In a batch analysis, the "Show Output" sub-window displays the geometric mean and arithmetic mean of the backcalculated modulus values for the selected points. For example, the arithmetic and geometric means of the backcalculated modulus of layer No. 1 are shown in Equations 3-5 and 3-6:

$$Arithmetic\ Mean\ (E_1) = \frac{1}{n}(E_{11} + E_{12} + \cdots + E_{1n}) \qquad (3-5)$$

$$Geometric\ Mean\ (E_1) = (E_{11} \times E_{12} \times \ldots \times E_{1n})^{1/n} \qquad (3-6)$$

where n is the number of test points selected for backcalculation. E_{11}, E_{12} and E_{1n} are the backcalculated modulus of layer 1 at test points 1, 2, and n.

The “Show Output” sub-window also contains an “FWD File Image” button displaying the content of the FWD file loaded (if a file is loaded) and a “LEAF Output” button showing the results of a LEAF analysis.

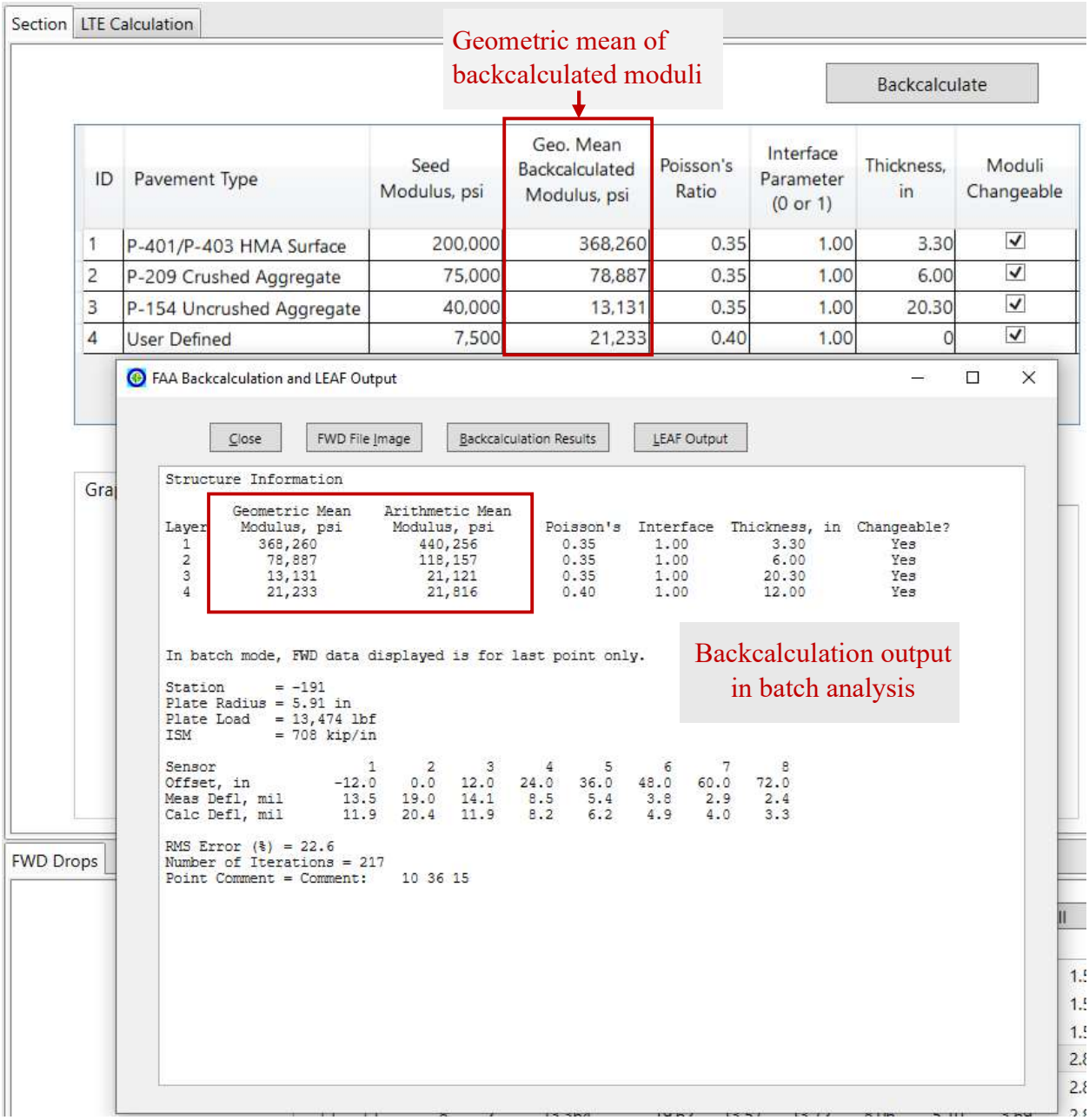


Figure 3-35. “Show Output” sub-window displaying backcalculation results of a batch analysis

3.8.8 Abort an In-Progress Backcalculation

When a backcalculation is being performed, the “Backcalculate” button will change to read “Stop Backcalculate”. The user can abort an in-progress backcalculation by clicking the “Stop Backcalculate” button (see Figure 3-34). Backcalculation results can still be found up to the point

of termination in the “Show Output” sub-window for a single analysis and in the .csv output file in a batch analysis.

3.9 Batch Graph

The “Batch Graph” function displays graphs showing either measured and calculated deflections or backcalculated layer moduli. The x-axis of the graphs is the DMI information of the selected test points. The sub-window is opened by clicking the “Display” drop-down from the tool bar and then selecting “Batch Graph”.

The “Deflections” tab displays the graph of the measured and calculated deflections for two sensors: 1) the sensor under the FWD load plate (e.g., D_0) and 2) the outermost sensor from the load plate (e.g., D_{72}). Select FWD drops in the “FWD Drops” module to view the measured deflections on the batch graph. Through use of the checkboxes in the tab, the user may choose to display either measured or calculated deflections or both simultaneously. The user may also toggle on and off any individual drop or individual sensor using the “Drop No.” and “Exclude Drop” boxes, respectively. It should be noted that the batch graph will only display information for drops selected in the “FWD Drops” module. For example, if there are four drops, 1 through 4, and only drops 3 and 4 are selected in the main interface, the batch graph will not be able to display drops 1 and 2, even if the user toggles them on using the checkboxes in the “Drop No.” box. The user may display measured deflections for any drop by closing the “Batch Graph” sub-window, altering the drop selections in the “FWD Drops” module, and reopening the “Batch Graph” sub-window. However, calculated deflections will only be displayed for drops included in the backcalculation analysis.

The “Modulus” sub-window displays the backcalculated modulus values for the drops and locations included in the backcalculation analysis in a logarithmic scale (for the vertical axis). The user can toggle off any layer of the pavement using the checkboxes in the “Layer” box.

Figure 3-36 shows an example of the “Deflections” tab in the “Batch Graph” sub-window. The example includes a backcalculation analysis performed for drop no. 6 at twelve stations (drop locations). Each drop and station are selected in the main window. The batch graph shows the measured and calculated deflections for sensors at 0-inch (D_0) and 72-inch offset (D_{72}) for all selected stations. Since the analysis used only drop no. 6, the batch graph shows the deflections only for that drop. Toggling on drop no. 2 will not change the graph in this example. Figure 3-37 shows the “Modulus” tab of the above example. The graph shows the backcalculated modulus values in logarithmic scale with all layers toggled on for display.

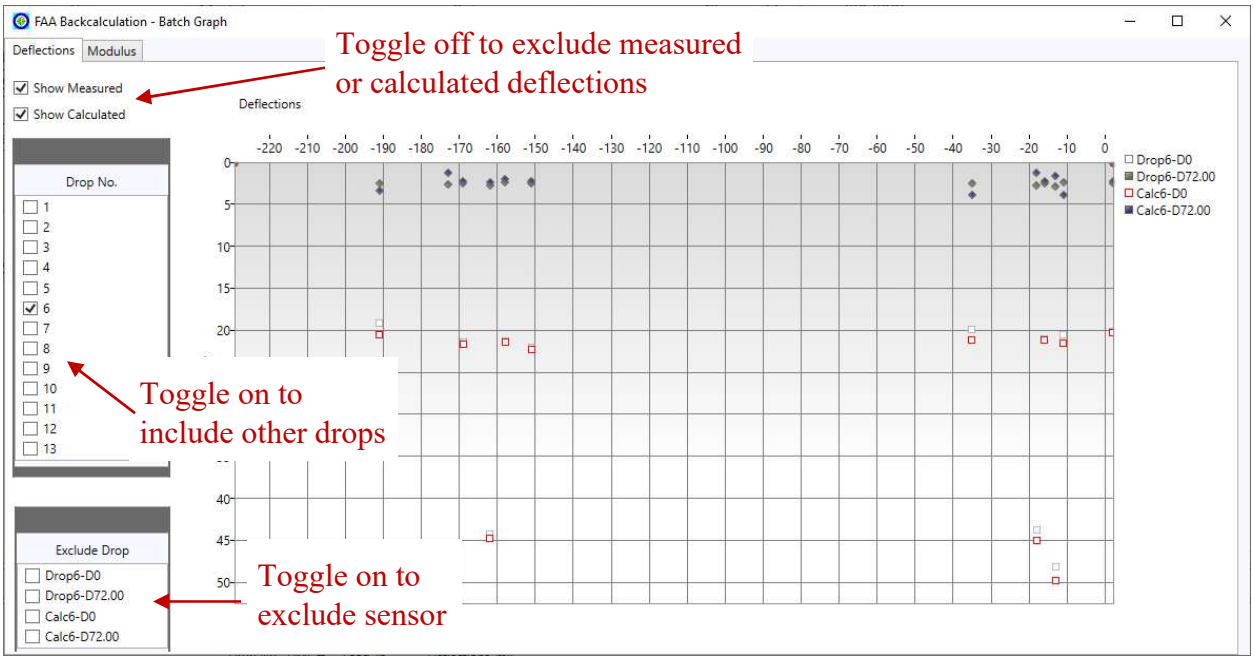


Figure 3-36. "Batch Graph" sub-window displaying measured and calculated deflections

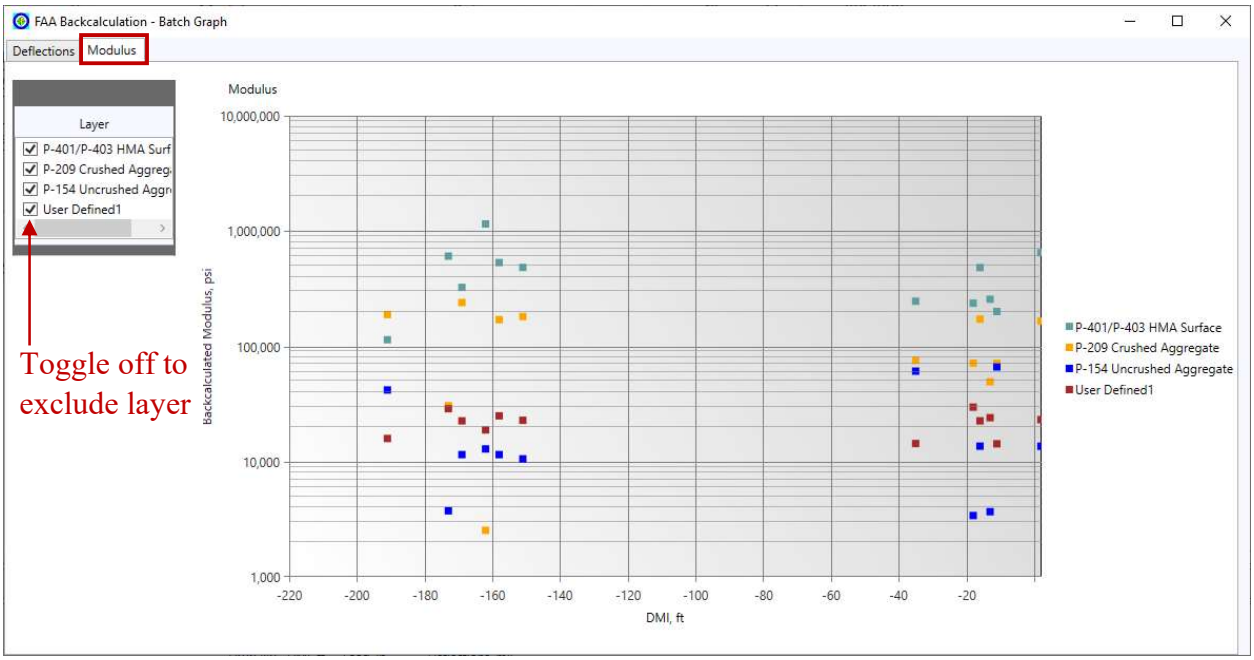


Figure 3-37. "Batch Graph" sub-window displaying backcalculated modulus values

3.10 Load Transfer Efficiency (LTE) Calculation

BAKFAA provides the user with the ability to calculate load transfer efficiency (LTE) between two jointed concrete slabs. For this, data from FWD tests performed at slab joints must be used. There are three LTE scenarios based on sensor configuration and layout during testing, each with

a unique LTE calculation. These are defined below and shown in the LTE case module (Figure 3-38).

1. Case 1 – a sensor at the negative offset (-12 inches), testing is conducted with the joint placed in between the load plate and the sensor at the first positive offset (12 inches).
    a. LTE (%): $\frac{d_2}{d_0} \times 100$
2. Case 2 – a sensor at the negative offset (-12 inches), testing is conducted with the joint placed in between the sensor at the negative offset (-12 inches) and the load plate.
    a. LTE (%): $\frac{d_1}{d_0} \times 100$
3. Case 3 – no sensor at the negative offset (-12 inches), testing is conducted with the joint placed in between the load plate and the sensor at the first positive offset (12 inches).
    a. LTE (%): $\frac{d_1}{d_0} \times 100$

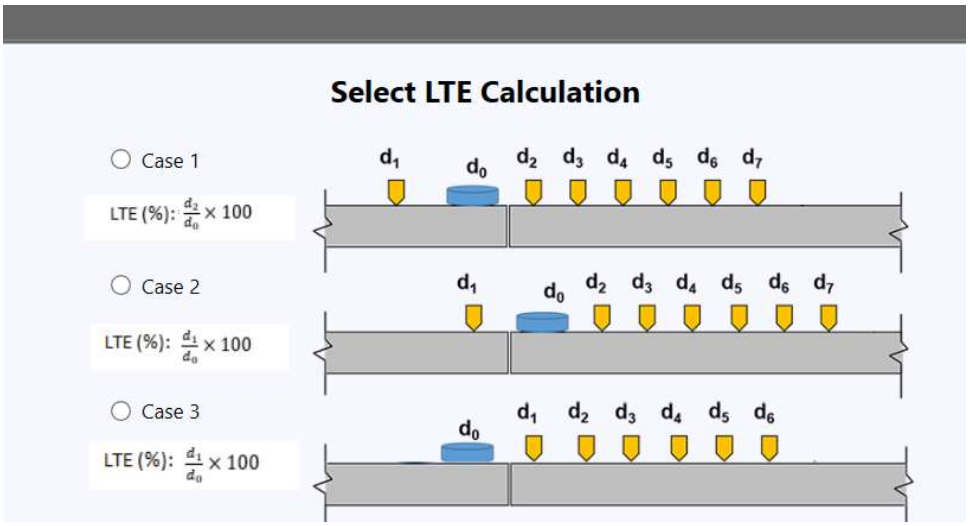


Figure 3-38. LTE cases

To perform the LTE calculation, follow steps listed below and reference Figure 3-39:

1. Load the desired FWD file and click on the "LTE Calculation" module.
2. Three LTE scenarios will be displayed. Between Cases 1, 2, and 3, select the one that corresponds to the FWD sensor configuration and layout during testing.
3. Click on an FWD drop. The calculated LTE will appear in the LTE tab.

BAKFAA will automatically disable Case 1 and 2 for FWD tests with no negative offset sensor or if the user excludes the negative offset sensor. BAKFAA disables Case 3 for FWD tests with a negative offset sensor. In batch analysis, the LTE values are saved in the .csv file.

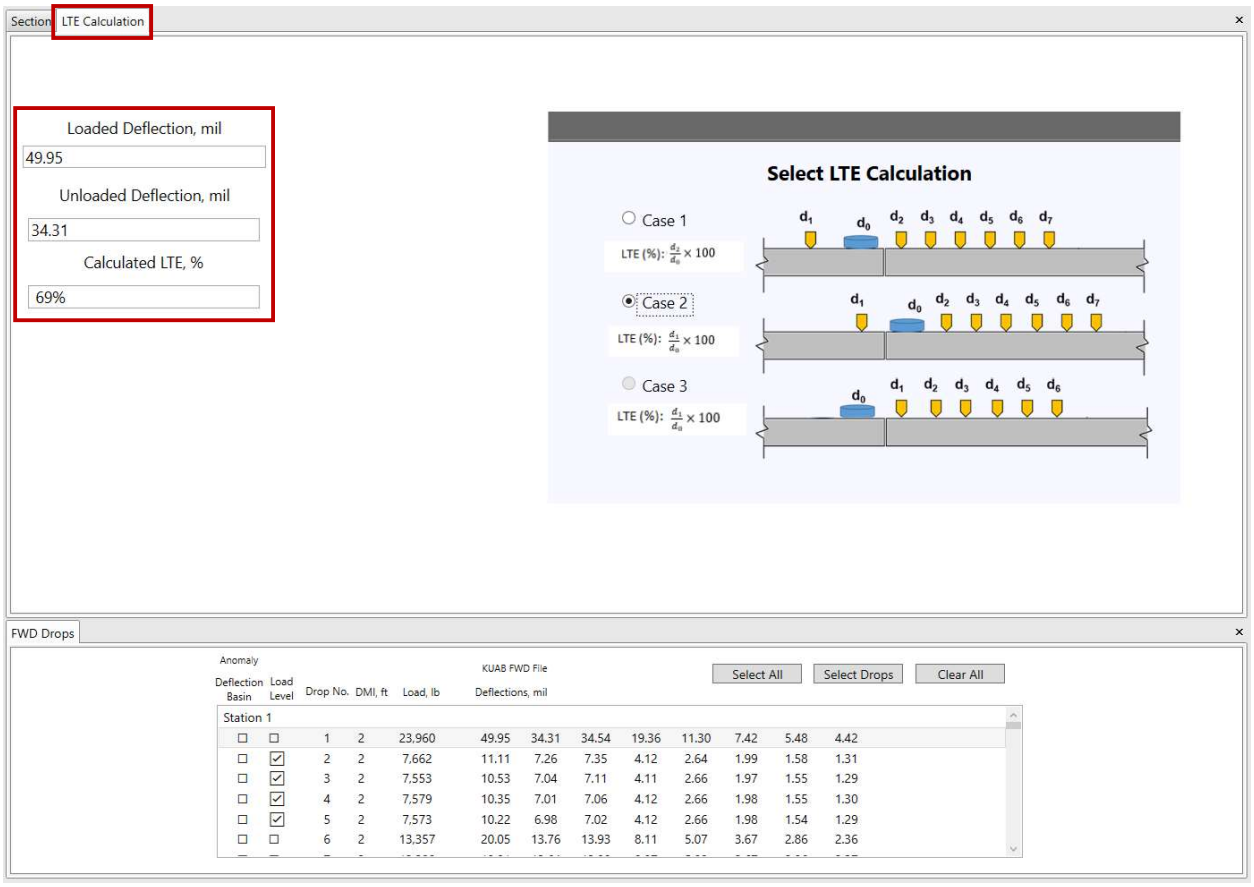


Figure 3-39. LTE calculation in BAKFAA

CHAPTER 4. SELECT LOAD AND RUN LEAF

The FAA Layered Elastic Analysis program (LEAF) is built into BAKFAA, allowing the user to compute pavement responses under various load types. LEAF calculates pavement stress, strain, and deflection values at different evaluation points within the pavement structure using a layered elastic model. Layered elastic models assume that each layer in a pavement structure is homogeneous, isotropic, and linearly elastic. The model is based on the Boussinesq mathematical model which assumes the following:

1. The pavement layers extend infinitely in the horizontal direction.
2. The bottom layer extends infinitely downward.
3. The materials are not stressed beyond their elastic ranges.

LEAF requires modulus of elasticity, Poisson's ratio, layer thickness, and layer interface condition values to perform the layered elastic analysis. LEAF retrieves the required inputs from the pavement structure defined by the user in the "Section" module. LEAF uses either the seed modulus values or the backcalculated modulus if a backcalculation analysis has been performed. The user can assign the depth of evaluation (the depth at which the pavement responses are calculated) in the "Design Options" module as shown in Figure 4-1.

Figure 4-1 illustrates the inputs required for a LEAF analysis. In this example, since no backcalculation analysis has been performed, LEAF will use the seed modulus values for analysis. Note that continuity in some of the pavement responses between two pavement layers depends on their interface condition. For fully bonded interfaces (interface parameter of 1), continuity exists for the horizontal strains, vertical stress, and deflection. For fully unbonded interfaces (interface parameter of 0), continuity in responses only exist in vertical stress and deflection. The evaluation depth must be assigned so that it represents the target layer. For example, the assigned evaluation depth is 9.29 inches in Figure 4-1 to calculate the pavement responses at the bottom of the P-209 layer, which is located at a depth of 9.30 inches.

To initiate the LEAF analysis, click the "Display" drop-down in the top toolbar and then click "Select Load and Run LEAF" after setting the required values.

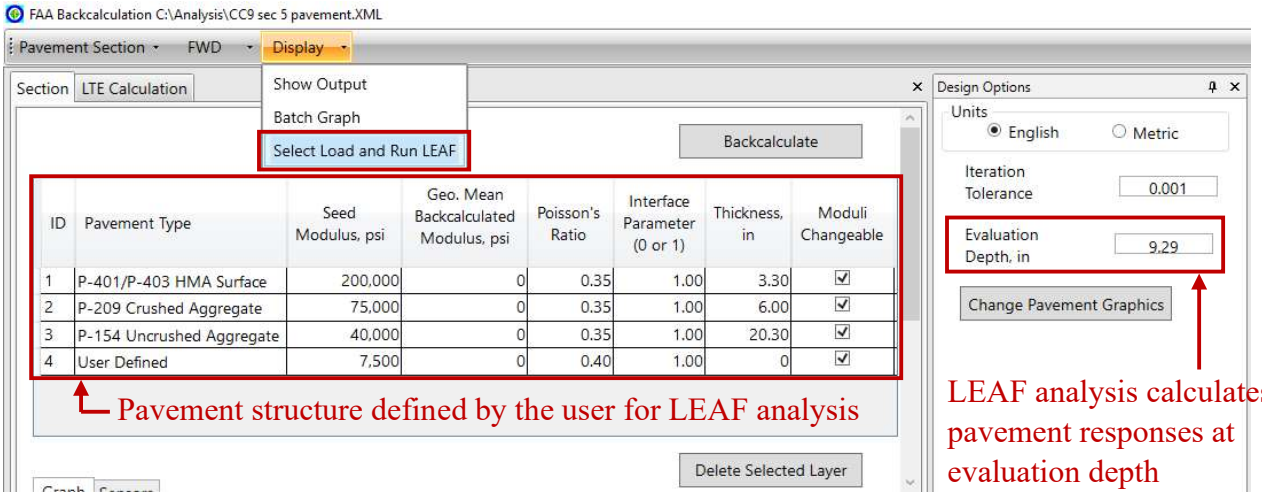


Figure 4-1. Pavement structure and evaluation depth settings for LEAF analysis

Table 4-1 shows the critical locations, including the desired response to be used and reason why, commonly used in the analysis of pavement structure response.

Table 4-1. Critical locations in analysis of pavement structure responses

Location	Response	Reason for Use
Pavement surface	Deflection	Imposing load restrictions and overlay design
Bottom of HMA layer	Horizontal tensile strain	Predict fatigue failure in the HMA
Top of intermediate layer (Base or Subbase)	Vertical compressive strain	Predict rutting failure in the base or subbase
Top of subgrade	Vertical compressive strain	Predict rutting failure in the subgrade

4.1 Load and Gear Configuration

After clicking "Select Load and Run Leaf", the "FWD Drops" module will be replaced with a "Select Load" module containing the LEAF load library (see Figure 4-2). The load library contains generic gear geometry, aircraft gear configurations, and FWD loading configurations. The user must select the desired load type to perform the LEAF analysis.

The information in the LEAF load library is contained in the file LEAFAircraft.Ext, located in the same directory as the application executable BAKFAA.exe (C:\Program Files\FAA\BAKFAA by default). The user can modify the LEAFAircraft.Ext file, including adding or removing user-defined landing gear geometry or modifying the X and Y coordinates of defined evaluation points. Figure 4-3 provides an example of items in LEAFAircraft.Ext file that define a B-747-400 landing gear within the program. Figure 4-4 shows a schematic of the B-747-400 landing gear parameter.

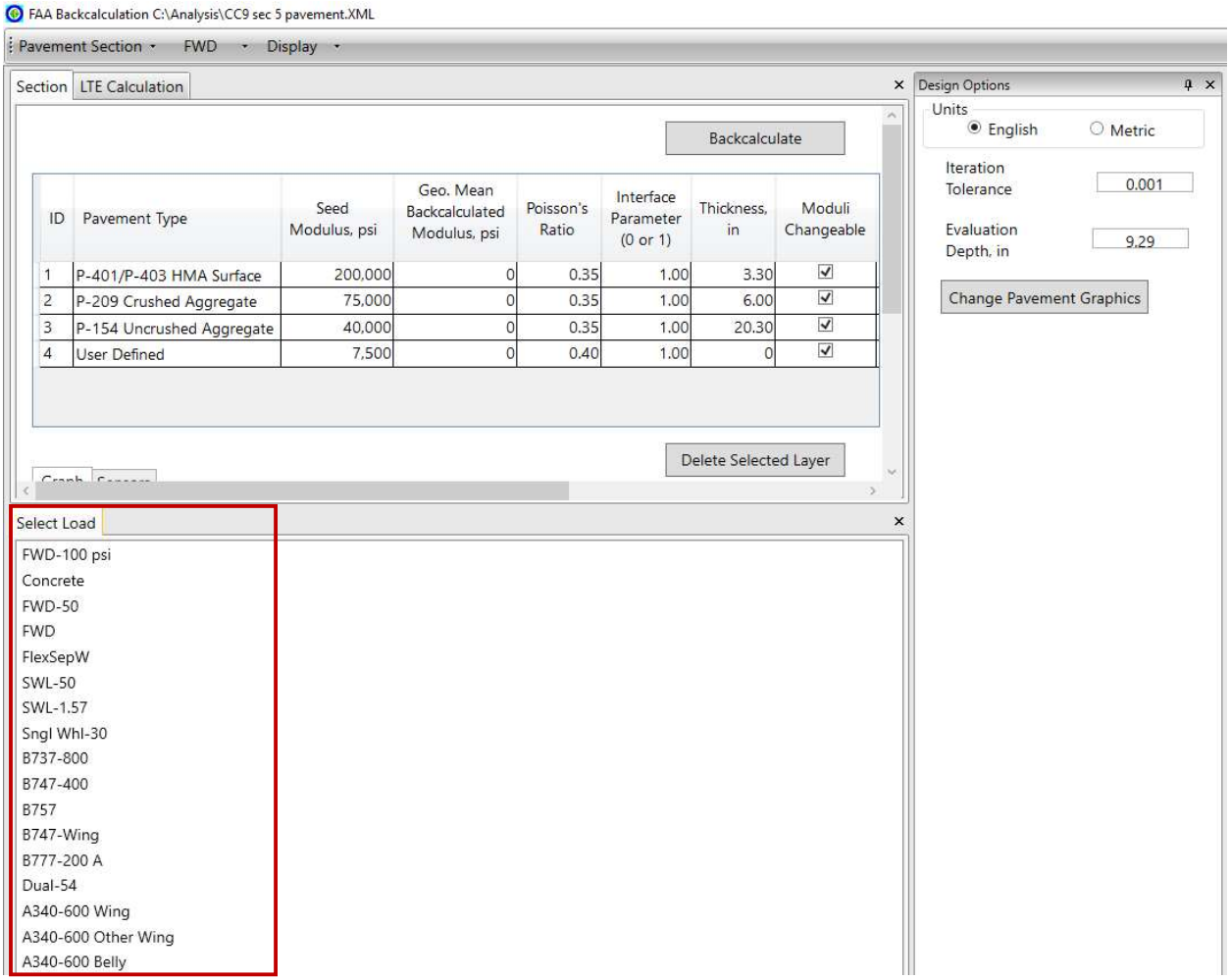


Figure 4-2. Load Library for LEAF analysis

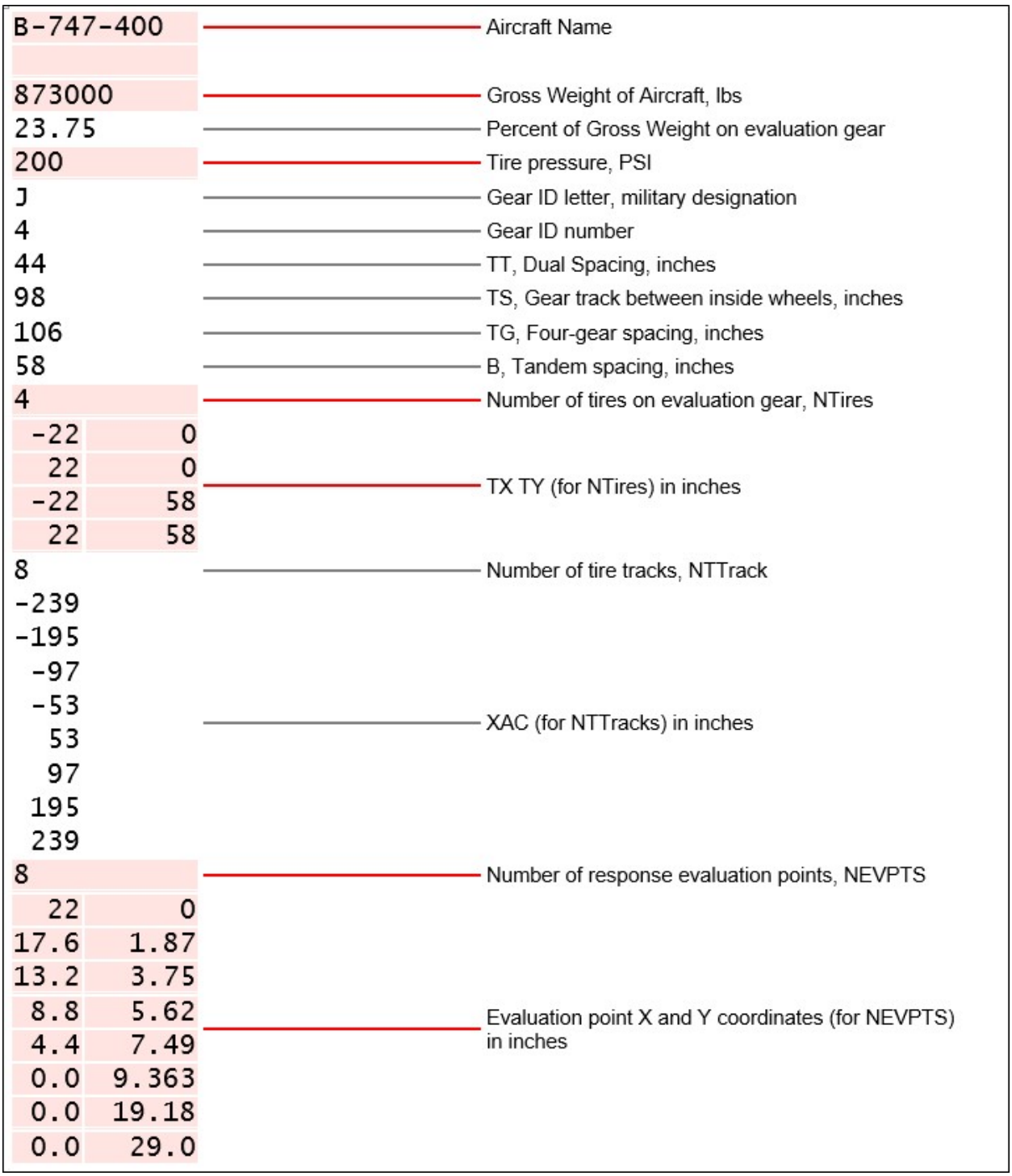


Figure 4-3. Summary of items in the LEAFAircraft.Ext file for a B-747-400 example

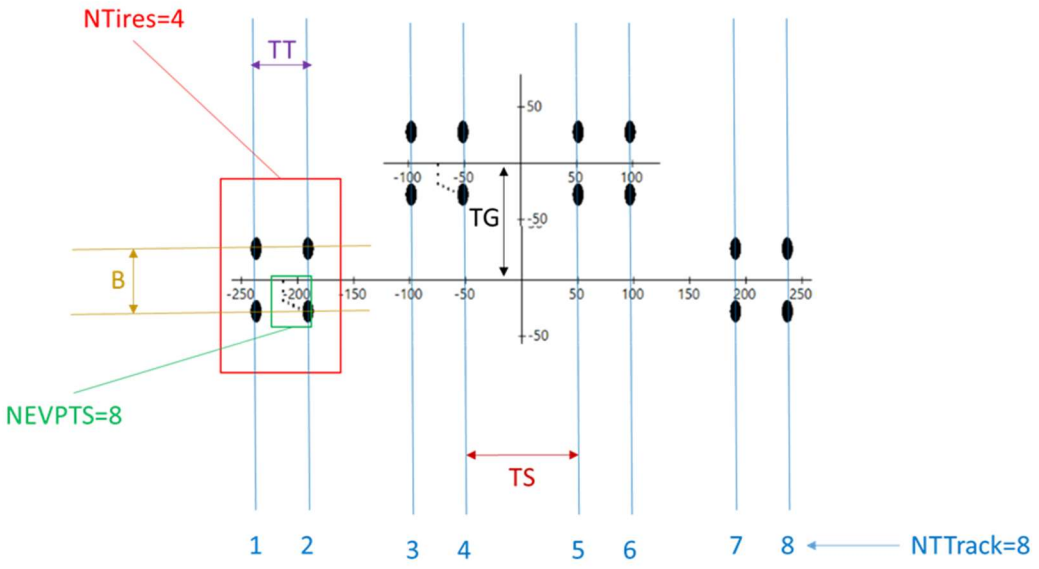


Figure 4-4. Schematic of B-747-400 Landing Gear Parameters in the LEAFAircraft.Ext file

4.2 LEAF Output

After defining the pavement layer structure, assigning the target evaluation depth, and selecting the load type from the library, the LEAF output will automatically open the "Show Output" sub-window to view the LEAF output. It can be manually opened by the user by clicking "Display", "Show Output", then "LEAF Output". The LEAF output is an ASCII text document containing:

A. Pavement layer structure table

- A.1 Layer no.
- A.2 Thickness
- A.3 Modulus of elasticity
- A.4 Poisson's ratio
- A.5 Interface condition

B. Aircraft table

- B.1 Aircraft gear data table from LEAFAircraft.Ext
- B.2 Aircraft Make/Mode
- B.3 Fraction of load on main gear, percent
- B.4 Gear load
- B.5 Number of tires

C. Tire data table from LEAFAircraft.Ext, with loads evenly divided among all tires

- C.1 Tire no.
- C.2 Tire radius
- C.3 Tire contact area

C.4 Tire contact pressure
C.5 Tire load
C.6 X-coordinate
C.7 Y-coordinate

D. Pavement responses at evaluation points, including eval. point no., layer no. (determined by evaluation depth), X-coord, Y-coord, and Z-depth

D.1 Stress
- D.1.1 Vertical
- D.1.2 Horizontal Y
- D.1.3 Horizontal X
- D.1.4 XZ shear
- D.1.5 YZ shear
- D.1.6 XY shear

D.2 Strain
- D.2.1 Vertical
- D.2.2 Horizontal Y
- D.2.3 Horizontal X
- D.2.4 XZ shear
- D.2.5 YZ shear
- D.2.6 XY shear

D.3 Displacement
- D.3.1 Vertical
- D.3.2 Horizontal X
- D.3.3 Horizontal Y

CHAPTER 5. SAMPLE BAKFAA ANALYSIS

This section contains sample analyses using BAKFAA, including backcalculations for three different pavement structures (flexible, rigid, and composite) and an LTE calculation. The data files used for the follow examples, including the FWD files and the saved pavement section .XML files, can be found in the BAKFAA 3.4 setup package, downloadable from the following link:

https://www.airporttech.tc.faa.gov/Products/Airport-Pavement-Software-Programs

To follow along with the examples or for further practice, these files should be downloaded to the computer running BAKFAA.

Also included in this section is a discussion of commonly encountered error messages.

5.1 Flexible Pavement Example

This example uses data from an HWD test conducted on a 3-layered flexible pavement. Table 5-1 shows the pavement structure. The HWD test drops include one seating drop of 36 kips (165 kN) and three drops of 12 kips, 24 kips, and 36 kips (55 kN, 110 kN, and 165 kN).

Table 5-1. Composition of pavement layers for the flexible pavement example

Material Type	Thickness (in)
P-401/P-403 Hot Mix Asphalt	6
P-209 crushed stone base	12
Subgrade with a CBR of 10	

To perform the backcalculation:

1. Assign the pavement layer structure provided in Table 5-1 manually or by loading the "Flexible Pavement Structure.xml" file by clicking "Pavement Section" in the toolbar then "Load Pavement Section" and selecting the desired file.
2. Assign seed modulus values for each layer. Factors influencing the selection of an appropriate seed modulus value are discussed below.

   The knowledge from the asphalt layer temperature can help in the estimation of the seed modulus for that layer. Other factors such as aging conditions and level of damage should also be considered. The FWD file shows the air and pavement surface temperatures as 40°F and 54°F, respectively. The asphalt concrete mid-depth temperature can be estimated using the BELLS equation in accordance with ASTM D7228-06a(2020), "Standard Test Method for Prediction of Asphalt-Bound Pavement Layer Temperatures". BELLS incorporates the pavement surface temperature recorded by the FWD machine, the time of the day the test was conducted and the average temperature over the past 24-hour period to estimate the

asphalt mid-depth temperature. For this example, the mean previous day temperature is equal to 68°F. Using the BELLS equation, the asphalt mid-depth temperature is 60.3°F.

The following exponential equation can be used to estimate the asphalt concrete seed modulus at the testing temperature using a reference modulus corresponding to a reference temperature.

$$E_T = E_{Ref} \times e^{C(T - T_{Ref})} \qquad (5-1)$$

where E_T = modulus at testing temperature

T = testing temperature at asphalt mid-depth

E_{Ref} = known reference modulus at reference temperature

T_{Ref} = reference temperature

C = coefficient

C is typically taken as -0.036 in US units or -0.065 in SI units. Assuming the reference modulus is 500,000 psi at 77°F, the estimated modulus at 60.3°F would be 912,153 psi. Using a seed modulus of 910,000 psi should be sufficient. Alternatively, any temperature correction relationship for similar asphalt concrete material can be used for this estimation.

Use the default seed modulus of 75,000 psi for the P-209 base layer.

Using the $E = 1{,}500 \times CBR$ (Heukelom & Klomp 1962) correlation, the subgrade seed modulus would be equal to 15,000 psi.

3. Load the saved "Flexible Pavement Example.fwd" file by clicking "FWD" in the ribbon then "Load FWD File" and selecting the file.
4. Select drop no. 4 and exclude the sensor at the -12 in. offset.
5. Start backcalculation by clicking the "Backcalculate" button.
6. Review the backcalculation results.

When reviewing the backcalculation results, use engineering judgment to determine whether the backcalculated modulus values are realistic and if the RMSE is acceptable.

Figure 5-1 shows the resulting backcalculation modulus values. While the RMSE is nearly acceptable, the backcalculated modulus values are not reasonable. The modulus of the asphalt layer is high and the modulus of the base layer is lower than that of the subgrade.

In this example, the following can be performed in an attempt to improve the backcalculation results:

- Change the seed modulus.
- Change the iteration tolerance.
- Add a stiff layer below the subgrade.

If the seed modulus values for the base and subgrade are changed to 60,000 psi and 10,000 psi, respectively, the backcalculated modulus values are more realistic, but the RMSE of 14.6% is relatively high (Figure 5-2). Further attempts should be made to improve the analysis.

In the next attempt, add a stiff layer with a seed modulus of 100,000 psi 10 ft. below the pavement surface. The backcalculation should be repeated by treating the stiff layer modulus as both changeable and not changeable (fixed) and the results compared. Figure 5-3 shows the backcalculation results where the stiff layer was fixed in the backcalculation analysis. In this case, the resulting backcalculated modulus values are realistic and the RMSE is less than 5%. The asphalt modulus of 1,113,820 psi at the test temperature 60.3°F is equivalent to a modulus of 611,000 psi at the reference temperature of 77°F. It is important to adjust all temperature-dependent materials to a reference temperature for reporting purposes. The backcalculated base modulus of 69,687 psi is reasonably consistent with the default modulus for the P-209 aggregate base material. Finally, the backcalculated subgrade modulus of 10,289 psi (equivalent to CBR of 6.7) is lower than the assumed subgrade CBR, but it is still acceptable.

Converted into ksi, the results of this analysis for all drops are presented in Table 5-2.

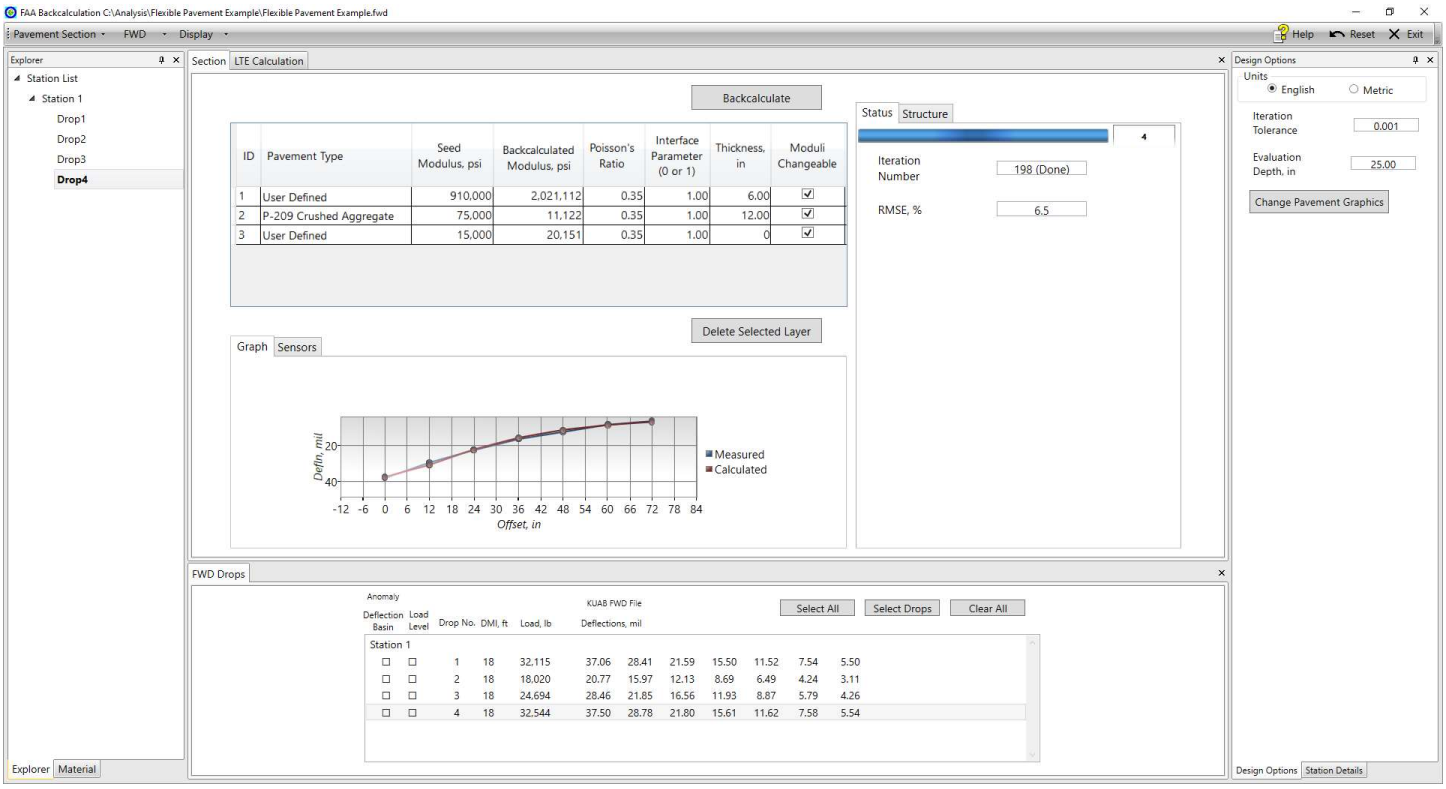


Figure 5-1. Backcalculation results for flexible pavement example (first attempt)

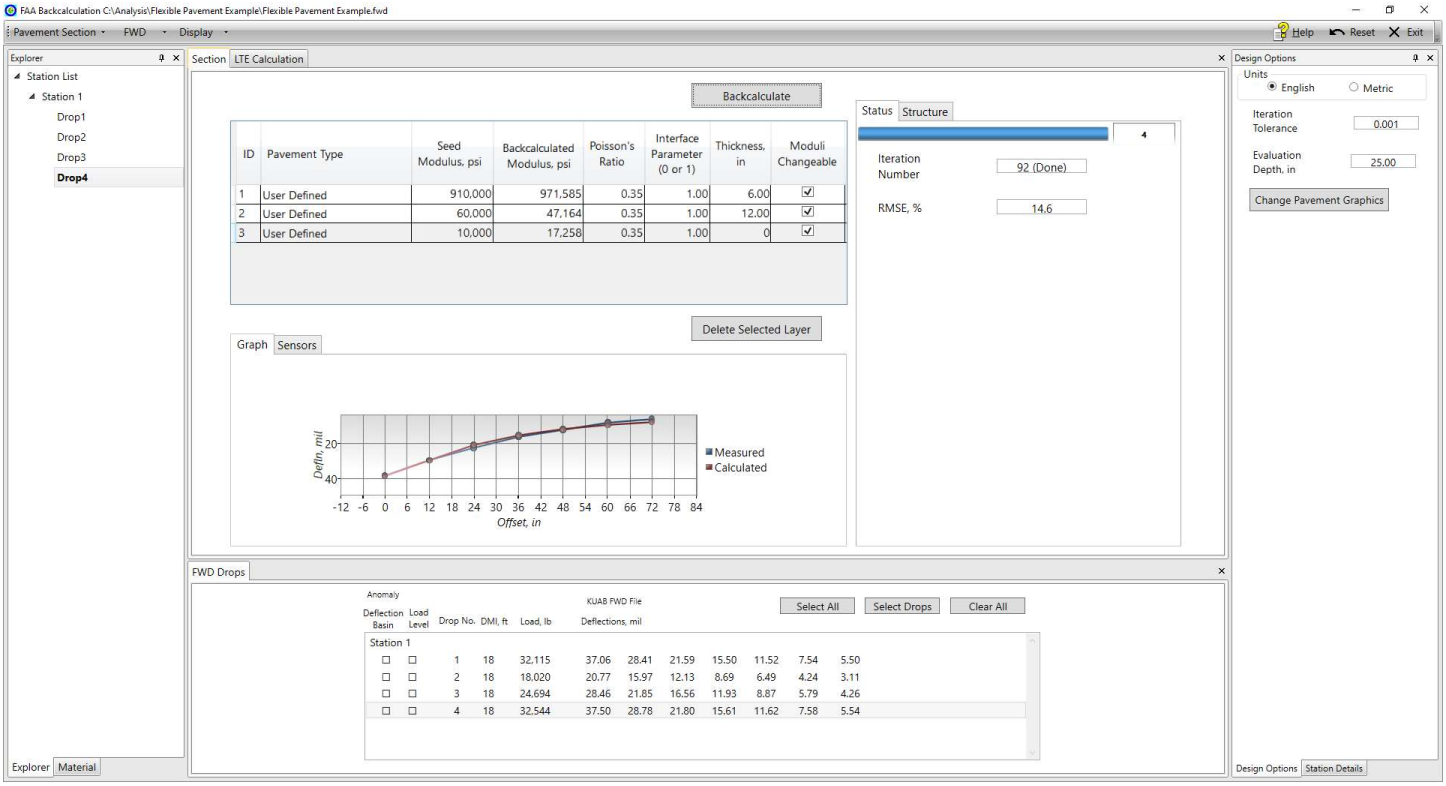


Figure 5-2. Backcalculation results for flexible pavement example (second attempt)

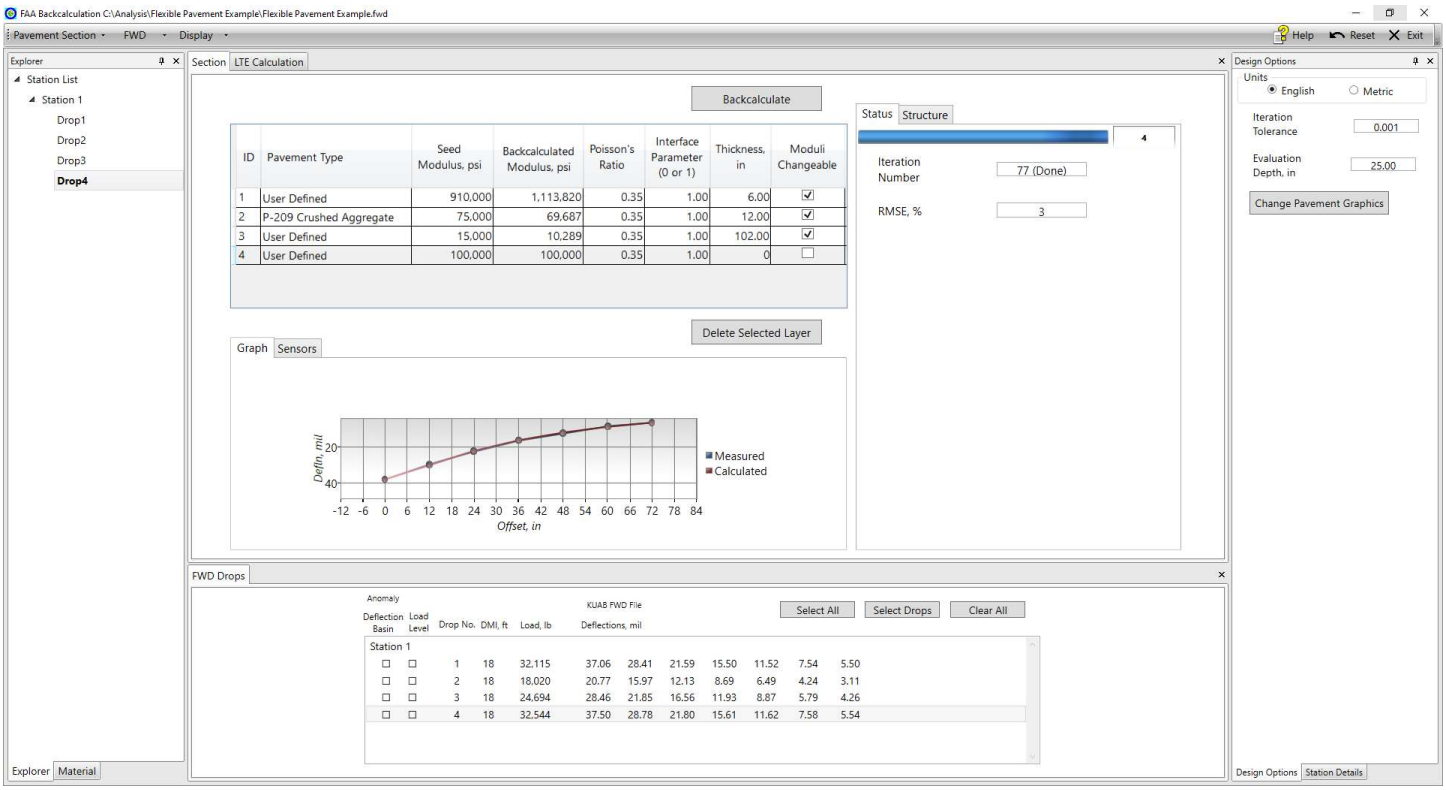


Figure 5-3. Backcalculation results for flexible pavement example (third attempt)

Table 5-2. Flexible pavement example backcalculated layer modulus values and statistics

Drop[1]	Asphalt Concrete (ksi)[2]	Aggregate Base (ksi)	Subgrade (ksi)	RMSE %
2	1,1150 [630]	68.3	10.2	2.8
3	1,122 [615]	69.8	10.2	2.9
4	1,115 [611]	69.8	10.3	3
Geometric Mean[2]	1,129 [619]	69.3	10.2	2.9
Standard Deviation[2]	1.01	1.011	1.0	0.1

1. Seating drop is not typically included in analysis.

2. At tested temperature and at reference temperature of 77°F between brackets.

5.2 Rigid Pavement Example

This example uses data from an HWD test conducted on a 3-layered rigid pavement and performs both a single analysis and a batch analysis. Table 5-3 shows the pavement structure. The HWD test drops are 30 kips, 32 kips, and 36 kips (133 kN, 142 kN, and 160 kN) with testing only performed at the center of the slabs.

Table 5-3. Composition of pavement layers for the rigid pavement example

Material Type	Thickness (in)
P-501 Cement concrete pavement	14
P-156 Cement Treated Subgrade	12
Subgrade with a CBR of 6	

To perform the single analysis backcalculation:

1. Assign the pavement layer structure provided in Table 5-3 manually or by loading the saved “Rigid Pavement Structure.xml” file by clicking “Pavement Section” in the toolbar then “Load Pavement Section” and selecting the desired file.
2. Assign seed modulus values for each layer as shown in Figure 5-4.
3. Load the saved “Rigid Pavement Example.FWD” file by clicking “FWD” in the toolbar then “Load FWD File” and selecting the file.
4. Select drop no. 3 at DMI location of 200 ft.
5. Start backcalculation by clicking the “Backcalculate” button.
6. Review the backcalculation results.

While the RMSE is low (1.0%), the backcalculated modulus values are not reasonable, as shown in Figure 5-4. After adding a stiff layer 10 ft. below the pavement surface, rerun the backcalculation. In this case, the backcalculated modulus values are reasonable and the RMSE is low (1.6%), as shown in Figure 5-5.

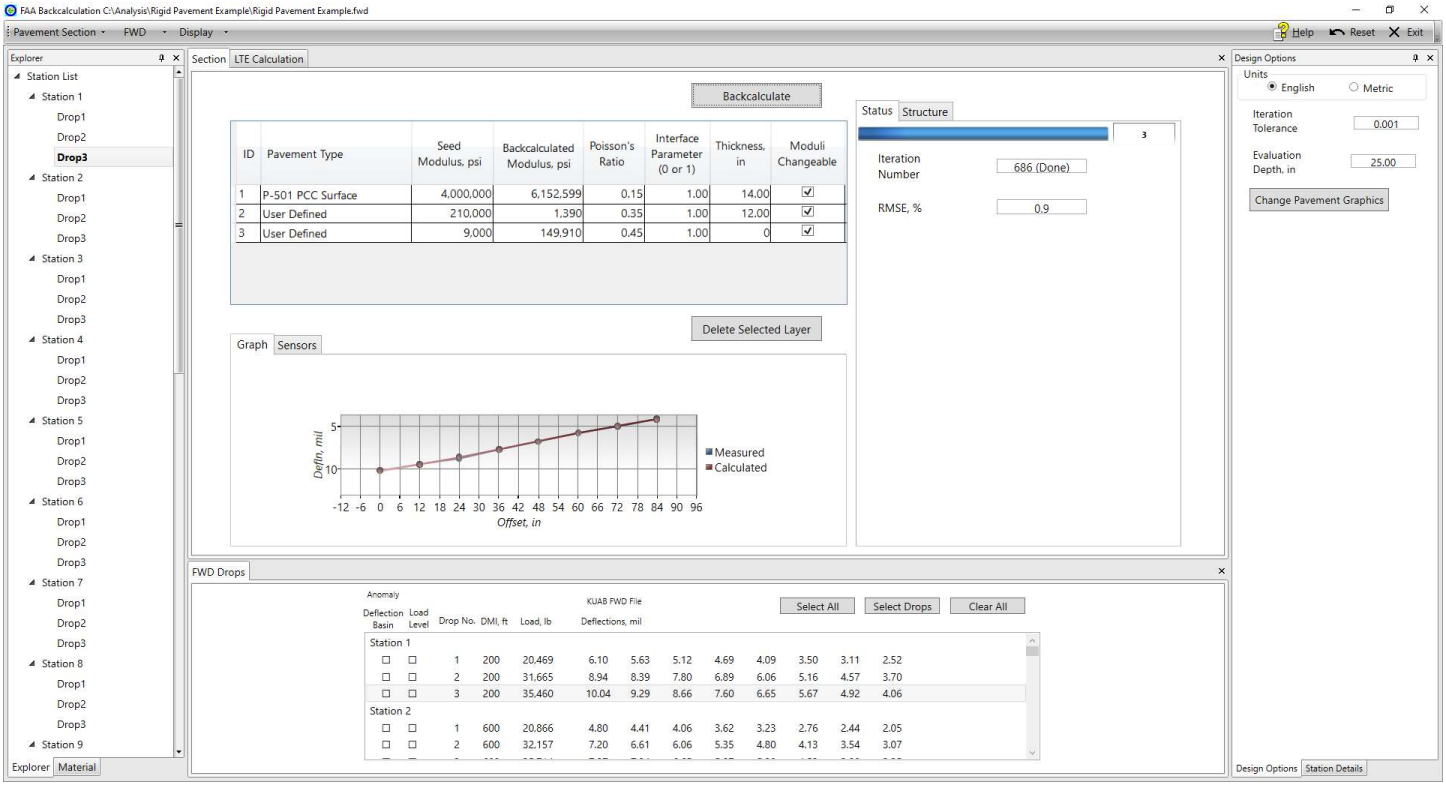


Figure 5-4. Backcalculation results for rigid pavement example (first attempt)

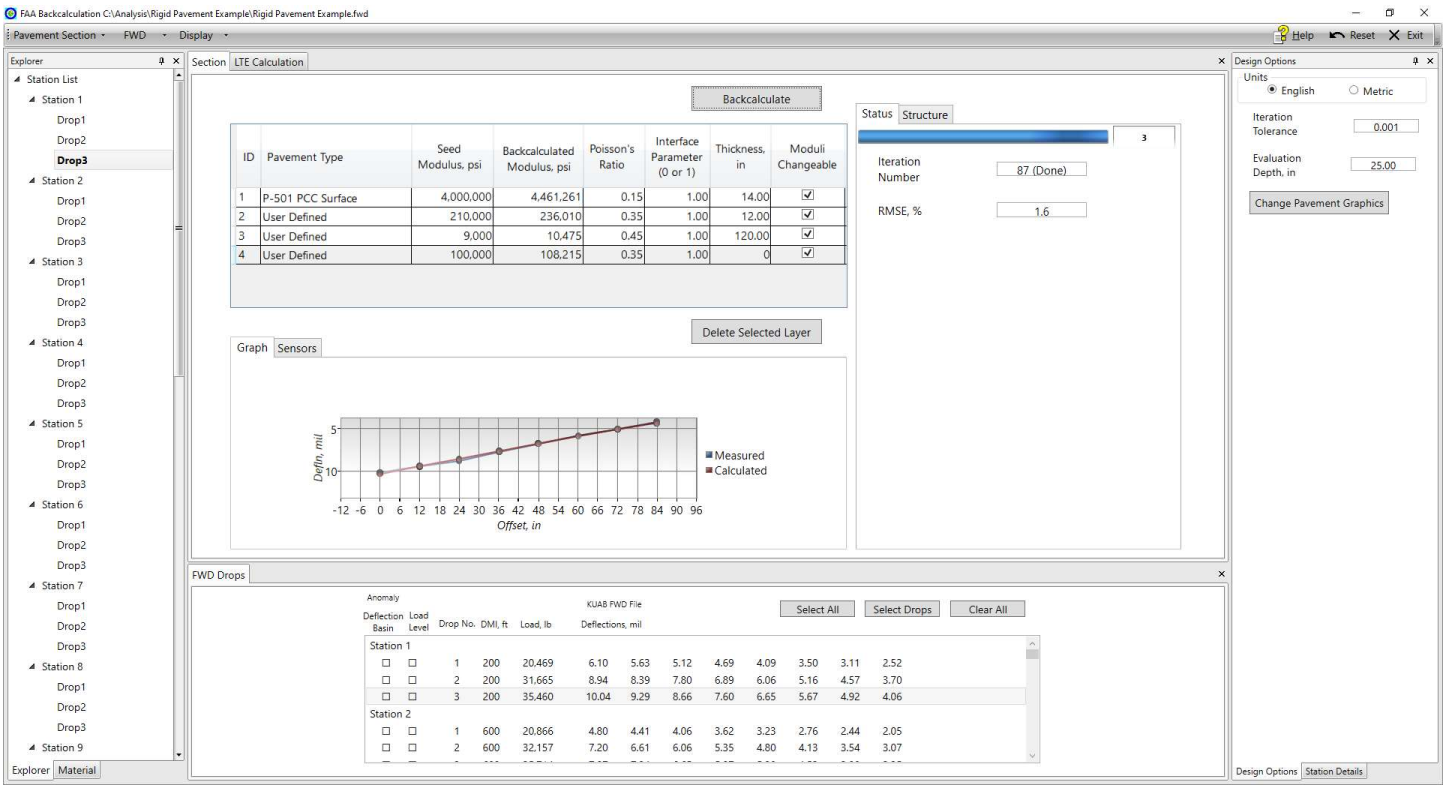


Figure 5-5. Backcalculation results for rigid pavement example (second attempt)

To perform the batch analysis backcalculation:

1. Assign the pavement layer structure using the refined structure from the single analysis backcalculation above, as shown in Figure 5-5. This can be done manually or by loading the saved "Rigid Pavement Structure_Stiff Layer.xml" file.
2. Assign seed modulus values for each layer as shown in Figure 5-5.
3. Load the saved "Rigid Pavement Example.FWD" file.
4. Select the third load level drops (drop no. 3) for all test locations by clicking the "Select Drop" button in the "FWD Drops" module. Select drop number "3" from the menu, then click "OK".
5. Start backcalculation by clicking the "Backcalculate" button.
6. When prompted, click "Yes" to performing a batch analysis and assign a file path to save the .csv file.
7. Click the "Display" drop-down then "Batch Graph" to display the backcalculated deflections and modulus values and review the results, as shown in Figure 5-6.

The test location at 6,600 ft. seems to be an outlier based on the very high backcalculated PCC modulus of 12,000 to 13,000 ksi. Inspection of the deflection basin at DMI of 6,600 (Figure 5-7) reveals an unusual deflection with minimal decrease in sensor deflections at offset distances greater than 48 in. This point can be removed from the backcalculation by three methods:

1. Delete the drops for that location from the FWD data "View FWD File" sub-window (Figure 5-8).

a. Click “FWD” in the toolbar then “View FWD File” to open the sub-window.
b. Under the “Delete Drop” column, check the boxes for the drops at the 6,600 ft. location.
c. Click the “Update Data” button. This will remove the FWD data for this location from the loaded FWD data used in the backcalculation. (Note: This will not affect the original FWD file saved to the computer.)
d. Rerun the backcalculation.

2. Deselect the drop for that location.
    a. In the “FWD Drops” module, deselect the highlighted drop. This will remove it from the backcalculation analysis, but will not remove it from the loaded FWD data.
    b. Rerun the backcalculation.
3. Delete the results for that drop and location manually in the output .csv file.

Converted to ksi, the backcalculated modulus values and associated statistics after removing the outlier test point are presented in Table 5-4.

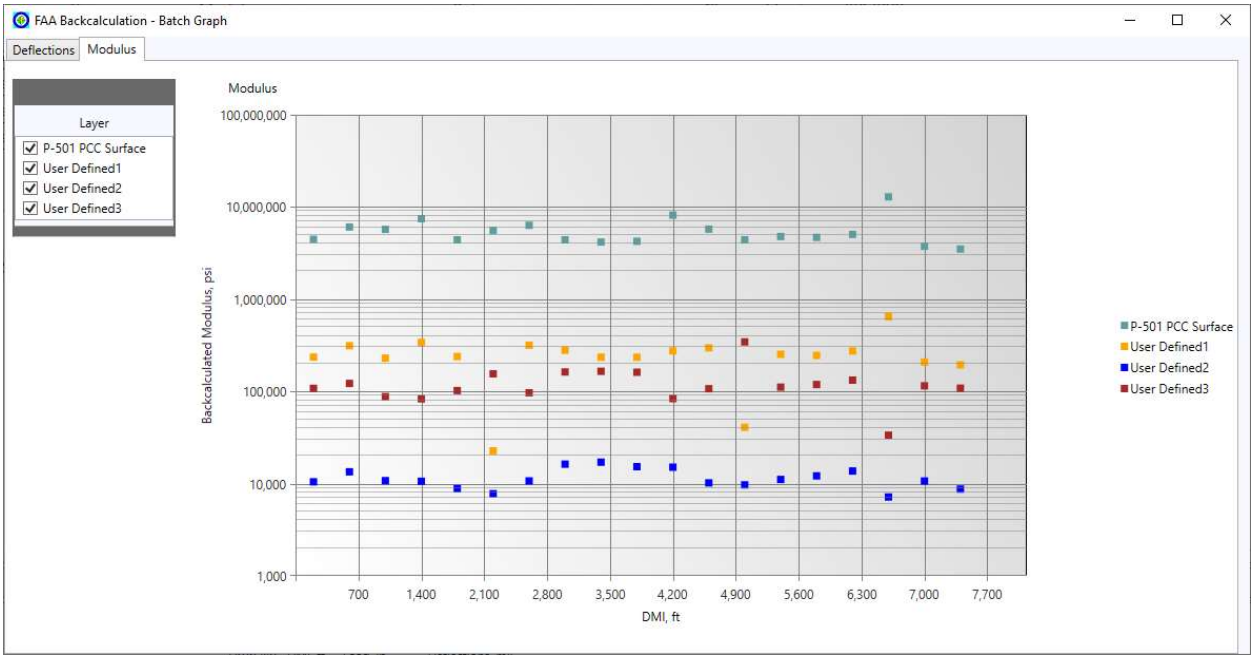


Figure 5-6. Backcalculated layer modulus values for all test points for drop no. 3 in rigid pavement example

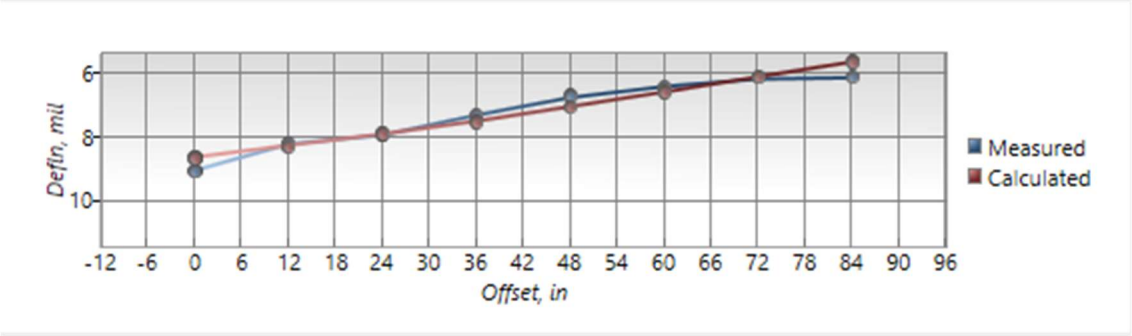


Figure 5-7. Deflection basin at DMI of 6,600 ft. showing unusual deflections in the rigid pavement example

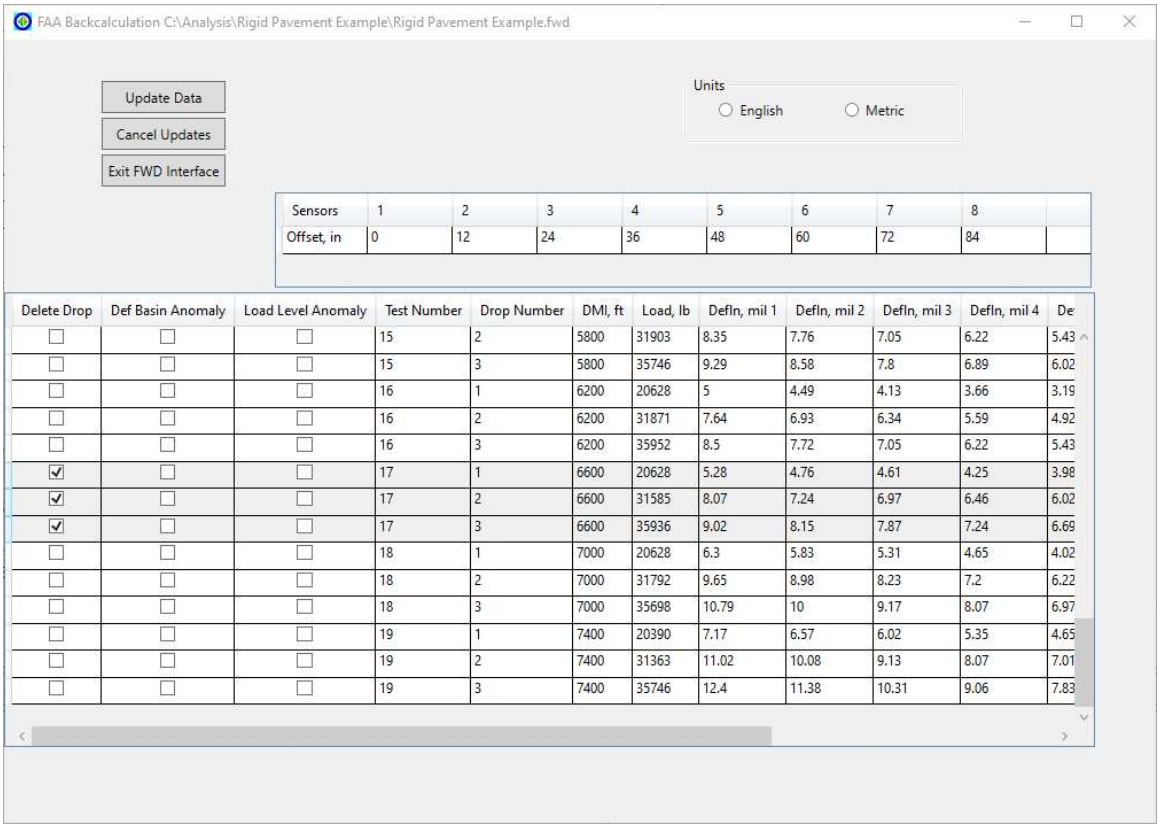


Figure 5-8. FWD data for DMI of 6,600 ft. removed using the "View FWD File" sub-window in the rigid pavement example

Table 5-4. Rigid pavement example layer modulus values and statistics

Parameter[1]	PCC (ksi)	Cement Treated Subgrade (ksi)	Subgrade (ksi)
Layer Modulus Geometric Mean	5,032	203	11.5
Geometric Standard Deviation Factor (STDF)	1.255	2.04	1.252
16th Percentile (Average/STDF)	4,009	99	9.2

1. Excluding outlier at DMI 6,600 ft.

5.3 Composite Pavement Example

This example uses data from an HWD test conducted on a 3-layered composite pavement. Table 5-5 shows the pavement structure. The HWD test drops include one unrecorded seating drop of 32 kips and three drops of 32 kips, 42 kips, and 54 kips (142 kN, 186 kN, and 240 kN). The FWD file shows the air and pavement surface temperatures ranging from 52°F to 57°F and 46°F to 50°F, respectively. This example will use three approaches: default seed modulus values, temperature-adjusted P-401 seed modulus value, and fixed P-401 modulus value.

Table 5-5. Composition of pavement layers for the composite pavement example

Material Type	Thickness (in)
P-401/P-403 Hot Mix Asphalt	2.25
P-501 Portland Cement Concrete	10
Subgrade with a CBR of 15	

To perform the batch analysis backcalculation:

1. Assign the pavement layer structure provided in Table 5-5 manually or by loading the saved “Composite Pavement Structure.xml” file by clicking “Pavement Section” in the toolbar then “Load Pavement Section” and selecting the desired file.
2. Assign seed modulus values for each layer as shown in Figure 5-9 (BAKFAA default values).
3. Load the saved “Composite Pavement Example.FWD” file by clicking “FWD” in the toolbar then “Load FWD File” and selecting the file.
4. Select drop no. 2 for all test points.
5. Start backcalculation by clicking the “Backcalculate” button.
6. When prompted, click “Yes” to performing a batch analysis and assign a file path to save the .csv file.
7. Review the results.

The first approach using default seed modulus values results in highly variable P-401 and P-501 layer backcalculated modulus values, as shown in Figure 5-10. Additionally, the geometric mean of the P-401 layer backcalculated modulus values (approximately 517 ksi) is very low when considering the low air and surface temperatures recorded during the HWD testing. These factors indicate that the backcalculation analysis should likely be modified to achieve better results.

The second approach is to adjust the asphalt layer seed modulus value for temperature. The previous day average air temperature is equal to 21°F. Using the BELLS equation, the asphalt mid-depth temperature in the range of 42°F to 46°F is calculated, with an average of 44°F. Assuming the reference modulus is 500 ksi at 77°F, the seed modulus at 42°F would be equal to 1,700 ksi using Equation 5-1. Using this value for the seed modulus of the P-401 layer and repeating the backcalculation analysis results in the backcalculated modulus values shown in Figure 5-11 and Figure 5-12. The layer modulus values for the majority of the test points are reasonable with relatively less variability compared to the results from the analysis using the default seed modulus values.

The third approach is to fix the modulus value of the asphalt. This can be done for composite pavement analysis due to the low stiffness of the surface layer relative to the P-501 layer. As seen in Figure 5-13, unchecking the "Moduli Changeable" box for the asphalt layer fixes the modulus value in the backcalculation. Fixing the asphalt modulus to 1,700 ksi results in layer modulus values similar to those in the second approach, as shown in Figure 5-14.

The backcalculated modulus values and statistics for the three approaches are shown in Table 5-6 using a log-normal distribution. The high variability in the backcalculated P-401 modulus values for the first approach is reflected by the high standard deviation factor of 2.326. The backcalculated layer modulus values for the second and third approaches are relatively similar, with those from the second approach (temperature-adjusted seed modulus values) being the most reasonable.

Of special note is the fact that the backcalculated subgrade modulus is higher than the target subgrade modulus (using a CBR of 15). This is a common problem when performing static backcalculation for stiff pavement structures such as rigid or composite pavements.

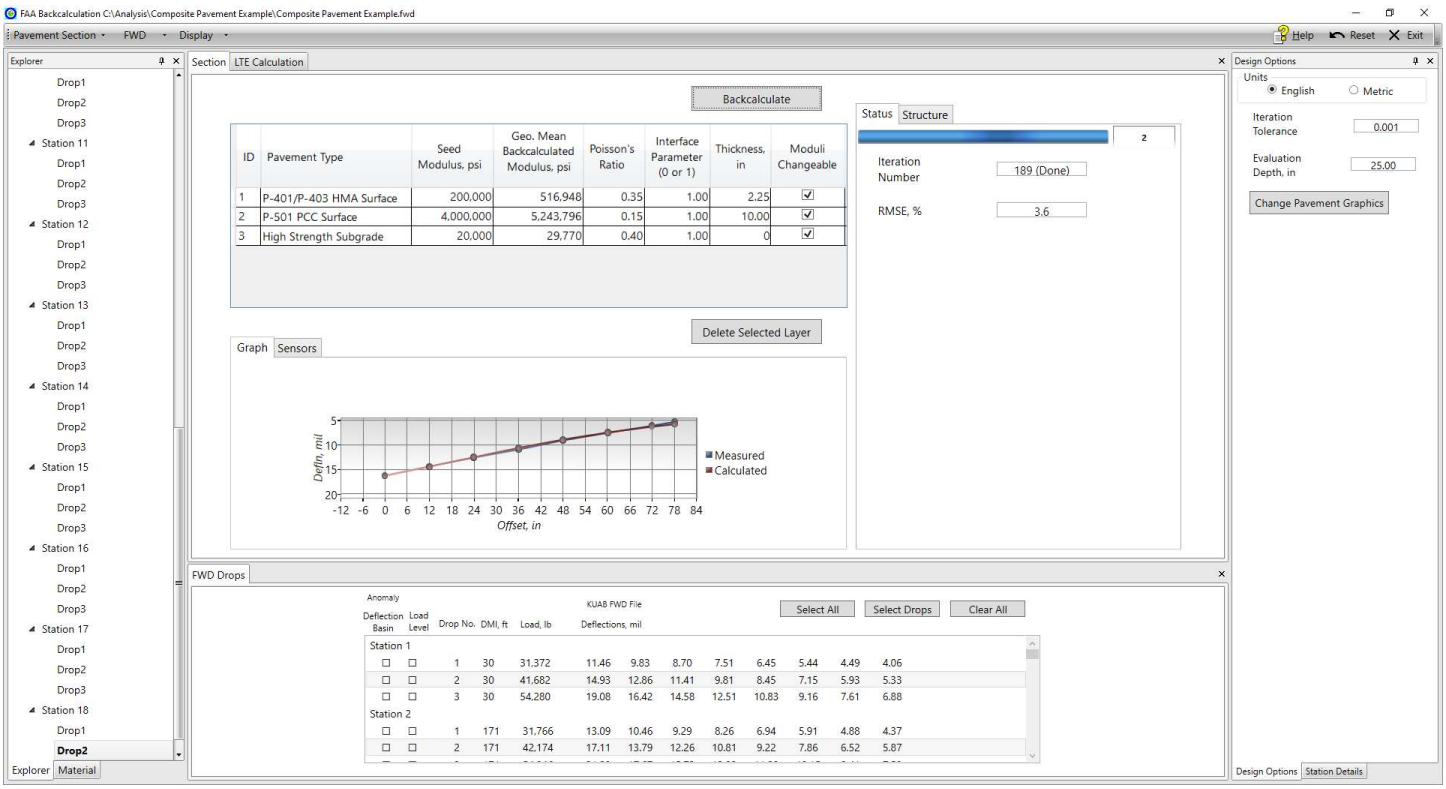


Figure 5-9. Backcalculation analysis using default seed modulus values in the composite pavement example

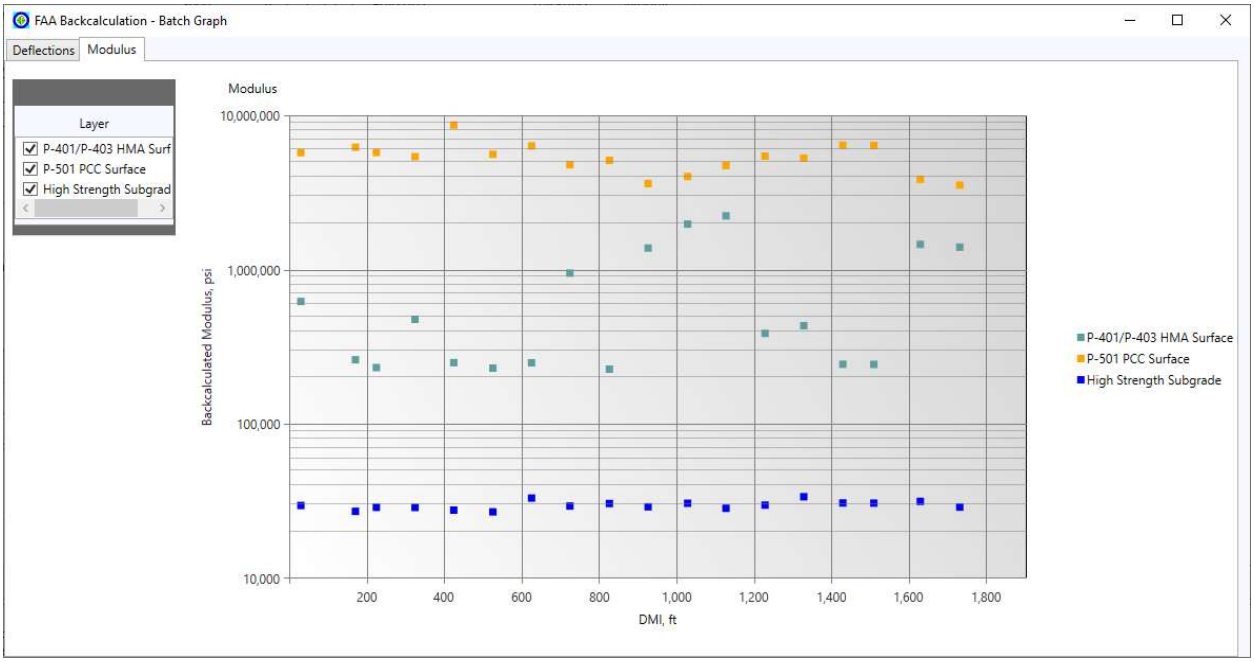


Figure 5-10. Backcalculated layer modulus values for drop no. 2 at all test points using default seed modulus in the composite pavement example

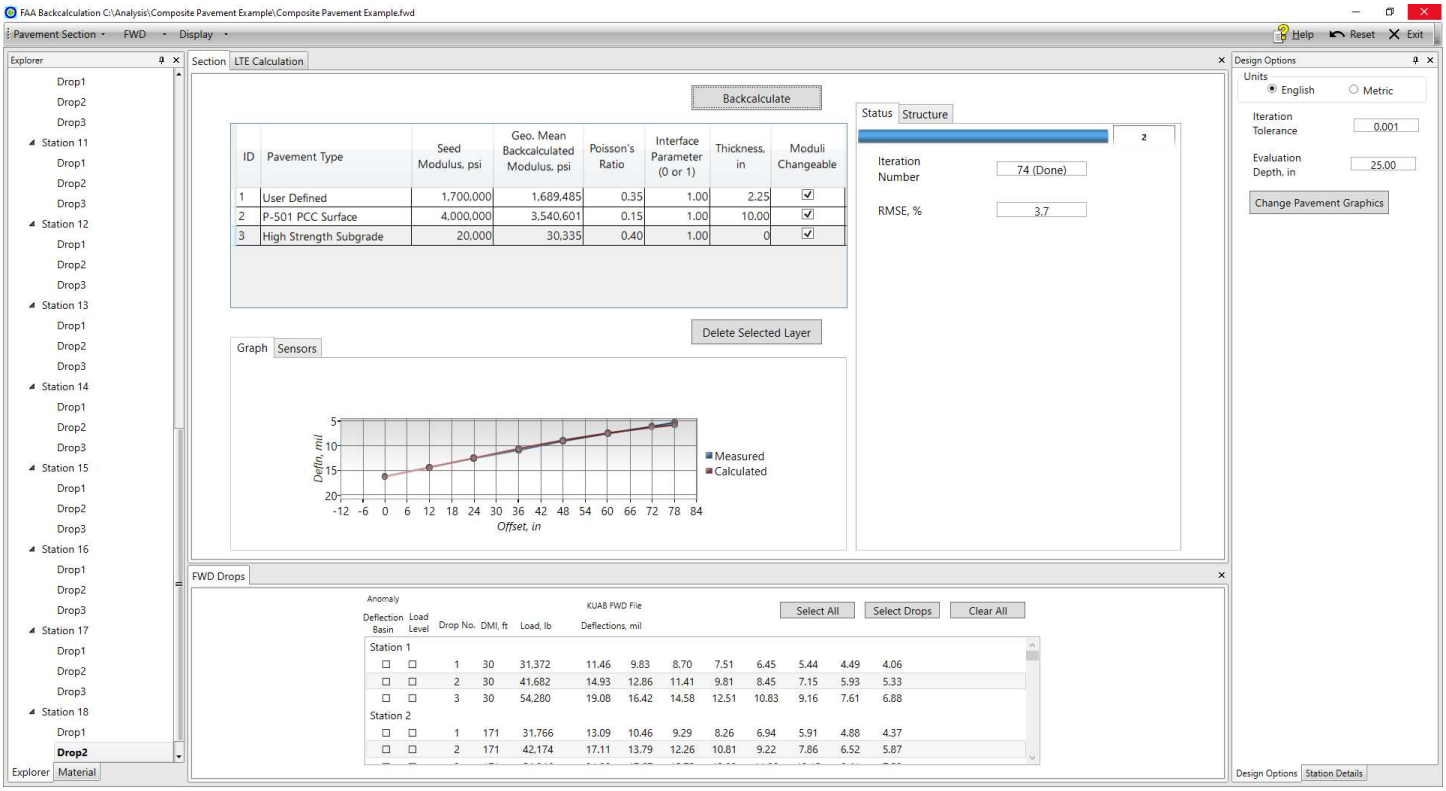


Figure 5-11. Backcalculation analysis using temperature-adjusted P-401 modulus in the composite pavement example

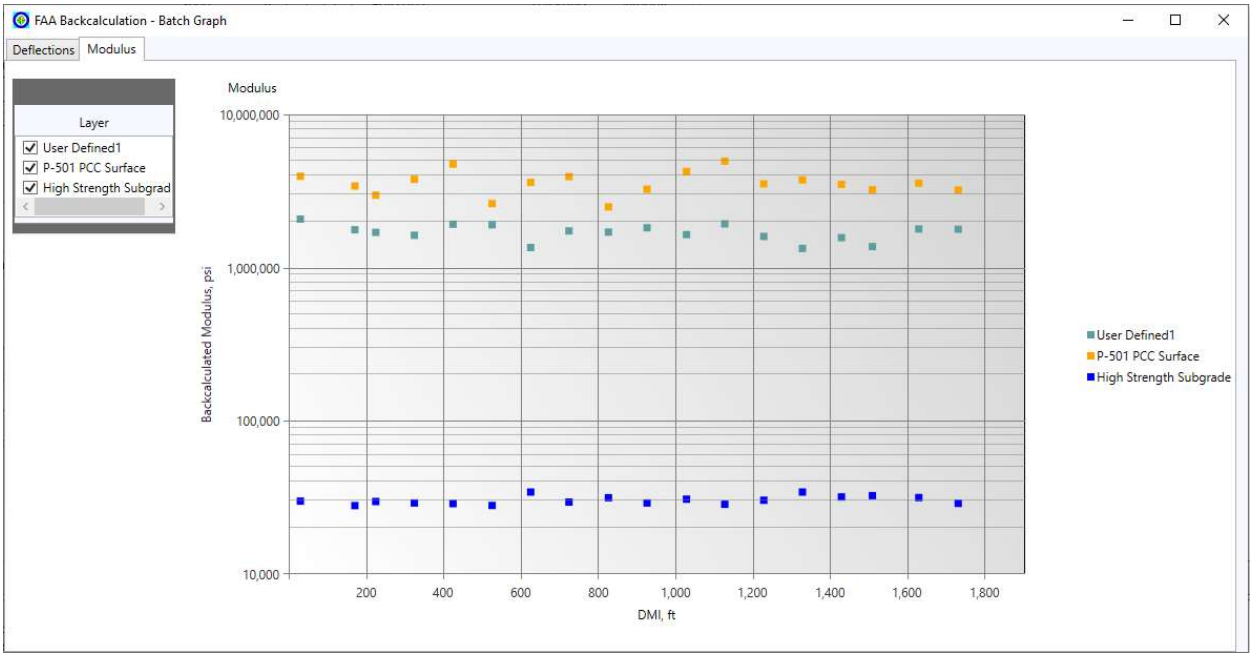


Figure 5-12. Backcalculated layer modulus values for drop no. 2 at all test points using temperature-adjusted P-401 seed modulus in the composite pavement example

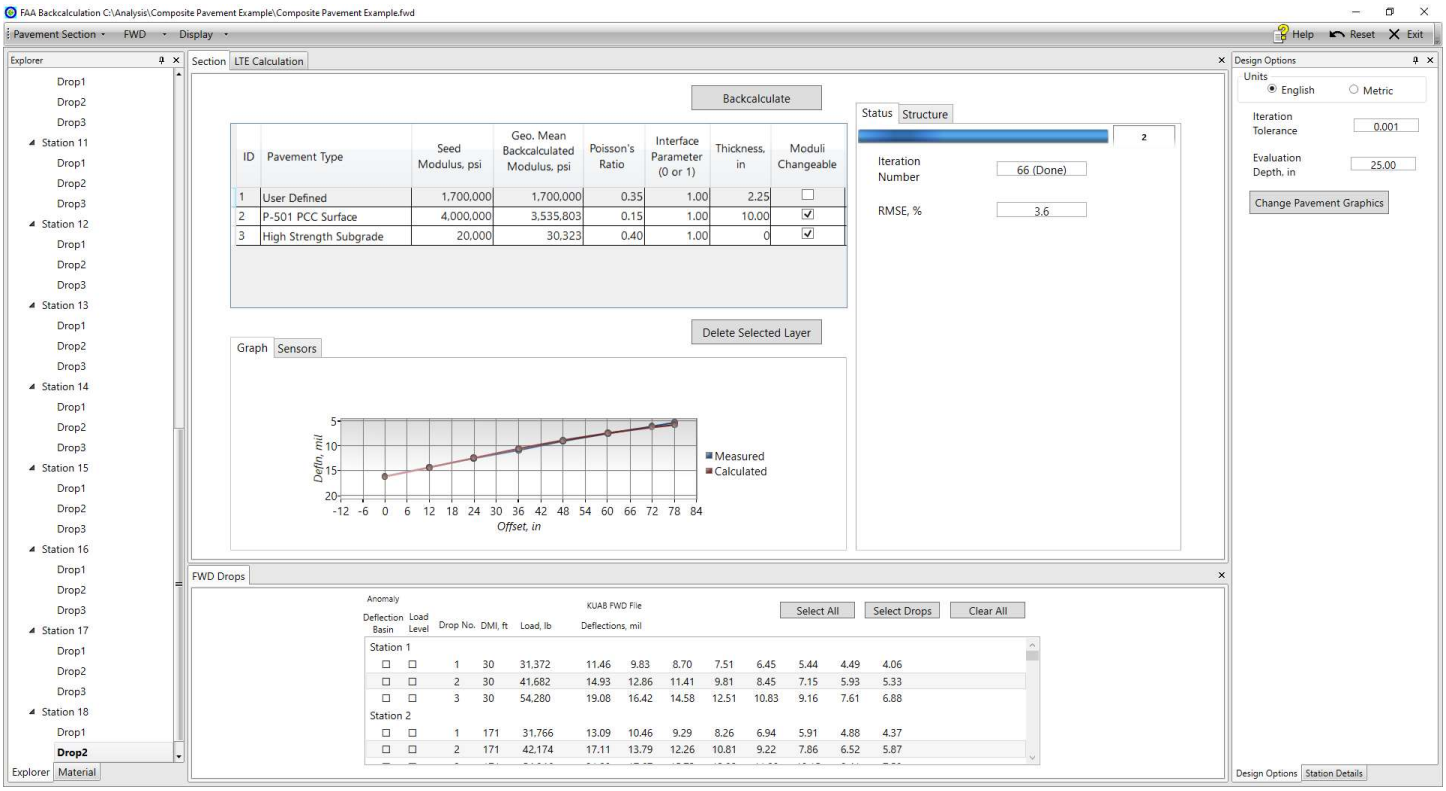


Figure 5-13. Backcalculation analysis using fixed P-401 modulus value in the composite pavement example

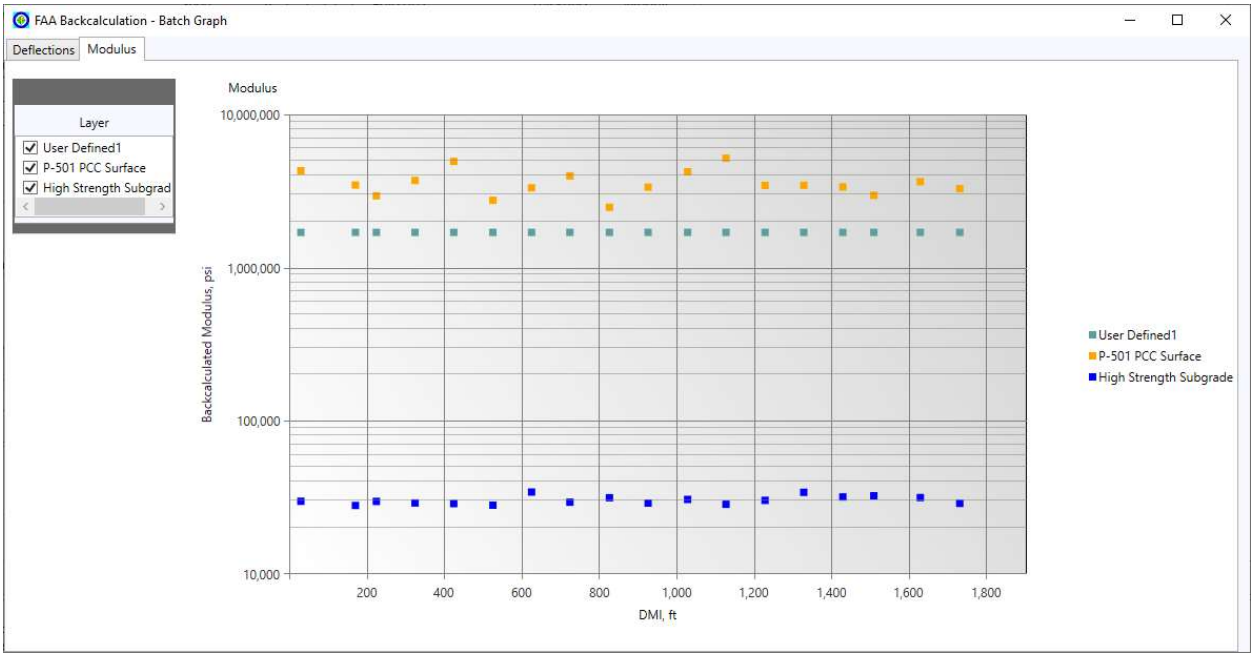


Figure 5-14. Backcalculated layer modulus values for drop no. 2 at all test points using fixed P-401 modulus value in the composite pavement example

Table 5-6. Composite pavement example backcalculated layer modulus values and statistics for three approaches

Case	Parameter[1]	Layer Modulus (ksi)		
		P-401	P-501	Subgrade
Default Seed Moduli	Average	517	5,244	29.8
	Standard Deviation Factor (STDF)	2.326	1.258	1.063
	16th Percentile (Average/STDF)	222	4,169	28.0
Adjusted P-401 Modulus	Average	1,689	3,541	30.3
	Standard Deviation Factor (STDF)	1.133	1.194	1.065
	16th Percentile (Average/STDF)	1,491	2,966	28.5
Fixed P-401 Modulus	Average	1,700	3,536	30.3
	Standard Deviation Factor (STDF)	1.000	1.209	1.064
	16th Percentile (Average/STDF)	1,700	2,923	28.5

1. Based on log-normal distribution.

5.4 LTE Calculation Example

This example uses data from an HWD test conducted on transverse joints of a jointed rigid pavement system. The HWD test configuration at the joints involves one sensor at the -12 in. offset placed on one slab and the load plate and other sensors placed on the adjacent slab. This corresponds to Case 2 as shown in Figure 5-15.

To perform the LTE calculation:

1. Load the saved “RW 16L34R Transverse Joint Tests.F25” file by clicking “FWD” in the toolbar then “Load FWD File” and selecting the file.
2. Select a drop and location to perform the LTE calculation (in this example, drop no. 3 at DMI location of 400 ft.).
3. Click the “LTE Calculation” module.
4. Select the appropriate case (in this example, Case 2).
5. Review the results in the LTE module, including loaded deflection, unloaded deflection, and the LTE percentage.

For this example, the LTE value is 86%.

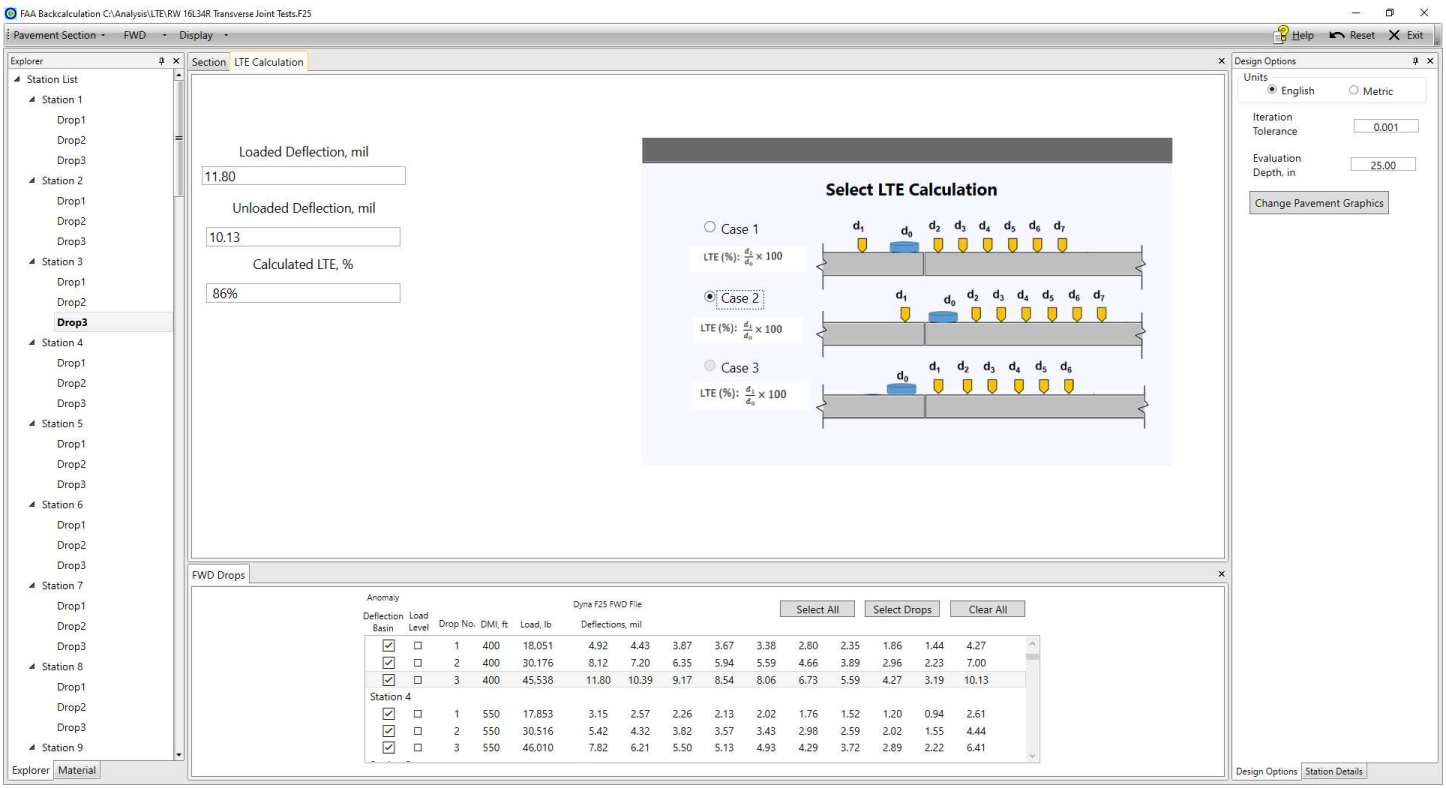


Figure 5-15. LTE calculation example

5.5 Error Messages

There are two types of error messages in BAKFAA: user input error (due to incorrect input by the user) and FWD file error (due to missing or misplaced data in FWD files).

5.5.1 User Input Errors

BAKFAA gives the following error messages related to input parameters:

1. Seed Modulus: The input for seed modulus value should be larger than zero.

   Error Message: Only values larger than 0 are allowed for Seed Modulus.

2. Poisson's Ratio: The input for Poisson's Ratio should be between 0 and 0.5.

   Error Message: Acceptable range for Poisson's Ratio is 0-0.5.

3. Interface Parameter: The input for Interface Parameter should be 0 or 1.

   Error Message: Acceptable values for Interface Parameter are 0 or 1.

4. Layer Thickness: the input for pavement layer thickness should be larger than zero, except for the last layer in the structure.

   Error Message: Zero is not acceptable for layer's thickness except for last layer (Subgrade)

   Error Message: Negative values are not acceptable for layer thickness.

5. Iteration Tolerance: the input for iteration tolerance should be between 0 and 1.

Error Message: Values less than or equal to 0 are not acceptable for iteration tolerance.

6. Deflection in “View FWD File” sub-window: The input for measured deflection should be larger than zero.

   Error Message: Only values larger than 0 are allowed for deflection.

7. Load in “View FWD File” sub-window: The input for load should be larger than zero.

   Error Message: Only values larger than 0 are allowed for load.

8. Pavement structure: Layers cannot be added below the subgrade.

   Error Message: <Material> layer cannot be presented below the Subgrade layer.

9. Pavement structure: More than one subgrade layer cannot be added.

   Error Message: Only one subgrade layer can be presented in the pavement structure.

5.5.2 FWD File Error

Depending on the type of corruption in the FWD file, an error message will pop up identifying the error type and the line in which the error occurred in the file.

CHAPTER 6. ADVICE AND REMINDERS

- The XML file generated by clicking “Pavement Section” then “Save Pavement Section” button can contain both pavement structure and FWD data or only pavement structure data, depending on the user selection.
- A subgrade layer has restrictions on where it can be placed in the structure, and only one subgrade layer can be present. If these restrictions are limiting user configuration, a “User Defined” layer with the same layer parameters does not have any restrictions and functions identically to the subgrade in BAKFAA.
- In a backcalculation analysis, there may exist many solutions (such as different combinations of layer modulus values) for a given pavement structure and measured FWD deflections. Backcalculation may be very sensitive to the input parameters such as seed modulus and iteration tolerance. Backcalculation often requires many iterations and use of engineering judgment to find the most reasonable results. Reasonable results are obtained when the backcalculated modulus values are both within an acceptable range and the RMSE is small (less than 5%).
- Backcalculation results are not always reliable for pavement systems with four layers or more. If possible, consider combining pavement layers with comparable modulus values and introduce them as one layer. In addition, backcalculation results for thin layers (less than 3 in. or 7.5 cm) may not be reliable. If possible, consider combining the thin layers with another layer.
- Adding a stiff layer (such as bedrock) as the last layer in the pavement system may improve the accuracy of the backcalculation analysis results. Where applicable, this should be based on construction records, geotechnical investigation, or knowledge of the site.
- Consider excluding sensors with negative offsets or sensors with errors from the backcalculation analysis.

CHAPTER 7. REFERENCES

American Association of State Highway and Transportation Officials. “Pavement Deflection Data Exchange: Technical Data Guide, Version 1.0”, 1998.

ASTM International. (2023), ASTM D5858-96(2020): Standard Guide for Calculating In Situ Equivalent Elastic Moduli of Pavement Materials Using Layered Elastic Theory.

ASTM International. (2023), ASTM D7228-06a(2020): Standard Test Method for Prediction of Asphalt-Bound Pavement Layer Temperatures.

Federal Aviation Administration Advisory Circular. (2011), Advisory Circular No. 150/5370-11B: Use of Nondestructive Testing in the Evaluation of Airport Pavements.

Federal Aviation Administration Advisory Circular. (2018), Advisory Circular No. 150/5370-10H: Standard Specifications for Construction of Airports.

Flannery, Brian P., Press, William H., Teukolsky, Saul A., Vetterling, William T. “Numerical Recipes in Fortran*”.* Cambridge University Press, 1992, Section 10.4.

Heukelom, W. and Klomp, A.J.G. (1962) Dynamic Testing as a Means of Controlling Pavement during and after Construction. Proceedings of the 1st International Conference on the Structural Design of Asphalt Pavement, Ann Arbor, Michigan, 20-24 August 1962, 667-685.

Rinde, J. A. "Poisson's ratio for rigid plastic foams." Journal of applied polymer science 14, no. 8 (1970): 1913-1926.

Samsonov, Gregory V. Handbook of the Physicochemical Properties of the Elements. Springer Science & Business Media, 2012.